

Writing the {Unwritable}: {Raveling} {Worlds}

Vulcan, Julie (2022.0)

ID do Documento: Writing-the-Unwritable-Raveling-Worlds

ABSTRACT ORIGINAL

“Are we on track to bridging the gap? Absolutely not.” We are still on target for a temperature rise in excess of three degrees by the end of this century. 3 On August 9, 2021 the human world is in its second year managing a slippery pandemic complicated by extreme weather events causing floods, wildfires, hurricanes, and cyclones on unprecedented scales. The Intergovernmental Panel on Climate Change (IPCC) releases the first contribution by Working Group I: Climate Change 2021: The Physical Science Basis for the Sixth Assessment Report. 4 The message is clear. We will most likely reach two degrees warming by the year 2050 unless global emissions are reduced now. The world as we know it seems to be unraveling. I engage raveling as a figuration to assist me and remind me that doing is also the process of undoing. 5 A ravel is a complication. Its etymological origins in weaving allow it to be at once an antonym and a synonym. To ravel is to entangle and disentangle. Humans ravel their beings and doings into the many beings and doings of the nonhuman world while simultaneously complicating the way the collective world is raveled. The greater effects of the more serious complications by some are now being keenly felt by most. The entangling and disentangling of things as we know them, for example a period of relative climate stability or the very notion of a thing, are being perceived as a kind of coming undone. It is a raveling-out that makes it plain the effects of human complications as we start to experience (individually and collectively) the climate and the world differently. What happens when the familiar is made unfamiliar? One definition for ravel out (although obsolete in current use) is to destroy, spoil or waste while to unravel can mean to free from obscurity or to reveal. 6 I engage the tension between these two intimately entangled meanings to explore the ways different entities can reveal themselves when other entities fall away. When what we know disintegrates and what remains is unfamiliar—a bush-land reduced to a negative space. 7 How might we imagine what can be possible when things feel precarious and unstable? Anna Tsing suggests to “live with precarity” requires a combination of noticing and stretching our imaginations into a “strange new world” to “grasp its contours” and the possibilities within it (2015: 2–3). In this way a disturbance like a wildfire, creates an opening. It beckons, drawing us closer into a place, rather than withdrawing. It encourages a “radical curiosity” and invites the “transformative encounter”(2015: 144, 152). To come closer might be to come alongside like a kind of walking-with or a journeying together. A journey that allows for indeterminacies rather than working to decipher, resolve or solve—processes determined to find an end. Walking-with, working-with, writingwith a companion that is not human, invites us into a new praxis. One that requires a letting go of preconceived notions of space and time attached to human notions of economy and resolution. A companion accompanies, sharing and assisting in company. Writing with a companion requires writing with “response-ability”—a Haraway term that recognizes the asymmetries with companions while allowing the opportunity to close gaps by cultivating a “capacity to respond”(Haraway 2016: 78). Paradoxical as it might seem in the context of this paragraph, a response-ability to ash might be through the simple action of inaction.

Careful {Speculations}: {Toward} a {Caring} {Science} of {Forensic} {Genetics} in {Colombia}

Olarte-Sierra, María Fernanda; Pérez-Bustos, Tania (2020.0)

ID do Documento: Careful-speculations-Toward-a-caring-science-of-forensic-genetics-in-Colombia

ABSTRACT ORIGINAL

158 Feminist Studies 46, no. 1. © 2020 by Feminist Studies, Inc. María Fernanda Olarte-Sierra and Tania Pérez-Bustos Careful Speculations: Toward a Caring Science of Forensic Genetics in Colombia Feminist Science and Technology Studies (STS) has recently opened up the question of care as a set of practices related to the sustainability of life.¹ The field of feminist studies more broadly has extensively 1. This literature mostly comes from Northern contexts and started to flourish with the seminal work of María Puig de la Bellacasa, "Matters of Care in Technoscience: Assembling Neglected Things," *Social Studies of Science* 41, no. 1 (2010): 85–106. This essay gathers scattered feminist STS reflections on these issues and looks at them through the lens of care, including authors such as Susan Leigh Star, Donna Haraway, Isabelle Stengers, Hilary Rose, and Annemarie Mol. The concept of care orienting this exercise of rereading these authors is proposed by Berenice Fisher and Joan Tronto, "Toward a Feminist Theory of Caring," in *Circles of Care: Work and Identity in Women's Lives*, ed. Emily K. Abel and Margaret K. Nelson (Albany: State University of New York Press, 1990), 35–62. Before and after that essay came out, Puig de la Bellacasa published several other papers related to this issue, including "Touching Technologies, Touching Visions. The Reclaiming of Sensorial Experience and the Politics of Speculative Thinking ," *Subjectivity* 28, no. 1 (2009): 297–315; "Nothing Comes without Its World': Thinking with Care," *The Sociological Review* 60, no. 2 (2012): 197– 216; "Ethical Doings in Naturecultures," *Ethics, Place & Environment* 13, no. 2 (2010): 151–69; "Making Time for Soil: Technoscientific Futurity and the Pace of Care," *Social Studies of Science* 45, no. 5 (2015): 691–716; and *Matters of Care: Speculative Ethics in More than Human Worlds* (Minneapolis: University of Minnesota Press, 2017). All these publications have been widely cited by other authors, particularly in a special issue of *Social Studies of Science* on the politics of care published in 2015 (volume 45, issue 5). Other examples María Fernanda Olarte-Sierra and Tania Pérez-Bustos 159 studied how care, and associated practices such as transnational care chains, are feminized and precarious—involving an ethos embodied by those who actively work to maintain or preserve another's life. Feminists have focused on the politics of care embedded in invisible jobs related to aesthetics, domestic work, child and elder care, and sexual activities and see all these tasks as related to the reproduction of life and culture.² Central to this conceptualization of care is the understanding that these jobs are typically women's work. There is a common association between this care and notions of "motherhood" as it relates to values such as compassion , love, and respect, which have been constructed as universally feminine . However, the main goal of feminist studies of care has been to deconstruct this association, particularly because it is imbued with the white, wealthy, heteronormative ideal of femininity that underlies a gendered division of labor. The reality is that class, race, nation, and gender determine who cares for whom, whether or not labor is paid, and to what extent care is exploitative. One strand of recent research problematizes both valorizing and criticizing care practices

on the grounds that they are based on a rather homogeneous understanding of the feminine. Instead, these scholars seek to resurrect valuable feminist work without relying on essentialism of authors contributing to this discussion on care as an ethos for science include Ruth Müller and Martha Kenney, "Agential Conversations: Interviewing Postdoctoral Life Scientists and the Politics of Mundane Research Practices," *Science as Culture* 24, no. 4 (2014): 537–59; Charis Thompson, *Good Science: The Ethical Choreography of Stem Cell Research* (Cambridge, MA: MIT Press, 2013); and Carrie Friese, "Realizing Potential in Translational Medicine: The Uncanny Emergence of Care as Science," *Current Anthropology* 54 (S7): S129-S138. 2. From Latin America, see, for example, Luz Gabriela Arango Gaviria, Adira Amaya Urquijo, Tania Pérez-Bustos, and Javier Pineda Duque, eds., *Género y cuidado: teorías, escenarios y políticas* (Bogotá: Universidad Nacional de Colombia, 2018); Pascale Molinier and Luz Gabriela Arango Gaviria, eds., *El trabajo y...*

Moving Crops and the Scales of History

Bray, Francesca; Hahn, Barbara; Lourdusamy, John Bosco; Saraiva, Tiago (2023.0)

ID do Documento: Moving-Crops-and-the-Scales-of-History

ABSTRACT ORIGINAL

A bold redefinition of historical inquiry based on the “cropscape”—the people, creatures, technologies, ideas, and places that surround a crop. Human efforts to move crops from one place to another have been a key driving force in history. Crops have been on the move for millennia, from wildlands into fields, from wetlands to dry zones, from one imperial colony to another. This book is a bold but approachable attempt to redefine historical inquiry based on the “cropscape”: the assemblage of people, places, creatures, technologies, and other elements that form around a crop. The cropscape is a method of reconnecting the global with the local, the *longue durée* with microhistory, and people, plants, and places with abstract concepts such as tastes, ideas, skills, politics, and economic forces. Through investigating a range of contrasting cropscaapes spanning millennia and the globe, the authors break open traditional historical structures of period, geography, and direction to glean insight into previously invisible actors and forces. © 2023 by Yale University. All rights reserved.

On the {Ground}: {Terrestrial} {Theopoetics} and {Planetary} {Politics}

van Horn, O'neil (2023.0)

ID do Documento: On-the-Ground-Terrestrial-Theopoetics

ABSTRACT ORIGINAL

A bold, theoretical, and pragmatic book that looks to soil as a symbol for constructive possibilities for hope and planetary political action in the Anthropocene. Climate change is here. Its ravaging effects will upend our interconnected ecosystems, and yet those effects will play out disproportionately among the planet's nearly 8 billion human inhabitants. *On the Ground* explores how one might account for the many paradoxical tensions posed by the Anthropocene: tensions between planetarity and particularity, connectivity and contextuality, entanglement and exclusion. Using the philosophical and theological idea of "ground," Van Horn argues that ground—when read as earth-ground, as soil—offers a symbol for conceiving of the effects of climate change as collective and yet located, as communal and yet differential. In so doing, he offers critical interventions on theorizations of hope and political action amid the crises of climate change. Drawing on soil science, theopoetics, feminist ethics, poststructuralism, process philosophy, and more, *On the Ground* asks: In the face of global climate catastrophe, how might one theorize this calamitous experience as shared and yet particular, as interconnected and yet contextual? Might there be a way to conceptualize our interconnected experiences without erasing critical constitutive differences, particularly of social and ecological location? How might these conceptual interventions catalyze pluralistic, anti-racist planetary politics amid the Anthropocene? In short, the book addresses these queries: What philosophical and theological concepts can soil create? How might soil inspire and help re-imagine forms of planetary politics in the midst of climate change? *On the Ground* thus roots us in a robust theoretical symbol in the hopes of producing and proliferating intersectional responses to climate change. © 2024 Fordham University Press. All rights reserved.

Caring for the environment: {Plastic} waste management and environmental concerns in {Colombian} coffee-growing communities

Arboleda Muñoz, Germán Antonio; Palacios, Lily Marcela; Portela Guarín, Hugo; Villada Castillo, Héctor Samuel (2024.0)

ID do Documento: Caring-for-the-environment-Plastic-waste

ABSTRACT ORIGINAL

A study was conducted with coffee growers from the municipalities of Caldon (Cauca) and Pitalito (Huila) in Colombia. Workshops, participant observation and farm visits were used to identify elements to understand (a) practices to the care of the environment, (b) the perception of the use of plastic materials, and (c) perspective toward biodegradable packages in their practices. Among the main results are an awareness for the care of the water resource and the care of the trees as its main contributions to the protection of its environment. In both communities, the problem is perceived due to the management of plastic waste and its influence on environmental deterioration due to poor management. They also recognize that the development of biodegradable bags can contribute to caring for their environment and make their economic activity more sustainable. © 2024 American Anthropological Association.

“Now Live from Lagos, Tehran and Oceanside”: Three B7CC Leaders Reflect on Strategies for Cooperation After the Anthropocentric Purge

Müller, Franziska (2023.0)

ID do Documento: Now-Live-from-Lagos-Tehran-and-Oceanside

ABSTRACT ORIGINAL

A videoscript from a documentary aired in 2100, this piece narrates how a new political institution formed in the aftermath of the “anthropocentric purge,” a climate change induced catastrophe that left vast areas uninhabitable. Three leaders of the organization based in Lagos/Nigeria, Tehran/Iran, and Oceanside/Colofornia(fka USA/Abya Yala) reflect how cooperation and solidarity-namely a new norm called “random ubuntu” have again materialized, while facing ecological trauma and long-lasting ecological destruction. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

Designing “Down to Earth.” Lessons Learned from Transformative Social Innovation

Tassinari, Virginia; Manzini, Ezio (2023.0)

ID do Documento: Designing-Down-to-Earth

ABSTRACT ORIGINAL

ABSTRACT The consequences of anthropocentric ways of thinking, designing, producing, and consuming are becoming painfully clear. Moving from this observation, several designers have become aware that design culture needs to reorient itself beyond anthropocentrism. In this article, we start with Bruno Latour’s proposal of coming “down to Earth,” which acknowledges that we are ultimately “terrestrials,” and connect it to Maria Puig de la Bellacasa’s idea of “care.” These concepts are pillars of the concrete guidelines we propose here: they are foundations for designing in a regenerative way, stepping beyond anthropocentrism. This paper traces a convergence between this emergent account of design and some recent social innovations, which are, in our opinion, making what it might mean to get down to Earth tangible. This convergence becomes particularly evident when introducing the concept of the “quality of complexity”: in other words, a qualitative dimension characteristic of those experimentations in which people re-orient their daily lives towards reweaving the web of life. While designing can strengthen its ability to be regenerative by exploring how to further engage with practices with this specific qualitative dimension, it might also provide philosophy with some concrete examples of a praxis taking further concrete steps down to Earth. © 2023 Informa UK Limited, trading as Taylor & Francis Group.

The {Ruined} {Seascapes} of the {Wasteocene}

Vina, Michael A. (2025.0)

ID do Documento: The-science-question-in-alternative-agricultures

ABSTRACT ORIGINAL

Abstract Along the south-central coast of Ecuador, the shifting seascape and broader economic processes bind inhabitants with the unique sociomaterial qualities of fish and their commodified products, such as fishmeal. Based on ethnographic fieldwork, this article argues that the vitality contained within fish eggs is not located within the fish itself but rather within the relationship between fish bodies and the specific human knowledges that coproduce the subjective life of fish and its commodified forms. By examining the concealment of fish from industrial boats to fishers' households as a form of "marginal gain," this work connects hope and anxiety with fishers' notions of decline and indeterminacy. Fishers' reappropriation of fish forms a critique of fishmeal economies through shifting environmental histories of the sea that are sensed through nonhuman bodies and experienced simultaneously as beneficial and detrimental. Rather than viewing these human and nonhuman configurations as another example of human hubris in the Anthropocene, this article proposes that such relations are better understood within the Wasteocene, a logic that exposes the ways that human and nonhuman bodies are rendered physically and conceptually wasted through fishers and corporate instances of caring and killing.

An active disturbance rejection control approach to vibration control on flexible systems based on frequency response

Lin, Shuyang (2024.0)

ID do Documento: An-active-disturbance-rejection

ABSTRACT ORIGINAL

Active disturbance rejection control (ADRC) is considerably applied due to its advantage of focusing on merely dominant parameters. Research on flexible systems frequently confronting perplexing disturbances can utilize this method to simplify irrelevant items as a single variable. In this paper, we focused on vibration control problems in flexible systems with the application of ADRC and constructed a second-order system model under the guideline of fundamental principles of ADRC and an innovative algorithm for tuning feedforward compensation ADRC. During the simulation, we discussed three cases in which each solely one parameter varies while others are kept invariant. Time, open-loop frequency, and close-loop frequency responses were respectively analyzed in all cases as to determine the stability of the system. According to the simulation results, we arrived at the conclusion: we should choose the specification of a flexible system within an intermediate range and evade from critical system parameters to procure stability and efficiency. © 2024 John Wiley & Sons Ltd.

Ways of World Knowing: Local Knowledge, Coastal Communities, and Equitable Ocean Governance

Massimi, Michela; Brown, Abbe E.L.; Jaspars, Marcel (2026.0)

ID do Documento: Ways-of-World-Knowing

ABSTRACT ORIGINAL

After decades of long and thorny negotiations at the United Nations, the High Seas Treaty was finalised in 2023, to protect marine biodiversity in areas beyond national jurisdiction. This treaty is just one of many instruments that have been introduced as part of global policy efforts to protect oceans amid a climate crisis that is having devastating effects on marine life. Despite the prevalence of ocean knowledge in coastal communities around the world, local ways of knowing have often been marginalised in scientific narratives and legal provisions. *Ways of World Knowing* brings together philosophers of science, marine scientists, and lawyers to discuss the role and importance of local coastal community knowledge in order to better represent their voices in ocean governance. By analysing the epistemic value of varieties of local knowledges, this volume invites us to overcome the dichotomy often found in legal documents between marine scientific research and local knowledge. Here, the contributors use the term 'local knowledge' (rather than 'traditional knowledge') deliberately to refer to varieties of ways of knowing more broadly understood—spanning coastal communities from Scotland to Canada, from Brazil to New Zealand—whose distinctive features include their being non-written, artisanal and experiential in nature, and intergenerationally transmitted. Topics include the knowledge of Brazilian fishing communities and of past Hebridean kelpmakers, the cultural significance of herring spawning for the Squamish Nation in Canada, the Indigenous People's knowledge in Australian legal provisions, and arts-based research and practice in international governance spaces, among others. By bringing a situated knowledge approach to ongoing, timely, and thorny governance questions about the ocean, this volume is an innovative contribution not just for the 'blue humanities' but for environmental studies at large at the intersection of philosophy of science, marine science, and environmental law. © Oxford University Press 2026.

OntoFusionCrop: An Ontology Centric Approach for Crop Recommendation Based on Bagging and Semantic Alignment

Chandramouli, Aparna; Deepak, Gerard (2022.0)

ID do Documento: Ontofusioncrop

ABSTRACT ORIGINAL

Agriculture is a crucial source of livelihood. There are many farmers who take up farming as their main source of income. The major difficulty present among farmers is that their crop selection is not based on the soil and weather conditions. Thus, it is essential for farmers to know the necessary information regarding the different facets of crop production, best agricultural practices, suitable crops for a specific soil type to be grown in a particular weather, etc. A crop recommender system, which has accurate and precise information, serves as a means for the farmers to farm appropriate crops and get the best yield. This paper proposes an ontology-based recommender system to provide the appropriate details regarding crops and other information associated with them. The proposed OntoFusionCrop uses ontology cluster that includes soil ontology, crop ontology, geographical ontology and agricultural ontology. This strategy uses the crop recommendation dataset from Kaggle which is classified using bagging. The classified instances along with the ontology cluster are semantically aligned using spider monkey optimization algorithm from which we get facts after suitable verification. The query is asked by the user and the recommender system gives the output. The proposed model achieved 93.87% accuracy and A minimum of 0.05 FDR compared to the existing models. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

HIAS: Hybrid Intelligence Approach for Soil Classification and Recommendation of Crops

Palvannan, S.; Deepak, Gerard (2022.0)

ID do Documento: HIAS-Hybrid-Intelligence-Approach-for-Soil-Classification

ABSTRACT ORIGINAL

Agriculture is the largest industry in the world, and it is vital to any country's economic development. Farmers' failure to select the appropriate crop for cultivation is a significant and severe mistake that results in poor agricultural output. For the surplus of crop production, a soil classification and crop recommendation system were developed, which used a semantically driven hybrid intelligence method. In this paper, the Hybrid Intelligence Approach for Soil (HIAS) model was presented and tested with several baseline models using southern South Australian Soil datasets to evaluate the proper prediction. Besides, ontology is generated from the summarised content of agricultural and geological eBooks. Textbooks and then the query and Soil dataset are pre-processed. The semantic similarity is calculated using Jaccard similarity, Cosine similarity, and SemantoSim measure under the Squirrel search algorithm to ensure that the relevant result is optimised and set entities formed from a set of recommendations are optimised. The term enrichment is done using MediaWiki and WikiData, and then the LSTM classifier is used for classification. Finally, the precision, accuracy, recall, F-measure, and FNR for the soil classification and crop recommendation system are evaluated. The precision percentage of 99.78% and the lowest FNR of 0.01 is obtained from the HIAS model. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

Effectiveness of Crop Recommendation and Yield Prediction using Hybrid Moth Flame Optimization with Machine Learning

Gopi, Subbu Raman; Karthikeyan, Mani (2023.0)

ID do Documento: Effectiveness-of-Crop-Recommendation-and-Yield-Prediction

ABSTRACT ORIGINAL

Agriculture is the main source of income, food, employment, and livelihood for most rural people in India. Several crops can be destroyed yearly due to a lack of technical skills and changing weather patterns such as rainfall, temperature, and other atmospheric parameters that play an enormous role in determining crop yield and profit. Therefore, selecting a suitable crop to increase crop yield is an essential aspect of improving real-life farming scenarios. Anticipating crop yield is one of the major concerns in agriculture and plays a critical role in global, regional, and field decision-making. Crop yield forecasting is based on crop parameters and meteorological, atmospheric, and soil conditions. This paper introduces a crop recommendation and yield prediction system using a Hybrid Moth Flame Optimization with Machine Learning (HMFO-ML) model. The presented HMFO-ML method effectively recommends crops and forecasts crop yield accurately and promptly. The proposed model used a Probabilistic Neural Network (PNN) for crop recommendation and the Extreme Learning Machine (ELM) method for the crop yield forecasting process. The HMFO algorithm was used to improve the forecasting rate of the ELM approach. A wide-ranging simulation analysis was carried out to evaluate the HMFO-ML model, showing its advantages over other models, as it exhibited a maximum R2 score of 98.82% and an accuracy of 99.67%. © by the authors.

Experiments in Situ: Soil Repair Practices as Part of Place- Based Action for Change in El Salvador

Millner, Naomi (2023.0)

ID do Documento: Experiments-in-Situ-Soil-El-Salvador

ABSTRACT ORIGINAL

Agroecological networks have long valorized the experience of small-scale farmers and have cultivated processes of in situ experimentation to systematically evaluate the effectiveness of traditional techniques. In this chapter I suggest that this focus on experience and experimentation is vital within broader ecologies of 'mending' environmental damage, and to the ideas of ecological reparation put forward by this book. Specifically, I suggest that the loop of collective reflection on experience afforded by the examples I document in El Salvador allow for a supplementation of official histories with small stories; with purposefully censored collective memory; and with everyday expertise. The case presented in this chapter is engaged in experimental soil repair through composting practices and the creation of 'biofertilizers' – recipes that use local organic ingredients to boost soil health. I show how agroecological approaches to soil repair place campesino expertise at the heart of agricultural innovation and ecological reparation. Together, these fuller archives allow for a wider array of possible solutions to soil damage, while also enabling contestation to take place over stories told over land, ownership, lives, and communities. © 2023 Bristol University Press.

Experiments in Situ: Soil Repair Practices as Part of Place- Based Action for Change in El Salvador

Millner, Naomi (2023.0)

ID do Documento: Experiments-in-Situ-Soil-Repair-Practices

ABSTRACT ORIGINAL

Agroecological networks have long valorized the experience of small-scale farmers and have cultivated processes of in situ experimentation to systematically evaluate the effectiveness of traditional techniques. In this chapter I suggest that this focus on experience and experimentation is vital within broader ecologies of 'mending' environmental damage, and to the ideas of ecological reparation put forward by this book. Specifically, I suggest that the loop of collective reflection on experience afforded by the examples I document in El Salvador allow for a supplementation of official histories with small stories; with purposefully censored collective memory; and with everyday expertise. The case presented in this chapter is engaged in experimental soil repair through composting practices and the creation of 'biofertilizers' – recipes that use local organic ingredients to boost soil health. I show how agroecological approaches to soil repair place campesino expertise at the heart of agricultural innovation and ecological reparation. Together, these fuller archives allow for a wider array of possible solutions to soil damage, while also enabling contestation to take place over stories told over land, ownership, lives, and communities. © 2023 Bristol University Press.

Take Care of Soils: Toward a Pluralistic Integral Soil Ethics

Grunwald, Sabine (2023.0)

ID do Documento: Take-Care-of-Soils-Toward-a-Pluralistic-Integral-Soil-Ethics

ABSTRACT ORIGINAL

Although land ethics and ethics focused on agriculture have been formalized, an explicit general soil ethics is lacking. The human approach to nature has been one of ethical relativism. Human-centric ethics that justify in one way or another the primacy of human desires, needs, values, and benefits have caused soil degradation and some serious destruction of the soil environment and planet Earth. This chapter provides an overview of normative ethical strands applied to soils/nature/environment that include (1) natural ethics (Natural Law ethics) grounded either in (a) entelechy, vital natural principle or (b) religious moralities (ethics of Divine Commands); (2) virtue ethics associated with valuation of soils/nature/Earth; (3) deontological ethics that is duty, obligation, or rule ethics subdivided into (a) act deontological ethics, (b) rule deontological ethics, and (c) Kantian ethics; (4) consequentialist ethics focused on the consequences (such as soil carbon sequestration; well-being and flourishing of people) with (a) ethical egoism, (b) act utilitarianism, and (c) rule utilitarianism; (5) intuitionist theory that combines duty and consequential ethics; and (6) relational ethics with (a) stewardship ethics and (b) ethics of care. A novel integrative soil ethics-Pluralistic Integral Soil Ethics (PISE)-is presented. PISE synthesizes multiple ethical pillars into a coherent soil ethics that informs the use, management, preservation, conservation, security, health, quality, and sacredness of soils. This integrative soil ethics features three pillars. The first pillar is focused on soil and environmental valuation and people's moral standards toward soil. This pillar explores the meaning of virtue, which is the quality of being morally good ("minimizing harm") in regard to soil and environment. The second pillar is focused on soil and environmental literacy (i.e., observable data, facts, knowledge, maps, and models of soils) and assessment of the consequences of soil use and management through understanding of cause-effect relations. The third pillar is focused on soil and environmental competency and awareness. This pillar represents ethics of care that stresses the empathic subjective and intersubjective intimate relations with soil/nature. Soil care is associated with people's closeness to personalize soils and making them one's own through sensing, feeling, and experiences of soil and nature. PISE is an integrative ethics that reframes "soil care" as a noun into an active process, that is, "to take care of soils," which is deliberately intentional to minimize soil degradation and honor soils in terms of their beauty and naturalness. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

"what is the Topic of the Group, Please?" on Migration, Care and the Challenges of Participation in Design

Krüger, Max; Weibert, Anne; De Castro Leal, Debora; Randall, Dave; Wulf, Volker (2021.0)

ID do Documento: what-is-the-Topic-of-the-Group-Please-on-Migration-Care-and-the-Challenges-of-Participation-in-Design

ABSTRACT ORIGINAL

An increasing number of people around the world are forced to leave their homes due to the climate catastrophe, regional conflict or poverty. In their new host countries however, refugees and migrants are frequently met by a wide range of challenges, including wider societal participation. The difficulties migrants and refugees face have also increasingly become the topic of HCI and CSCW work. In this paper we report on a three year-long project, involving refugees, migrants and activist supporters in a co-design project to develop tools that aid the process of resettling. Several aspects have challenged equal participation in the project, including divergent motives, unequal power distribution and cultural heterogeneity. Despite these challenges the project outcomes are in use and maintained beyond the project runtime through voluntary actors. We reflect on this discrepancy between process and outcome drawing on the concept of care. © 2021 ACM.

Ecological Masculinities: Theoretical Foundations and Practical Guidance

Hultman, Martin; Pulé, Paul M. (2018.0)

ID do Documento: Ecological-Masculinities-Theoretical-Foundations-and-Practical-Guidance

ABSTRACT ORIGINAL

Around the globe, unfettered industrialisation has marched forth in unison with massive social inequities. Making matters worse, anthropogenic pressures on Earth's living systems are causing alarming rates of thermal expansion, sea-level rise, biodiversity losses in terrestrial and aquatic ecosystems and a sixth mass extinction. As various disciplines have shown, rich white men in the Global North are the main (although not the only) perpetrators of this slow violence. This book demonstrates that industrial/breadwinner masculinities have come at terrible costs to the living planet and ecomodern masculinities have failed us as well, men included. This book is dedicated to a third and relationally focused pathway that the authors call ecological masculinities. Here, they explore ways that masculinities can advocate and embody broader, deeper and wider care for the global through to local ('glocal') commons. Ecological Masculinities works with the wisdoms of four main streams of influence that have come before us. They are: masculinities politics, deep ecology, ecological feminism and feminist care theory. The authors work with profeminist approaches to the conceptualisations and embodiments of modern Western masculinities. From there, they introduce masculinities that give ADAM-n for Earth, others and self, striving to create a more just and ecologically viable planet for all of life. This book is interdisciplinary. It is intended to reach (but is not restricted to) scholars exploring history, gender studies, material feminism, feminist care theory, ecological feminism, deep ecology, social ecology, environmental humanities, social sustainability, science and technology studies and philosophy. © 2018 Martin Hultman and Paul M. Pulé.

Application of artificial intelligence and machine learning in agriculture

Marwaha, Sudeep; Deb, Chandan Kumar; Haque, Md. Ashraful; Naha, Sanchita; Maji, Arpan Kumar
(2023.0)

ID do Documento: Application-of-artificial-intelligence

ABSTRACT ORIGINAL

Artificial intelligence (AI) is the branch of science that deals with the development of machines to mimic human intelligence. Machine learning (ML) is a subdomain of AI where the machine can learn automatically from data without being explicitly programmed. Agriculture is constantly pressed upon to produce more with less resource. AI and ML techniques have the capacity to optimize resource utilization by analysing agricultural data. It has changed the present-day face of farming by predicting various input parameters and forecasting post-harvest life of a crop. This chapter discusses the different AI and ML techniques available and how they have been used in different phases of the agriculture life cycle. This chapter includes vast range areas in agriculture that requires AI and ML. It includes soil, irrigation, and disease managements. Importance of AI in the field of plant phenomics also included in this chapter. The probable use of geographic information system (GIS) and remote sensing coupled with AI are discussed in this chapter. © Springer Nature Singapore Pte Ltd. 2023. All rights reserved.

Human-nature relations in urban gardens: Explorations with camera traps

Vella, Kellie; Ploderer, Bernd; Brereton, Margot (2021.0)

ID do Documento: Human-nature-relations-in-urban-gardens-Explorations-with-camera-traps

ABSTRACT ORIGINAL

As cities grow, their people become increasingly distanced from nature except within private and public green spaces. Sensing technologies provide a means to harness curiosity about the animals living in these spaces, and possibly also connect interest to care. Yet little is known as to how people may use these technologies, or the implications for human-nature relations. To learn more, we gave commercial camera traps to ten adult participants to understand how they explored their gardens, what they wanted to learn, and what they did with this knowledge. We discovered trade-offs between control and care; the usefulness of different media and mystery; the temporalities of engaging in natural sensing practice; and a prevalence of sharing media within households. We discuss design for convivial cohabitation with the creatures in our garden. This research contributes to better human-nature relations through citizen sensing, as well as HCI for urban biodiversity conservation. © 2021 ACM.

Ethical issues involving long-term land leases: A soil sciences perspective

Timmermann, C.; Félix, G.F. (2023.0)

ID do Documento: Ethical-issues-involving-long-term-land-leases

ABSTRACT ORIGINAL

As populations grow and arable land becomes increasingly scarce, large-scale long-term land leases are signed at a growing rate. Countries and investors with large amounts of financial resources and a strong agricultural industry seek long-term land leases for agricultural exploitation or investment purposes. Leaders of financially poorer countries often advertise such deals as a fast way to attract foreign capital. Much has been said about the short-term social costs these types of leases involve, however, less has been said about the normative dimension of their long-term environmental impact. We therefore will focus on the likely impact such deals have for soil conservation, by: (1) briefly introducing the basics of longterm leasing arrangements by comparing land leases to the renting of buildings; (2) explaining from a soil sciences perspective the difficulties in assessing the current value of an estate and in calculating the damages of soil erosion and degradation; and (3) show how difficult it is to incentivize the conservation of soil quality when one cannot sufficiently and cost-effectively valorise existing environmental capital and eventual future damages. Attempting to oblige tenants through contracts to invest in sustainable stewardship has limited potential when liability payments do not reflect true costs and are hard to enforce. © Wageningen Academic Publishers The Netherlands, 2019. All rights reserved.

An ontology of soil properties and processes

Du, Heshan; Dimitrova, Vania; Magee, Derek; Stirling, Ross; Curioni, Giulio; Reeves, Helen; Clarke, Barry; Cohn, Anthony (2016.0)

ID do Documento: an-ontology-soil-properties

ABSTRACT ORIGINAL

Assessing the Underworld (ATU) is a large interdisciplinary UK research project, which addresses challenges in integrated inter-asset maintenance. As assets on the surface of the ground (e.g. roads or pavements) and those buried under it (e.g. pipes and cables) are supported by the ground, the properties and processes of soil affect the performance of these assets to a significant degree. In order to make integrated decisions, it is necessary to combine the knowledge and expertise in multiple areas, such as roads, soil, buried assets, sensing, etc. This requires an underpinning knowledge model, in the form of an ontology. Within this context, we present a new ontology for describing soil properties (e.g. soil strength) and processes (e.g. soil compaction), as well as how they affect each other. This ontology can be used to express how the ground affects and is affected by assets buried under the ground or on the ground surface. The ontology is written in OWL 2 and openly available from the University of Leeds data repository: <http://doi.org/10.5518/54>. © Springer International Publishing AG 2016.

Urban food sharing: Rules, tools and networks

Davies, Anna R. (2019.0)

ID do Documento: Urban-food-sharing-Rules-tools-and-networks

ABSTRACT ORIGINAL

Available Open Access under CC-BY-NC licence. This book explores the history and current practice of food sharing. Illustrated by rich case studies from around the world, the book uses new empirical data to set an agenda for research and action. The book will be an important resource for researchers, policy makers and sharing innovators to explore the impacts and sustainability potential of such sharing for cities. © Policy Press 2019.

Valuing plants in devalued spaces: Caring for Baltimore's Street trees

Shcheglovitova, Mariya (2020.0)

ID do Documento: Valuing-plants-in-devalued-space

ABSTRACT ORIGINAL

Baltimore City, MD is addressing its future with expansive sustainability initiatives. These include an aggressive tree planting campaign to double the city's tree canopy by 2037. While discourses of greening present tree plantings and related programs as a resolution for the legacies of racist housing market practices, these programs are themselves subject to the legacies of spatial inequalities in access to infrastructural care. Sustainability discourses present urban trees as inherently valuable economically and environmentally but these discourses are disconnected from trees' needs for ongoing care and maintenance. The daily material practices of caring for and maintaining trees are deprioritized in favor of planting more trees to gain these supposedly "inherent" benefits. In the spaces where trees were meant to bring economic and environmental vitality, their deaths reinforce the racist legacies they claim to correct. This paper examines these links and contradictions within the framework of relational urban political ecology. Through a lens of care, this paper shows how humans and non/humans actively co-construct urban space and how just spaces can come about through attention to the needs of humans and non/humans. © The Author(s) 2019.

Making soil in the Plantationocene

Ofstehage, Andrew (2024.0)

ID do Documento: Making-soil-in-the-Plantationocene

ABSTRACT ORIGINAL

Based on 14 months of ethnographic research, this paper analyzes soil management within the plantation model of farming in order to understand the extent to which life on large-scale monocultural farms can be controlled and directed toward extractive production. Transnational soy farmers in Western Bahia Brazil 'correct' soils in the region to make them productive and marshal this agronomic work to claim that they have added value to the land by 'building it up'. Still, the permeability of the plantation keeps transnational farmers from achieving their dreams of control. © 2023 Informa UK Limited, trading as Taylor & Francis Group.

Wine Is Our Bread: Labour and Value in Moldovan Winemaking

Ana, Daniela (2022.0)

ID do Documento: Wine-Is-Our-Bread

ABSTRACT ORIGINAL

Based on ethnographic work in a Moldovan winemaking village, *Wine Is Our Bread* shows how workers in a prestigious winery have experienced the country's recent entry into the globalized wine market and how their productive activities at home and in the winery contribute to the value of commercial terroir wines. Drawing on theories of globalization, economic anthropology and political economy, the book contributes to understanding how crises and inequalities in capitalism lead to the 'creative destruction' of local products, their accelerated standardization and the increased exploitation of labour. © 2022 Daniela Ana. All rights reserved.

Funding priorities and climate change: Lessons from agriculture and action-oriented science

Shaw, Jamie (2025.0)

ID do Documento: Funding-priorities-and-climate-change

ABSTRACT ORIGINAL

Billions of dollars get spent every year on climate change research-whether trying to understand its mechanisms, or how to intervene to prevent some of its catastrophic consequences. While the enormous industry of climate change science leaves plenty of room for pursuing a wide array of projects, there remain pertinent questions about identifying the most promising potential disruptive innovations with regards to climate change. To make things more complicated, the context of application is usually several degrees of separation away from where most climate change science gets pursued. This disconnect between research and action makes the problem of identifying research priorities especially challenging in this context. The purpose of this chapter is to make progress on this topic by interpreting the funding situations behind two case studies on disruptive innovations on climate change. First, the chapter considers holistic management or a series of techniques for rotating livestock which arguably has the potential to create more effective carbon sinks. Second, it considers some of the projects funded by the Paul K. Feyerabend Foundation related to environmental understanding and action. Finally, the chapter considers the various ways in which funding agencies have been successful or hampered in supporting important disruptive innovations, showing how this can relate to institutional demarcation criteria about what counts as 'science.' © 2026 Donald S. Maier, Justin Donhauser, and Michael Weber. All rights reserved.

Industrializing Bacterial Work: Microbiopolitics, Biogas Alchemy, and the French Waste Management Sector

Daniel, François-Joseph (2023.0)

ID do Documento: Industrializing-Bacterial-Workus

ABSTRACT ORIGINAL

Biological waste recycling has recently attracted widespread interest and investment. Large industrial plants that use microbiological engineering to process municipal waste and produce biogas have been established in different countries including Germany, France, Portugal, Brazil, Canada, and China, to name a few. These biowaste facilities are not simply classical energy infrastructures, as they are commonly described, but rather rely on the power of bacteria, archaea, and fungi at several levels to accomplish the work of waste metamorphosis. Such an appropriation of microbes' vital force is based on specific and complex human–microbe relations, or microbiopolitics, that rely on practices of attention, care, and proximity with waste material. However, in these industrial attempts of upgrading the metabolic work of bacteria, the need for more hands-on daily care of waste materials and biological processes is being superseded by the automation of waste processing. Close examination of the French context shows that this shift produces ignorance regarding the growth and evolution of bacterial colonies and reduces humans' attention and proximity to the industrial process, thereby depriving the microbes of elements that hitherto kept them domesticated. © The Author(s) 2022.

Rhythm Matters! How Rhythm Analysis Bridges Architecture and Sociology

Sefkatli, Pinar (2024.0)

ID do Documento: Rhythm-Analysis-Bridges-Architecture-and-Sociology

ABSTRACT ORIGINAL

By bringing the social dynamics in cities to the surface, rhythm analysis offers novel insights into the spatio-temporal notions that characterise urban life and experiences. The research presented in this contribution proposes a rhythm analysis framework for bridging architecture and social sciences to facilitate design interventions that are fine-tuned to the social context in cities. The results identify three aspects of city rhythms: (1) rhythms in the outdoor spaces and around the buildings, conceptualised as “rhythm zones”, (2) rhythms in the care services offered by societal organisations that are hosted in public and semi-public buildings, conceptualised as “rhythm-scapes” and (3) rhythms of ordinary activities that shape the urban social life and interactions, conceptualised as “rhythm spheres”. Each aspect includes different methods for analysing and documenting rhythms, enabling novel possibilities for spatial practices. Focusing on the dynamic quality of physical urban spaces, relations between actors and institutions and interactions between communities creates a bridge between academic research and spatial practice at a level of analysis of the urban social life that is straightforward, replicable and inventive. © (2023), (Lifelong Learning EU). All Rights Reserved.

Walking with rocks – with care

Rantala, Outi; Valtonen, Anu; Salmela, Tarja (2020.0)

ID do Documento: Walking-with-rocks-with-care

ABSTRACT ORIGINAL

By way of combining insights from the emerging geosocial literature and feminist new materialism, the chapter develops a situated, 'down-to-earth' embodied perspective to investigate the lively relationalities of rocks and humans, and the way care is implicated in them. The analysis demonstrates that care is manifested in these relationalities in complex ways, taking the form of both detachment and entanglement, which do not exclude one another but instead contest an assumption of a harmonic relation between human and more-than-human subjects. Hence, the study provides a different story from that which is usually told about the Arctic area, of which the region studied is a part. © Anu Valtonen, Outi Rantala and Paolo Davide Farah 2020. All rights reserved.

Community economies and climate justice

Roelvink, Gerda (2018.0)

ID do Documento: Community-economies-and-climate-justice

ABSTRACT ORIGINAL

Climate change appears to act as a metanarrative of concern linking different areas of political-economic activism with environmental justice movements. Community economies' scholars would certainly agree with Naomi Klein's call for action and appreciate the devastating impact of human life on the planet, yet they significantly depart from the politics that comes with capitalocentric thinking on climate change. The generational yardstick offers an alternative view to "big" governmental change and also gets activists away from the paralysis that comes from focusing on the need to change a "capitalist" system. An entry point to collective change making in response to climate change, has been developed through recent work on the commons. The commons is a central theme within the climate justice movement with several different meanings. Another community economies entry point to thinking about economic life in a time of climate change, and in particular exchanges with the more-than-human world, is perhaps surprisingly though research on the species-being of humans. © 2018 selection and editorial matter, Stefan Gaarsmand Jacobsen; individual chapters, the contributors.

Caring for strangers: Alterity, alliances, and reptile conservation in the “gecko garden refuges” in manapany-les-bains, la réunion

Krieg, Lisa Jenny (2020.0)

ID do Documento: Caring-for-strangers-Alterity-alliances-and-reptile-conservation

ABSTRACT ORIGINAL

Commonly seen as either dangerous or unsocial, reptiles represent a kind of alterity that is often deemed categorically different from warm-blooded mammals. In the village of Manapany-les-Bains, however, on the island of La Réunion in the Western Indian Ocean, care for the endangered gecko species *Phelsuma inexpectata* has taken an unexpected turn: initiated by a local NGO, village residents have declared their private gardens a gecko-friendly zone. Engaging concepts of multispecies care and alterity, this article explores what caring for a reptile means. Based on ethnographic field research in Manapany-les-Bains, I will discuss how the gecko's otherness remains an ambivalent trope, being both reduced and capitalized on, and that new alliances are exclusionary and fragmented. © Franz Steiner Verlag, Stuttgart 2020

Clustering of meteorological data to improve agricultural decisions: a case study with SIMAGRO-RS

De Oliveira, Marcos Antonio; Varone, Flavio A.; Fraisse, Clyde W.; Araújo, Ricardo Matsumura;
Cavalheiro, Gerson Geraldo H. (2024.0)

ID do Documento: Clustering-of-meteorological-data-to-improve-agricultural-decisions

ABSTRACT ORIGINAL

Context: Climate variability plays a fundamental role in shaping agricultural outcomes, influencing crop growth, yield, and resource management strategies. Problem: Identifying meaningful patterns in meteorological datasets to optimize agricultural practices is a challenging and complex task. Solution: This paper explores the use of clustering techniques on SIMAGRO-RS weather data to uncover meaningful patterns for agricultural applications. IS Theory: This study addresses the challenges of Information Systems applied to Agriculture, especially related to General Systems Theory (TGS). Method: The data was, first, extracted, then, preprocessed, filtered, aggregated, and standardized. Clustering was performed using K-means, K-medoids, and Hierarchical Agglomerative Clustering algorithms to cluster data from 19 meteorological stations in groups of stations with similar climate behavior. To define the number of clusters, the Elbow Plot and Silhouette Score methods were used. After clustering, the clusters were evaluated to identify the best result in terms of intracluster similarity. Supervised learning was used to design a specific metric for this problem, using an RMSE error measure. Summary of Results: From the experiments, the number of clusters $k = 2$ was found as the optimal number for the dataset. Regarding the clustering algorithm, the K-medoids algorithm presented better performance, with lower prediction RMSE (0.4990). IS Contributions and Impact: Based on the results, ways of interpreting clustering are suggested, in order to make it useful in the planning of agricultural practices in the state of Rio Grande do Sul. © 2024 ACM.

Analysis of the Use of Mobile Application to Advance Agricultural Traceability

Reffatti, Leonardo; Porto, Josiane Brietzke; Barbosa, Jorge Luis Victoria (2022.0)

ID do Documento: Analysis-of-the-Use-of-Mobile-Application-to-Advance-Agricultural-Traceability

ABSTRACT ORIGINAL

Context: Nova Aliança has been industrializing the production of grapes by 410 families of winegrowers in Serra Gaúcha, since 2011. Grapes, the basis for beverages, are traced from the producer to the industry. Problem: In Brazil, traceability of plant products is a requirement of the market and became mandatory with Joint Normative Instruction ANVISA/SDA No. 2 of 07/02/2018, aiming at monitoring pesticide residues. However, little is known about the potential of digital tools for agricultural traceability. Solution: This article presents results of the transformation of physical documents to digital of the production areas and the use of harvest lots to improve the tracking of fruit loads, through a mobile application, developed under Design Science Research. IS Theory: It deals with challenges of Information Systems applied to Agriculture, under the Design Theory and Design Science paradigm. Method: The method is characterized by observational, analytical, and exploratory technological research, operationalized by Design Science Research, with initial evaluation of the artifact via action research in this real context of use and quantitative analysis of the results. Summary of Results: The mobile application was used by 127 families, representing 30.97% in the 20/21 harvest. It replaced the records in physical documents completely, made it possible to improve the traceability of production in 20 million kilograms of grapes and minimized problems regarding cargo compliance. Contributions and Impacts in the IS Area: This solution has been widely used by farmers and the results point to improved traceability and reinforce viability for agriculture. © 2022 ACM.

RESPONSIBILITY BEYOND GROWTH: A Case for Responsible Stagnation

de Saille, Stevienna; Medvecky, Fabien; van Oudheusden, Michiel; Albertson, Kevin; Amanatidou, Effie; Birabi, Timothy; Pansera, Mario (2020.0)

ID do Documento: RESPONSIBILITY-BEYOND-GROWTH-A-Case-for-Responsible-Stagnation

ABSTRACT ORIGINAL

Critically assessing growth-based models of innovation policy, this enlightening study sparks new debate on the role and nature of responsible innovation. Drawing on insights from economics, politics, and science and technology studies, it proposes the concept of 'responsible stagnation' as an expansion of present discussions about growth, degrowth, responsibility and innovation within planetary limitations. This important intervention explores real-world relationships between the political economy, innovation policy and concepts of responsibility, and will be an invaluable resource for individuals and civil society organizations who seek to promote responsible innovation. © Bristol University Press 2020.

‘Thinking with Kaipara’: A decolonising methodological strategy to illuminate social heterogeneous nature–culture relations in place

Makey, Leane (2022.0)

ID do Documento: Thinking-with-Kaipara

ABSTRACT ORIGINAL

Decolonising methodologies continue to be critically developed to disrupt the marginalising approaches to knowledge production. By privileging Indigenous ways of knowing, being and doing, relations with nature are a more-than-human entanglement and a relational pursuit. Ecosystems such as estuaries and rivers are connected through kin-based relationships and treated as (or are) ancestors and family members. Such embodiment connects the body–mind–spirit to maintain relations with the mauri of ancestral beings and Deities. Within this ontology, nature is indistinguishable from culture. Our research responds to the call of how method might proceed to de-centre the human and make bodily and material encounters with the nonhuman matter. ‘Thinking with Kaipara’ is a methodological strategy that is a deliberate attempt to pursue embodied ways of producing knowledge. To work with situated knowledges, place and social difference to address the crisis of representation of such in ecosystem-based management, the problematising of ecosystem degradation and restoration practices. Research data produced is founded on human and nonhuman collaboration, an ethic of care, diverse epistemic nature–culture relations, social/nature/gender justice and equality, which makes for empirical evidence not enjoyed by scientific and technocratic methods utilised to inform and shape ecosystem-based management decisions and policy. Geo-creative practices are used by co-researchers/co-participants to recount their lived experiences and knowledges of ecosystem degradation and restoration. Storytelling, poetics, and painting, sculpting, whaikorero, waiata and writing were practices used. We argue that such geo-creative practices challenge the normative spaces and practices of disciplinary knowledge-making and enable the examination of social heterogeneous nature–culture relations in settler-colonial societies. © The Author(s) 2021.

Can caring democracy be a political concept for transformation?; [Kann sorgende demokratie eine politische theorie der transformation sein?]

Tronto, Joan C. (2016.0)

ID do Documento: Can-caring-democracy-be-a-political-concept-for-transformation

ABSTRACT ORIGINAL

Departing from debates in the 1980s among feminist Marxists in Germany and American feminist theorists of care, the article situates the similar origins of ongoing debates regarding women's larger exclusion from social life. The argument compares the discussion on production/reproduction to the caring democracy approach currently being developed. Here, care is oriented toward relationships, while reproduction/production focuses on capacities. The article then provides an epistemological grounding for care as a type of expressive-collaborative moral theory according to terminology used by Margaret Walker. It develops and defends that the logic of care is ultimately incompatible with capitalism, yet also recognizes its limits.

The Stuff of Soil Belowground Agency in the Making of Future Climates

Granjou, Céline; Salazar, Juan Francisco (2019.0)

ID do Documento: The-Stuff-of-Soil-Belowground-Agency-in-the-Making-of-Future-Climates

ABSTRACT ORIGINAL

Despite soil's vital ecological importance, its significance as a belowground tridimensional living world remains under-theorized in social and cultural research. Drawing on the reading of scientific literature and a series of interviews with scientists working at the juncture of soil and climate research, this article pursues a picture that highlights soil's capacities to shape future climates, including by fostering major planetary tipping points; we elaborate on the cultural and ethical significance of that picture for opening up alternative stories in which agency and change are not human-only prerogatives. We develop a critical stance on the growing expectations of storing more carbon into soils and argue for a better consideration of the situated, heterogeneous, and volatile dynamics of carbon within soils. We eventually call for more responsible ways of thinking about, and caring for, the myriad conglomerates of living, decaying, and dead matter that basically make up the stuff of soil. © Berghahn Books

Mountain Bike Trail Building, “Dirty” Work, and a New Terrestrial Politics

Cherrington, Jim; Black, Jack (2020.0)

ID do Documento: Mountain-Bike-Trail-Building-Dirty-Work-and-a New-Terrestrial-Politics

ABSTRACT ORIGINAL

Dirt is evoked to signify many important facets of mountain bike culture, including its emergence, history, and everyday forms of practice and affect. These significations are also drawn on to frame the sport's (sub)cultural and counterideological affiliations. In this article we examine how both the practice of mountain biking and, specifically, mountain bike trail building, raises questions over the object and latent function of dirt, hinting at the way that abjection can, under certain circumstances, be a source of intrigue and pleasure. In doing so, we suggest a resymbolization of our relationship with dirt via a consideration of the terrestrial. © 2019, © 2019 Taylor & Francis Group, LLC.

Ontology of Crop Pest Control

Onkov, Kolyo (2020.0)

ID do Documento: ontology-crop-pest-control

ABSTRACT ORIGINAL

Domain ontology of crop pest control consists of hierarchically structured biological and chemical information and concepts on crops, pests, pest control measures and relations among them. Despite vertical relations in hierarchies, the knowledge about crop protection measures leads to horizontal relations between classes biological and chemical objects. There is analogy between class objects in biological classification of crops, pests and pest control measures from one hand and from the other hand class objects and instances of object oriented programming. The developed domain ontology has characteristics of task ontology because it leads to building analytical models, data analysis and solving practical problems. Classification of tasks and applications based on the domain ontology is in the scope of this work, as well. The main aim of the ontology is the development of intelligent computer based systems intended to satisfy specific informational needs of the professionals and practitioners in agronomy, crop protection, plant medicine, economics and business. © 2020 ACM.

‘When you heal the soil...’: Environmental racism and socioecological repair in contemporary urban agriculture

Shostak, Sara (2022.0)

ID do Documento: Environmental-racism

ABSTRACT ORIGINAL

Drawing on data from an ethnographic study of urban agriculture in Massachusetts, this paper investigates the multiple meanings of soil for contemporary urban farmers and gardeners. I first consider how urban farmers speak for and with the soil in their neighborhoods to call attention to historical and ongoing environmental racism. These narratives highlight how racialized social processes—including redlining, blockbusting, white flight and disinvestment—have harmed the health of both people and the environment in urban communities of color. I then describe how urban farmers and gardeners articulate the importance of soil for health and well-being, especially for people whose relationships with the earth have been disrupted by capitalism, colonialism and racism. These narratives draw on both scientific and spiritual frameworks to highlight the healing potential of re-establishing direct relationships with nature, reclaiming ancestral knowledge about the healing properties of plants, and reconnecting with the ancestors themselves. Analysis of these interlinked narratives contributes to an emerging cross-disciplinary scholarship on the situatedness of ways of conceptualizing and interacting with soils, calling attention especially to the role of racialized inequities in the creation of harmful soil materialities and the possibilities of socioecological repair. © 2022 Informa UK Limited, trading as Taylor & Francis Group.

‘When you heal the soil...’: Environmental racism and socioecological repair in contemporary urban agriculture

Shostak, Sara (2022.0)

ID do Documento: When-you-heal-the-soil-Environmental-racism

ABSTRACT ORIGINAL

Drawing on data from an ethnographic study of urban agriculture in Massachusetts, this paper investigates the multiple meanings of soil for contemporary urban farmers and gardeners. I first consider how urban farmers speak for and with the soil in their neighborhoods to call attention to historical and ongoing environmental racism. These narratives highlight how racialized social processes—including redlining, blockbusting, white flight and disinvestment—have harmed the health of both people and the environment in urban communities of color. I then describe how urban farmers and gardeners articulate the importance of soil for health and well-being, especially for people whose relationships with the earth have been disrupted by capitalism, colonialism and racism. These narratives draw on both scientific and spiritual frameworks to highlight the healing potential of re-establishing direct relationships with nature, reclaiming ancestral knowledge about the healing properties of plants, and reconnecting with the ancestors themselves. Analysis of these interlinked narratives contributes to an emerging cross-disciplinary scholarship on the situatedness of ways of conceptualizing and interacting with soils, calling attention especially to the role of racialized inequities in the creation of harmful soil materialities and the possibilities of socioecological repair. © 2022 Informa UK Limited, trading as Taylor & Francis Group.

Alter-transitional justice; transforming unjust relations with the more-than-human

Celermajer, Danielle; O'Brien, Anne Therese (2021.0)

ID do Documento: Alter-transitional-justice-transforming-unjust-relations-with-the-more-than-human

ABSTRACT ORIGINAL

Drawing on the emerging field of multispecies justice, this article seeks to understand how the idea of transitional justice, capaciously understood, might be put to work to transform unjust relations between humans and the more-than-human. Reflecting on concerns in the literatures on animals and the environment concerning the cogency of addressing past wrongs against the more-than-human by using a justice framework, the article sets out a foundational agenda for transitional justice and a conceptual framework responsive to the ontological diversity of beings and communities other than humans. Focusing on soil specifically, the article explores the problem of developing transitional justice approaches for transforming relations that involve systemic violence where such violence is not acknowledged because the harmed being - soil - is not recognized as the type of community to which justice might be owed. To illustrate prototransitional justice, the article considers both the work of regenerative farmers and emergent collaborations between farmers and visual artists to explore how engagements with the arts of relating to the more than-human might move the as yet private transformations of relations with soil into a more public, albeit incipient, process of justice. © Edward Elgar Publishing 2021.

The right to the sustainable smart city

Heitlinger, Sara; Bryan-Kinns, Nick; Comber, Rob (2019.0)

ID do Documento: The-right-to-the-sustainable-smart-city

ABSTRACT ORIGINAL

Environmental concerns have driven an interest in sustainable smart cities, through the monitoring and optimisation of networked infrastructures. At the same time, there are concerns about who these interventions and services are for, and who benefits. HCI researchers and designers interested in civic life have started to call for the democratisation of urban space through resistance and political action to challenge state and corporate claims. This paper contributes to an emerging body of work that seeks to involve citizens in the design of sustainable smart cities, particularly in the context of marginalised and culturally diverse urban communities. We present a study involving co-designing Internet of Things with urban agricultural communities and discuss three ways in which design can participate in the right to the sustainable smart city through designing for the commons, care, and biocultural diversity. © 2019 Copyright held by the owner/author(s).

Towards an ontology-based soil information system

Shu, Yanfeng; Liu, Qing (2015.0)

ID do Documento: Towards-ontology-based

ABSTRACT ORIGINAL

Environmental information is critical to the sustainable use and management of the world's resources. Soils are a fundamental part of the environmental information requirement, and appropriate soil data and information are crucial to support evidence-based policy, planning and resource management decisions. For data to be useful, one basic requirement is that they be interpretable. Sufficient information should be provided to allow data to be unambiguously interpreted and used. Examples of such information include the location at which the soil was sampled, the property that was measured, the unit of measure, and quality assurance and quality control information. Furthermore, data should be easily integrated with other data sources, which is required in many modelling applications. For example, simulation of crop production may require, besides soil data, also weather, crop and fertilizer data. To meet these requirements, we have developed a soil ontology for modelling soil information. In this paper, we focus on the design of the ontology and its potential applications. We describe the use of the ontology to facilitate data access by mapping data to the ontology and making them available as Linked Data. We also discuss applications of the ontology for data integration, data classification and data validation. © 2020 Proceedings - 21st International Congress on Modelling and Simulation, MODSIM 2015. All rights reserved.

The Influence of Nonhuman Assemblage Interactions on Small Farmers' Perceptions of Weather in Oregon: A Social Network Analysis

Parks, Melissa (2022.0)

ID do Documento: The-Influence-of-Nonhuman-Assemblage-Interactions

ABSTRACT ORIGINAL

Farmers' perceptions of weather play an important role in shaping their responses to climate change. I combine assemblage theory, which focuses on the interactions between nonhuman and human entities as they co-function in space, with social network analysis to explore how small farmers' interactions with nonhuman plants and animals inform their perceptions of extreme and variable weather across Oregon. While assemblage theory and qualitative methods provide in-depth details on farmers' nonhuman interactions, survey and social network analysis highlight broader patterns across farmers' assemblages. My results illustrate the critical role that human-nonhuman interactions play in shaping common perceptions of weather extremes and variability among farmers, regardless of their beliefs about climate change, and the importance of diverse nonhuman interactions for shaping such perceptions. © 2022, The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature.

Design of Multiple Ontology Based Agro Knowledge Mining Model

Abdullah, Azween; Murali, E.; Sreeji, S.; Balusamy, Balamurugan; Rajashree, S. (2023.0)

ID do Documento: Design-of-Multiple-Ontology-Based-Agro-Knowledge-Mining-Model

ABSTRACT ORIGINAL

Farming is regarded as a major industry in India, accounting for 17% of the country's GDP growth. Agriculture employs 60% of the population hence it is considered an important sector in India. The important factors for agriculture are pest management, disease prevention, irrigation management, soil mineral composition, crop management, location, and the season in which the crop is grown. Hence all this information along with the techniques are well known only by the experienced farmers. Hence it is important to create an agro knowledge management system. As a result, this work makes an attempt to develop a multiple ontology-based agro knowledge management system. The designed system consists of agriculture information related to attributes of soil mineral, moisture, season, location, crop type, and temperature. It consists of multiple ontologies such as soil ontology, crop ontology, location ontology, and crop season ontology to provide agronomy knowledge. Soil ontology is premeditated to classify the soil type in a hierarchical order while crop ontology classifies the crop type, location ontology classifies locations suitable for different crop types and finally, crop season ontology classifies the season that is suitable for different crops. A rule base is built to develop the knowledge base and to validate the truthfulness of the knowledge base. Visualization of a knowledge base is carried out for better understanding and decision-making. © 2023 International Journal on Recent and Innovation Trends in Computing and Communication. All rights reserved.

Sharing the harvest: {Transformative} artful and activist methodologies for urban agroecology

Haide, Ella von der; Orrù, Anna Maria; Dyck, Barbara Van; Solomon, Debra; Ujuaje, Mama D.; Woods, Deirdre (Dee); Halder, Severin; Grey, Robin (2021.0)

ID do Documento:

Sharing-the-harvest-Transformative-artful-and-activist-methodologies-for-urban-agroecology

ABSTRACT ORIGINAL

Food is an entanglement of materiality and social practices, power relations, communities, histories and visions. The feminist approaches people use are string figure, situated knowledge, fictocriticism and storytelling, and bodily practice, such as makingthinking. The experimental is about how people bring together art, research, activism, collective learning, sciences and society into a collective experiment. The paradigmatic changes that urban agroecology necessitates in terms of science-society relations, humans and human settlements positioning within food systems are too important, too entangled, too political to be left to the reified depoliticised practices of (neoliberal) scientific practice. In each case, situated knowledge is viewed as entangled 'action possibilities' stemming from embodied practices. Situated knowledge is a broad umbrella under which other forms of knowledge are also present and particular to each case. Film-making is an art, and alternative media is a form of activism.

IoT-Enabled Soil Nutrient Analysis and Crop Recommendation Model for Precision Agriculture

Senapaty, Murali Krishna; Ray, Abhishek; Padhy, Neelamadhab (2023.0)

ID do Documento: IoT-Enabled-Soil-Nutrient-Analysis

ABSTRACT ORIGINAL

Healthy and sufficient crop and food production are very much essential for everyone as the population is increasing globally. The production of crops affects the economy of a country to a great extent. In agriculture, observing the soil, weather, and water availability and, based on these factors, selecting an appropriate crop, finding the availability of seeds, analysing crop demand in the market, and having knowledge of crop cultivation are important. At present, many advancements have been made in recent times, starting from crop selection to crop cutting. Mainly, the roles of the Internet of Things, cloud computing, and machine learning tools help a farmer to analyse and make better decisions in each stage of cultivation. Once suitable crop seeds are chosen, the farmer shall proceed with seeding, monitoring crop growth, disease detection, finding the ripening stage of the crop, and then crop cutting. The main objective is to provide a continuous support system to a farmer so that he can obtain regular inputs about his field and crop. Additionally, he should be able to make proper decisions at each stage of farming. Artificial intelligence, machine learning, the cloud, sensors, and other automated devices shall be included in the decision support system so that it will provide the right information within a short time span. By using the support system, a farmer will be able to take decisive measures without fully depending on the local agriculture offices. We have proposed an IoT-enabled soil nutrient classification and crop recommendation (IoT-SNA-CR) model to recommend crops. The model helps to minimise the use of fertilisers in soil so as to maximise productivity. The proposed model consists of phases, such as data collection using IoT sensors from cultivation lands, storing this real-time data into cloud memory services, accessing this cloud data using an Android application, and then pre-processing and periodic analysis of it using different learning techniques. A sensory system was prepared with optimised cost that contains different sensors, such as a soil temperature sensor, a soil moisture sensor, a water level indicator, a pH sensor, a GPS sensor, and a colour sensor, along with an Arduino UNO board. This sensory system allowed us to collect moisture, temperature, water level, soil NPK colour values, date, time, longitude, and latitude. The studies have revealed that the Agrinex NPK soil testing tablets should be applied to a soil sample, and then the soil colour can be sensed using an LDR colour sensor to predict the phosphorus (P), nitrogen (N), and potassium (K) values. These collected data together were stored in Firebase cloud storage media. Then, an Android application was developed to fetch and analyse the data from the Firebase cloud service from time to time by a farmer. In this study, a novel approach was identified via the hybridisation of algorithms. We have developed an algorithm using a multi-class support vector machine with a directed acyclic graph and optimised it using the fruit fly optimisation method (MSVM-DAG-FFO). The highest accuracy rate of this algorithm is 0.973, compared to 0.932 for SVM, 0.922 for SVM kernel, and 0.914 for decision tree. It has been observed that the overall performance of the proposed algorithm in terms of accuracy, recall, precision, and F-Score is high compared to other methods. The IoT-SNA-CR device allows the farmer to maintain his

field soil information easily in the cloud service using his own mobile with minimum knowledge. Additionally, it reduces the expenditure to balance the soil minerals and increases productivity. © 2023 by the authors.

LOVE AND CARE FOR PLACE IN AN ARCTIC COMMUNITY: Place development in Vardø, Norway

Larsen, Janike Kampevold (2023.0)

ID do Documento: LOVE-AND-CARE-FOR-PLACE-IN-AN-ARCTIC

ABSTRACT ORIGINAL

Historically the richest fishing town in the Norwegian north, the now post-Arctic city of Vardø has lost half its population due to the 1980s crises in the fishing industry and later, a failed policy on fishing quotas. The city's on-land fishing infrastructure has fallen into ruin, and the city has become emblematic of a shrinking postindustrial community in Norway. Cultural entrepreneurs have been working intensely to save the city, starting with its architectural heritage as a material condition for development. While the Norwegian north is subject to national policies that favor petroleum extraction, the Northern Sea Route and tourism as strategies for development, the population in Vardø insists on basing its development on local resources such as fish, birding tourism and architectural heritage. Bottom-up efforts and collectives initiatives aim to build a robust local economy, one that is not dependent on external funding. This article argues that people in Vardø demonstrate place development based in love and performed as care. A new greenhouse is the result of a two-year collaborative building process, initially between local entrepreneurs and the Oslo School of Architecture and Design but ultimately by a dedicated group of locals who exercise care work at a profound level. By an adaptation of feminist care theory to a unique civic place work, the article argues that a practice of care may alleviate or even substitute a policy that is human-centered and profit-motivated. People in Vardø offer attention to things that cannot take care of themselves. They sacrifice free time and sleep to political activism, to build, to entertain and educate politicians from the south and to exercise a care ethics that can be read as ethical obligations that are recreated through everyday doings. The article builds upon several years of research and collaboration with local entrepreneurs in Vardø under the auspices of the research project Future North, 2012–2018. For the past three years, work has unfolded through a collaboration between Tine Hegli, circular architectures specialist AHO, and the author, who is a landscape researcher at AHO and their respective teams, under the auspices of a new project: Common Resources. Strategies for a circular, balanced and shared management of areas under pressure (2020–). © 2024 selection and editorial matter, Leena Cho and Matthew Jull; individual chapters, the contributors. All rights reserved.

Technological Innovations in Agriculture: A Philosophy and Sociology of Science Approach

Kendig, Catherine; Thompson, Paul B. (2025.0)

ID do Documento: Technological-Innovations-in-Agriculture

ABSTRACT ORIGINAL

How do human interventions into the environment motivated by different aims transform agriculture in ways that create new causal relationships between organisms above and below ground? We provide a conceptual framework for a philosophical and sociological approach to agricultural biotechnology and its multiple impacts on agricultural systems. We begin with a brief account of the history of the philosophy of the agricultural sciences and the early reluctance of philosophers of science to engage in the philosophy of the applied sciences. Applied sciences in general and the agricultural sciences in particular rely on contingencies in social practice. Here, one would expect diversity among different disciplines and goals. Instead of the basic sciences' aim of discovering a univocal reality, applied sciences were reconciled to the possibility of incompatible prescriptions. Following this, we introduce some concepts and approaches central to discussions of agricultural biotechnologies including social epistemology and social ontology. Our coinage, engineered agricultural ecologies is meant to echo debate over novel ecosystems in conservation biology, where attempts to intervene in ecosystems damaged by species loss and climate change are advocated by some, opposed by others. By insisting that farms are engineered ecologies we hope to stimulate some crosstalk with environmental philosophers and conservation ecology. © The Author(s) 2025.

More-than-social movements: {Politics} of matter, autonomy, alterontologies

Ghelfi, Andrea; Papadopoulos, Dimitris (2022.0)

ID do Documento: More-than-social-movements

ABSTRACT ORIGINAL

How do social movements respond to the ecological crisis? In this chapter we aim to develop an understanding of contemporary socio-ecological movements starting from their capacity to instigate material transformations. We reframe social movements as 'more-than-social movements' to highlight that many contemporary mobilisations do much more than targeting recognised social institutions and political governance; they are practically transforming eco-societies within and with both the human and the nonhuman world. What constitutes the core of more-than-social movements action is the capacity to set up alternative forms of mundane existence and alternative politics of matter. By transforming the ways through which heterogenous entities relate to each other, these movements generate alternative ontologies of existence and generative and reparative practices of justice. We engage with the imaginaries and practices of agroecology, AIDS treatment activism and permaculture to rethink what autonomy might look like in more than human worlds.

Ungovernable Earth: Resurgence, Translocal Infrastructures and More-than-Social Movements

Ghelfi, Andrea; Papadopoulos, Dimitris (2022.0)

ID do Documento: Ungovernable-Earth-Resurgence

ABSTRACT ORIGINAL

How do social movements respond to the ecological crisis? In this paper, we reframe social movements as 'more-than-social movements' to highlight the fact that many contemporary mobilisations do much more than target recognised social institutions and political governance; indeed, they are practically transforming eco-societies with and within both the human and the nonhuman world. What constitutes the core of more-than-social movements' action is the capacity to set up alternative ecologies of existence, or 'alterontologies', as we call them in the paper. In what follows, we engage with the imaginaries and practices of agroecology, AIDS treatment activism and permaculture in order to rethink what autonomy and justice might look like in the context of today's ecological crisis. © 2021 The White Horse Press.

Semantics about soil organic carbon storage: DATA4C+, a comprehensive thesaurus and classification of management practices in agriculture and forestry

Fujisaki, Kenji; Chevallier, Tiphaine; Bispo, Antonio; Laurent, Jean-Baptiste; Thevenin, François; Chapuis-Lardy, Lydie; Cardinael, Rémi; Le Bas, Christine; Freycon, Vincent; Bénédet, Fabrice; Blanfort, Vincent; Brossard, Michel; Tella, Marie; Demenois, Julien (2023.0)

ID do Documento: Semantics-about-soil-organic-carbon-storage

ABSTRACT ORIGINAL

Identifying the drivers of soil organic carbon (SOC) stock changes is of the utmost importance to contribute to global challenges like climate change, land degradation, biodiversity loss, or food security. Evaluating the impacts of land use and management practices in agriculture and forestry on SOC is still challenging. Merging datasets or making databases interoperable is a promising way, but still has several semantic challenges. So far, a comprehensive thesaurus and classification of management practices in agriculture and forestry has been lacking, especially while focusing on SOC storage. Therefore, the aim of this paper is to present a first comprehensive thesaurus for management practices driving SOC storage (DATA4C+). The DATA4C+ thesaurus contains 224 classified and defined terms related to land management practices in agriculture and forestry. It is organized as a hierarchical tree reflecting the drivers of SOC storage. It is oriented to be used by scientists in agronomy, forestry, and soil sciences with the aim of uniformizing the description of practices influencing SOC in their original research. It is accessible in Agroportal (<http://agroportal.lirmm.fr/ontologies/DATA4CPLUS>, last access: 24 March 2022) to enhance its findability, accessibility, interoperability, and reuse by scientists and others such as laboratories or land managers. Future uses of the DATA4C+ thesaurus will be crucial to improve and enrich it, but also to raise the quality of meta-analyses on SOC, and ultimately help policymakers to identify efficient agricultural and forest management practices to enhance SOC storage. © Copyright:

Careful Interventions: On Care and Participation in Design for Migration and Arrival

Krüger, Maximilian (2024.0)

ID do Documento: Careful-Interventions-On-Care

ABSTRACT ORIGINAL

In 2015 and 2016 more than a million people sought refuge from conflict and hardships in Germany. While the response was ambivalent, across the country large networks of supporters formed, offering various kinds of assistance to the arriving persons. This book takes these developments as a starting point. It reports on an HCI project, building on the emancipatory traditions of Participatory Design and the Socio-Informatics approach, aiming to develop assistive technologies for forced migrants as well as their supporters. It focusses on the design process and reflects on the nature of intervening through participatory design in matters of migration and arrival. The work draws on feminist conceptualisations of care and outlines how care and participation interacted in this context to support the longterm maintenance of the design results, but also how care relations between the different participant groups undermined the democratic aims of participation. It draws out preliminary lessons for an interventionist HCI practice. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Fachmedien Wiesbaden GmbH, part of Springer Nature 2024.

A Survey on Organic Agro Data Towards Agriculture Using Data Mining

Murali, E.; Vignesh, R.; Deepa, D.; Priyanka, N.; Hemalatha, S.; Rajashree, S. (2022.0)

ID do Documento: A-Survey-on-Organic-Agro-Data-Towards-Agriculture-Using-Data-Mining

ABSTRACT ORIGINAL

In India, 60.6% of the land has been used for agricultural purposes. Farmer invests much in chemical fertilizers, manual labor, weed collection, pesticide management, and at last, they gain low productivity. In this paper, different data mining techniques are going to be analyzed for improving crop productivity and the use of organic farming. Here nutrient content of different food products is also going to be analyzed by comparing organic and inorganic food products. Ill effects of Inorganic farming are listed. Food produced using organic farming will bring out the health and well-being of human beings for a better future. Seasonal crop recommendations will also be done by making a complete study on an attribute such as PH, water holding capacity, erosion, etc. Weather data are collected to bring out the prediction of rainfall in a particular region. The accuracy of different data mining algorithm are also listed © 2022 IEEE.

Seeing copiapósols: anthropogenic soils, strategic unknowing, and emergent taxonomies in northern Chile

Ureta, Sebastián; Otaegui, Alvaro (2021.0)

ID do Documento: Seein-copiaposols-anthropogenic-soils

ABSTRACT ORIGINAL

In recent decades, anthropogenic soils have become so ubiquitous that for some authors they should be taken as the “golden spike” signaling the start of the Anthropocene. Despite their prominence, leading soil taxonomies have resisted calls to recognize them as a proper kind of soil. Such omission has importantly limited the ways in which soil practitioners can account and deal with the sociopolitical aspects embedded in soil formation. Approaching the issue from a sociomaterial perspective, this paper studies the effects of such omission on the work of soil scientists working in northern Chile. By contrasting their usage of the USDA/NRCS soil typology with the realities found in the field, the work of strategic unknowing that such typologies achieve becomes evident. To challenge such situation, the paper concludes exploring the notion of emergent taxonomies, classifications that are sensitive to local configurations of materials, living beings and power out which soils emerge. © 2021, The Author(s), under exclusive licence to Springer Nature B.V.

Relationships of regeneration in Great Plains commodity agriculture

Snorek, Julie; Freidberg, Susanne; Smith, Geneva (2024.0)

ID do Documento: Relationships-of-regeneration-in-Great-Plains-commodity-agriculture

ABSTRACT ORIGINAL

In recent years regenerative agriculture has attracted growing attention as a means to improve soil health and farmer livelihoods while slowing climate change. With this attention has come increased policy support as well as the launch of private sector programs that promote regenerative agriculture as a form of carbon farming. In the United States many of these programs recruit primarily in regions where large-scale commodity production prevails, such as the Great Plains. There, a decades-old regenerative agriculture movement is growing rapidly, but not due to the incentives offered by companies' carbon programs. On the contrary, farmers are adopting regenerative practices to cut their dependence on corporate agrochemical inputs and expertise, and to thereby achieve technology sovereignty. These practice changes often strain farmers' existing social relationships while drawing them into new and previously neglected ones, including the more-than-human relations necessary for building soil health. These new relationships and the knowledge they generate may in turn lead farmers to think differently about their own autonomy. These findings provide insight into farmers' skepticism of private sector carbon farming programs, and highlight the value of attention to the multiple types of relationship change that accompany and facilitate regenerative transitions. © The Author(s), under exclusive licence to Springer Nature B.V. 2024.

Relationships of regeneration in Great Plains commodity agriculture

Snorek, Julie; Freidberg, Susanne; Smith, Geneva (2024.0)

ID do Documento: Relationships-regeneration-Great-Plains

ABSTRACT ORIGINAL

In recent years regenerative agriculture has attracted growing attention as a means to improve soil health and farmer livelihoods while slowing climate change. With this attention has come increased policy support as well as the launch of private sector programs that promote regenerative agriculture as a form of carbon farming. In the United States many of these programs recruit primarily in regions where large-scale commodity production prevails, such as the Great Plains. There, a decades-old regenerative agriculture movement is growing rapidly, but not due to the incentives offered by companies' carbon programs. On the contrary, farmers are adopting regenerative practices to cut their dependence on corporate agrochemical inputs and expertise, and to thereby achieve technology sovereignty. These practice changes often strain farmers' existing social relationships while drawing them into new and previously neglected ones, including the more-than-human relations necessary for building soil health. These new relationships and the knowledge they generate may in turn lead farmers to think differently about their own autonomy. These findings provide insight into farmers' skepticism of private sector carbon farming programs, and highlight the value of attention to the multiple types of relationship change that accompany and facilitate regenerative transitions. © The Author(s), under exclusive licence to Springer Nature B.V. 2024.

Sea ice out of time: Reckoning with environmental change

Yip, Julianne (2022.0)

ID do Documento: Sea-ice-out-of-time

ABSTRACT ORIGINAL

In September 2007, Arctic sea ice plummeted to a shocking record minimum at the time. The amount of ice lost that summer was equal to that lost over the previous 25 years. As Arctic sea ice escapes scientists' predictions, scholars in the social sciences and humanities have critically interrogated "nature"/"culture" divides that treat the time of nature as unchanging and distinct from human beings. This essay examines what concept of time emerges through Arctic sea ice as an analytic lens. By this, I mean scientific knowledge of sea ice and the conceptual possibilities for thinking ice temporalities and environmental time-reckoning that it opens up. Attending to these possibilities suggests different kinds of "clocks" to help reckon the time of environmental changes in the form of (1) climate anomalies (e.g., deviations in ice thickness), which offer a different way of telling environmental time that attends to the physical specificity of substances; and (2) the Arctic Oscillation, a semi-periodic world weather pattern that emerges from the thick of relationships among ice, atmosphere, ocean, and now humans, generating a collective planetary time. Finally, I argue that the relational human–nonhuman production of planetary time shifts the focus in social studies of time from collective time-reckoning, which assumes entities have a socioculturally determined concept of time, toward temporal coordination as a less anthropocentric mode of ordering shared realities. Coordination decenters "the Human" as an epistemic ordering principle and enlarges ordering to include a diversity of nonhuman ways of being. Through temporal coordination, environmental prediction would be the ordering of a collective reality that a multiplicity of human and nonhuman ways of being make together rather than the search for a more precise clock, or development of better technoscientific means to capture nonhuman temporalities external to human beings. © The Author(s) 2022.

Ontology in Hybrid Intelligence: A Concise Literature Review

Pileggi, Salvatore Flavio (2024.0)

ID do Documento: Ontology-in-Hybrid-Intelligence

ABSTRACT ORIGINAL

In the context of the constant evolution and proliferation of AI technology, hybrid intelligence is gaining popularity in reference to a balanced coexistence between human and artificial intelligence. The term has been extensively used over the past two decades to define models of intelligence involving more than one technology. This paper aims to provide (i) a concise and focused overview of the adoption of ontology in the broad context of hybrid intelligence regardless of its definition and (ii) a critical discussion on the possible role of ontology to reduce the gap between human and artificial intelligence within hybrid-intelligent systems, as well as (iii) the identification of possible future research directions in the field. Alongside the typical benefits provided by the effective use of ontologies at a conceptual level, the conducted analysis has highlighted a significant contribution of ontology to improving quality and accuracy, as well as a more specific role to enable extended interoperability, system engineering and explainable/transparent systems. Additionally, an application-oriented analysis has shown a significant role in present systems (70+% of cases) and, potentially, in future systems. However, despite the relatively consistent number of papers on the topic, a proper holistic discussion on the establishment of the next generation of hybrid-intelligent environments with a balanced co-existence of human and artificial intelligence is fundamentally missed in the literature. Last but not the least, there is currently a relatively low explicit focus on automatic reasoning and inference in hybrid-intelligent systems. © 2024 by the author.

Transformative Transdisciplinarity: An Introduction to Community-Based Philosophy

Ludwig, David; El-Hani, Charbel N. (2025.0)

ID do Documento: Transformative-Transdisciplinarity

ABSTRACT ORIGINAL

In the face of planetary crises - from biodiversity loss to climate change to food security - transdisciplinary methods promise effective and just responses through equal collaborations. However, transdisciplinarity also creates complex challenges by bringing together different actors with different frameworks, like scientists, Indigenous and local communities, and policy makers. Successful collaboration among such actors requires navigating different forms of knowledge, worldviews, values, and positions of power. In *Transformative Transdisciplinarity*, David Ludwig and Charbel N. El-Hani synthesize insights from the philosophy of science and empirical action research to address these challenges through a framework of partial overlaps. On the one hand, the framework highlights the overlapping concerns and perspectives of actors that provide common ground for collaboration and mutual understanding. On the other hand, it emphasizes partialities that require navigating differences and tensions between actors. This book addresses the fundamental epistemological, ontological, and political questions of transdisciplinarity through this framework of partial overlaps, aiming for a transformative vision of collaborative science in the face of planetary crises. This is an open access title available under the terms of a CC BY-NC-ND 4.0 International licence. It is free to read on the Oxford Academic platform and offered as a free PDF download from OUP and selected open access locations. © Oxford University Press 2025. All rights reserved.

Ferments and the AIDS virus: Interspecies counter-conduct in the history of AIDS

Linds, Justin Abraham (2019.0)

ID do Documento: Ferments-and-the-AIDS-virus-Interspecies-counter-conduct-in-the-history-of-AIDS

ABSTRACT ORIGINAL

In the first three decades after AIDS started infecting people in the USA and Canada, before, during and after the emergence of anti-retroviral therapies, numerous "alternative and holistic treatments" for AIDS were debated, tested, circulated, written about and taught. This paper, taking a narrow focus, examines documents that reveal how some people with AIDS developed a logic of care predicated on intimate interactions with microscopic lifeforms-the AIDS virus and the bacteria involved in fermentation, in particular. Focusing on the writings of Jon Greenberg and Sandor Katz, two former members of ACT UP/NY, I show that the men did not just dissent from management by biomedical authority but found new authority about how to care for themselves as people with AIDS from their interactions with non-human microscopic life. The practices and writings of both men demonstrate that Foucault's theory of counter-conduct exists in the history of AIDS as an interspecies process in which microscopic existents lead humans. From Katz and Greenberg, I argue there is an interspecies dimension to counter-conduct that exists as a frame for understanding people who find in non-human life a guide towards unconventional forms of care, revised forms of human behaviour and philosophies for persisting with illness. © Author(s) (or their employer(s)) 2019. No commercial re-use. See rights and permissions. Published by BMJ.

Trans-temporal pigs: humans, spirits and the temporal multiplicity of pigs in {Ifugao}, the {Philippines}

Remme, Jon Henrik Ziegler (2021.0)

ID do Documento:

Trans-Temporal-pigs-Humans-spirits-and-the-temporal-multiplicity-of-pigs-in-Ifugao-the-Philippines

ABSTRACT ORIGINAL

In the highland province of Ifugao, the Philippines, humans and spirits exist together but belong nevertheless to two different temporal dimensions. Pigs are central for enacting relations between humans and spirits, but their exact role as temporal mediators have not yet been thoroughly explained. In this article, I ask therefore how Ifugao pigs work to connect and disconnect humans and spirits across these temporal divides. I suggest an approach to pigs that sees them as 'trans-temporal hinges' that enable the transformations of relations between multiple disparate but still co-existing temporalities. Revolving around the disputes and tensions created by the pigs I planned to butcher at a farewell party celebrating the end of my fieldwork, the article outlines the different ways in which pigs operate to engender trans-temporal relations. Looking at pigs involvement in specific human-spirit relational assemblages, I show how pigs can inhibit the actualization of a future inheritance of a house and how they can potentially enable an authorization of prestige by the ancestors. Describing their role in sacrificial animals, I demonstrate how they also can set in motion both conjunctions and disjunctions of trans-temporal differentiations, and I show how conversion to Protestant Christianity rejection of sacrificed pigs as they may put converts in touch with a demonic past. Eliciting thus the various ways in which pigs contribute to the temporalization of social life, I argue that Ifugao pigs must be understood as inherently temporally multiple.

Making {Time} for and with {Honeybees}

Phillips, Catherine (2023.0)

ID do Documento: Making-time-for-and-with-honeybees

ABSTRACT ORIGINAL

In the last few years, pronouncements of 'insectageddon' or the 'insect apocalypse' in media and academe have raised the alarm about the loss of insect biodiversity. Such reports have engendered dispute and action, with much of each orienting around honey bees. In this chapter, I explore more-than-human temporalities through examination of how time becomes made among honey bees and their keepers. Beginning with a reflection on spending time with bees, the spectres of life and death provide a guide through the stories told of (among other things) catastrophe and loss, slow violence and quick kills, and learning to coordinate tempos and rhythms. Such livings and tellings travel not only with bees and keepers, but resound in broader considerations of human-nonhuman relations and responsibilities.

The Work of Agroecological Transition; [Il lavoro della transizione agroecologica]

Ghelfi, Andrea (2023.0)

ID do Documento: The-Work-of-Agroecological-Transition

ABSTRACT ORIGINAL

In this article the concept of work comes into tension with some agroecological experiences that populate the Genuino Clandestino network. Three vectors of analysis are developed through an investigation of peasant practices: the unstable borders between productive and reproductive labour in favour of a “practicability of life”; the decentralisation of human exceptionalism in favour of the experimentation of agronomic practices that recognise the active role of multiple more-than-human entities; the commons as a community of practices in which social transformation is inseparable from material regeneration. Through the analysis of the agroecological practices developed by the commoners of Mondeggi Bene Comune, in this article the author situates the challenge of ecological transition within the work and the practices of care that characterise the back to the land movement in Italy. Copyright © FrancoAngeli.

Environmental STS

Baya-Laffite, Nicolas; Boudia, Soraya; Granjou, Céline (2025.0)

ID do Documento: Environmental-STS

ABSTRACT ORIGINAL

In this chapter, the authors have chosen to structure our discussion around three key analytical avenues, without aiming for exhaustiveness. Each avenue highlights a foundational approach and demonstrates how it is applied to specific objects and case studies. The first avenue, likely the earliest lens through which STS approached environmental issues, stems from the centrality of materiality, particularly as influenced by feminist studies and the various propositions that emerged from them. The second avenue introduces another significant STS approach, which is the denaturalization of the global environment. The third avenue stems from the body of work devoted to environmental justice, and it shifts the focus downward, toward collective mobilizations in response to the social, racial and epistemic inequalities that shape our societies. The chapter briefly highlights the questions raised by environmental STS within the broader context of debates on the Anthropocene. © ISTE Ltd 2025.

THE CARING CITY: Ethics of Urban Design

Davis, Juliet (2022.0)

ID do Documento: THE-CARING-CITY

ABSTRACT ORIGINAL

In this important contribution to urban studies, Juliet Davis makes the case for a more ethical and humane approach to city development and management. With a range of illustrative case studies, the book challenges the conventional and neoliberal thinking of urban planners and academics, and explores new ways to correct problems of inequality and exclusion. It shows how a philosophy of caring can improve both city environments and communities. This is an original and powerful theory of urban care that can promote the wellbeing of our cities' many inhabitants. © Bristol University Press 2022.

Futuristic e-governance security with deep learning applications

Kumar, Rajeev; Abdul Hamid, Abu Bakar; Binti Ya'akub, Noor Inayah; Gaur, Madhu Sharma; Kumar, Sanjeev (2024.0)

ID do Documento: Futuristic-e-governance-security

ABSTRACT ORIGINAL

In today's rapidly advancing digital world, governments are increasingly relying on technology to enhance security systems and streamline governance. However, this growing reliance on digital platforms and data collection also presents significant challenges. Cybersecurity threats and privacy concerns pose large risks to sensitive information and can potentially leading to inaccuracies or breaches within deep learning models. There is a pressing need for comprehensive solutions that address these security issues and protect valuable data in the realm of e-governance. Futuristic e-Governance Security With Deep Learning Applications is a timely and indispensable resource that offers a holistic approach to tackling the security challenges of the digital era. The book presents a global perspective on the integration of intelligent systems with cybersecurity applications, highlighting cutting-edge techniques and methodologies to safeguard deep learning models from security attacks and privacy vulnerabilities. By exploring the latest advances and countermeasures in deep learning, this book equips scholars, researchers, and industry experts with the knowledge and tools they need to address security concerns and develop robust e-governance systems. This comprehensive volume not only sheds light on the current state-of-the-art methods but also delves into future trends and challenges. From skill development and tools for intelligence systems to deep learning, machine learning, blockchain, IoT, and cloud computing, the book covers a wide range of topics essential to understanding and implementing secure e-governance systems. With its practical insights and interdisciplinary approach, this book serves as a vital resource for academics, researchers, and professionals seeking to navigate the complex landscape of e-governance security and leverage deep learning applications to protect valuable data and ensure the smooth functioning of government operations. © 2024 by IGI Global. All rights reserved.

Sustainability conventions in a local organic consumer cooperative in Norway: Hope and trouble of participants

Torjusen, Hanne; Vittersø, Gunnar (2023.0)

ID do Documento: Sustainability-conventions-in-a-local-organic

ABSTRACT ORIGINAL

In two lines of inquiry, this chapter aims at analysing how Consumer Food Initiatives, here exemplified by the Vestfold Kooperativ, may contribute to food system sustainability. First, the chapter analyses the importance of shared conventions related to environmental, economic, and social sustainability as a common basis for the outcomes of the activities within the CFI. The second line of inquiry is related to process, by investigating the functioning and development of Vestfold Kooperativ. The chapter draws on conventions theory to analyse how shared conventions (hope) and contested issues (trouble) are perceived within the cooperative and how troubles eventually might be overcome. The chapter concludes that, in spite of the troubles that Vestfold Kooperativ faced, there are important lessons to be learnt from their experiences. These are related to the holistic approach to sustainability that was shared within the cooperative and, when promoted externally, may contribute to broadening the debate about possible and sustainable transition pathways in the food system. © 2023 selection and editorial matter, Oona Morrow, Esther Veen, and Stefan Wahlen; individual chapters, the contributors.

Futuring a world without disease: visualising the elimination of hepatitis C

Lancaster, Kari; Rhodes, Tim (2022.0)

ID do Documento: Futuring-a-world-without-disease

ABSTRACT ORIGINAL

Informed by work on futurity in science and technology studies, we trace how global disease elimination targets perform a world without disease through their translations in visual advocacy campaigns. Treating disease elimination targets and their visualisations as performative, we take the case of hepatitis C elimination to interrogate how futuring practices in public health govern the present and make effects. We focus specifically on how World Health Organization targets in the Global Health Sector Strategy on Viral Hepatitis entangle with visual resources produced by the World Hepatitis Alliance NOhep advocacy campaign. Targets and their visual representations in campaigns perform a disease elimination future which is set apart from the present, and yet urges action in-the-now. It enacts global health citizens but separates them from localised experiences of living with, and being cured of, disease. This disease elimination future relies heavily on instrumental rationalities and logics of the present, including the privileging of biomedical technoscientific knowledge, implementation science and global health governance, to the exclusion of other matters of concern, flattening out complexity to perform its certain achievability. These enactments raise political questions about how disease elimination futures might be made in a different mode. © 2020 Informa UK Limited, trading as Taylor & Francis Group.

Intergenerational Justice and Innovation for Long-Term Agricultural Sustainability

Timmermann, Cristian (2020.0)

ID do Documento: Intergenerational-Justice-and-Innovation-for-Long-Term-Agricultural-Sustainability

ABSTRACT ORIGINAL

Innovation in agriculture brings about a number of positive and negative externalities. In this chapter I will focus on one particular externality, which is how innovation affects the consumption of non-renewable or slowly renewable resources that are essential for securing the human right to food in the future. The reproduction and massive use of some of these innovations require the right to destroy the effectiveness of resources that were not created by the inventor nor those buying the inventions. The use of pesticides leads to a loss of their effectiveness due to biological resistance, leading to the destruction of the resource (i.e. of its effectiveness) and genetic pollution (i.e. an increase in resistant biological organisms). Similarly, the use of high-yield crop varieties leads to the loss of soil fertility. While the destruction of these resources is inevitable when using these inventions, there are a number of measures that can be taken to prolong the active life of these resources. Users of these resources can comply with strict usage regulations and exclusive rights holders can assert substantial pressure to make sure users generally comply with such policies. © 2020, Springer Nature Switzerland AG.

‘Back to nature’ or ‘full throttle with science and technology’ for sustainable agriculture?

Straatsma, Gerben (2025.0)

ID do Documento: Back-to-nature-or-full-throttle

ABSTRACT ORIGINAL

Introduction: The transformation of agriculture is essential to addressing global sustainability challenges. Any transformation first requires a perspective. Our view of agriculture is linked to our view of naturalness, artificiality, and unnaturalness. A contrast exists between those who prioritize naturalness as a guiding principle and those who advocate for science and technology. This study examines the philosophical and psychological foundations of these perspectives to help scientifically trained readers navigate the public discourse on sustainable agriculture. Naturalness: The emphasis on naturalness has several roots. Naturalness is associated with purity, authenticity, and moral virtue, shaping preferences in agriculture, environmental policies, and medicine. Philosophical and psychological work suggests the role of intuitive reasoning in shaping these constructs, which can contribute to cognitive biases that influence perceptions of sustainability. Science and Technology: A science-driven approach to sustainability emphasizes rational analysis, technological innovation, and evidence-based decision-making. The scientific community often underestimates how normative values and biases shape public acceptance of technological solutions, which may be perceived as unnatural. Two Cases: The contrast between naturalness and unnaturalness is illustrated through the approaches of organic agriculture and sustainable intensification. Organic agriculture is rooted in commitments to naturalness, but faces challenges such as lower yields. Sustainable intensification emphasizes science and artificial, technological solutions but encounters scepticism in shaping agriculture. Final remark: Bridging a philosophical and psychological contrast is essential for formulating a comprehensive sustainability strategy. An inclusive discourse, informed by awareness of perspectives, can help reconcile competing visions for the future of agriculture. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025.

Animality/coloniality: COVID-19 and the Animal question

Isaacs, Jenny R.; Otruba, Ariel (2021.0)

ID do Documento: Animality-coloniality-COVID-19-and-the-Animal-question

ABSTRACT ORIGINAL

Joining efforts to “decolonize geography”, we offer ‘animality/coloniality’ as a heuristic device for analyzing asymmetrical power relations between humans and nonhumanity. The case of COVID-19 is used to demonstrate the complexity, problems, and dangers of defining and deploying the category of ‘animal’. By illustrating how competing ideas of animality are worked through each other in understandings of and responses to zoonotic viruses, we suggest that delinking animality/coloniality should begin with an embrace of the ‘alien’ Other as a pathogenic constituency. While the perennial Animal question remains an important driver of new research in animal geographies, we caution that it should be taken up with reflexive, critical attention to one’s political purpose and situated location within (colonial) knowledge production. © Alice Hovorka, Sandra McCubbin and Lauren Van Patter 2021.

Towards sustainable agriculture: A blueprint for European KPI-based farm-level assessment

Bar■■■, B.T.; Marselis, S.M.; Ros, G.H.; van Doorn, A.; Erisman, J.W. (2025.0)

ID do Documento: Towards-sustainable-agriculture

ABSTRACT ORIGINAL

Key Performance Indicator (KPI) methodologies have emerged as potent tools for estimating sustainability performance in agriculture. With the proliferation of data availability, there is a growing opportunity to conduct sustainability assessments at the farm level and transition from traditional practice-based policies to performance-based policies. In this paper, we propose a blueprint for a Europe-wide KPI-based farm level sustainability performance assessment and monitoring system. By drawing insights from literature on farm-level sustainability assessment systems and tools, we synthesized eight guiding principles and a 5-step process framework for developing and operationalizing a European Agricultural KPI system built around the potential advantages of performance-based policies. The guiding principles include defining sustainability, involving stakeholders, ensuring applicability across sectors and countries, using normative benchmarking, and implementing decision support approaches. The proposed process framework can guide public and private stakeholders in Europe through the establishment of sustainability goals, modeling linkages between objectives and available data, KPI development, online user interface creation, and adaptive management. By leveraging advanced analytics and the integration of diverse agricultural data sources, our system aims to provide policymakers with the opportunity to instrumentalize performance-based financial support mechanisms to incentivize the adoption of sustainable agricultural practices among farmers. The essential components of a policy structure supporting the effective and ethical implementation of the proposed KPI system are discussed. © 2025 The Author(s)

Knittedness: On the expressivity of knitting as metaphoric process

Kilpi, Linnea (2023.0)

ID do Documento: Knittedness

ABSTRACT ORIGINAL

Knitted fabric is a part of everyday life for many, whether it be worn, made or metaphorically referenced. It can also be, when critically considered, an expressive practice with philosophical merit. The ostensibly simple looping of yarns into a continuous surface suggests conceptual depth through poetic comparisons to processes of living and storytelling, among others. The practice-led research project 'Stranded colour-work: Meaning-making through experimental knitting practices' worked to define the intrinsic qualities of knitting that make it meaningful, and to uncover its expressive potential as an experimental artistic practice. Through the creation of a series of figurative machine-knit artworks, the confluence of knitting and narrative created an opportunity to study the implications of stitched life. This article elucidates the theoretical framework behind the project and focuses on knittedness, a concept proposed through the research. Knittedness refers to the specific aesthetic and technical qualities inherent to the process, and how these qualities can become symbolically significant both in art making and in finding meaning and connection in life. The idea of pixelness is also proposed as an integral part of knitting and other pixelated processes, where the process and aesthetic quality is informed by a repetitive technical structure. These concepts can be expanded to advocate for a notion of craftedness, suggesting the value of determining the idiosyncratic expressivity of individual creative practices, and how a maker might use this knowledge to create technique-focused work and find new appreciation for the process. © 2023 Intellect Ltd.

Words Are Essential, but Underexamined, Research Tools for Microbes and Microbiomes

Szymanski, Erika (2021.0)

ID do Documento:

Words-Are-Essential-but-Underexamined-Research-Tools-for-Microbes-and-Microbiomes

ABSTRACT ORIGINAL

Language constitutes an essential set of scientific construction tools, not only for communicating knowledge, but for conceptualizing the world. Metaphors in particular, as conventions that guide and reproduce analogical reasoning, merit attention that they largely do not receive. My research addresses this deficit by examining how metaphors for handling microbes shape possibilities for working with yeast and bacteria in synthetic biology, microbiome research, and other fields that reconfigure what microbes can be. Though poised to reexamine assumptions, these fields routinely rest on metaphors and other language tools that quietly embed ways of thinking that may work against wider aims—for example, imagining bacteria as imperfect machines that should therefore be rendered increasingly passive and controllable. Researchers, therefore, need to examine how language tools structure their observations and expectations so that the tools they choose are appropriate for the work they want to do. © 2021 Szymanski.

Creative Practices of Care: The Subjectivity, Agency, and Affective Labor of Preparing Seeds for Long-term Banking

Chacko, Xan Sarah (2019.0)

ID do Documento: Creative-Practices-of-Care-The-Subjectivity-Agency-and-Affective-Labor-of-Preparing-Seeds-for-Long-term-Banking

ABSTRACT ORIGINAL

Laying aside the question of whether saving seeds in freezers is the most promising long-term solution to prevent the loss of plant biodiversity and secure our access to food in a troubled future climate, this article draws attention to the conditions of possibility that scaffold the seed bank world. Oft relegated to “tech” work that is unworthy of observation, this article focuses on the labor practices of seed curators as they prepare the seeds for their ultimate storage at the largest seed bank of wild plants—the Millennium Seed Bank Partnership in West Sussex, England. Contributing to the growing scholarship on care in technoscientific practice, I investigate how scientists summon their bodies, imaginations, and feelings to clean, screen, and count seeds, all the while producing knowledge that renders the seeds legible in the bank. By following the seeds through the experimental care practices espoused by scientists involved from the moment seeds arrive at the bank until they are ready for storage, I study how seemingly mundane tasks radically influence how “life” is being prepared for the future. [seed banking, gene banking, Millennium Seed Bank Partnership, care practice, laboratory studies, affective labor]. © 2019 by the American Anthropological Association.

Uncertainty and purpose of measurements of carbon sequestration in soil

Watkins, Aja (2026.0)

ID do Documento: Uncertainty-and-purpose-measurements-carbon-sequestration-in-soil

ABSTRACT ORIGINAL

Many philosophers of measurement agree that measurements and measurement uncertainty need to be evaluated by their adequacy for a particular purpose. This paper uses an analysis of measurements of carbon in soil, which are used to support carbon trading and carbon offsetting, to test the adequacy-for-purpose method of measurement evaluation. In particular, evaluating the soil carbon case requires not just matching measurement practices to aims, but also being able to evaluate the aims of carbon trading or offsetting and being able to tell whether the measurement practices are (in)adequate for those more-ultimate aims. © 2026 Elsevier Ltd

Waste Frontiers/War Enclosures: Decolonial Geosocial Analysis of Contaminated Military Land Conversions

Krupar, Shiloh (2024.0)

ID do Documento: Waste-Frontiers-War-Enclosures

ABSTRACT ORIGINAL

Military-industrial practices have left widespread contamination affecting land, water, air, and human and nonhuman bodies. This article uses decolonial analysis to examine the racial-colonial foundations that underlie contemporary efforts to reuse former U.S. military land for development projects. Interrelating scholarship on security and development with that of militarization and political ecology, I use “geosocial spectacle” to probe the material and pedagogical governance, frontier logics, and colonial aesthetics that emerge through land remediation and redevelopment. Featuring two military sites within U.S. base closure and realignment—the U.S. Front Range area of Colorado and former U.S. unincorporated territory of the Panama Canal Zone—the article delineates and expounds on two types of U.S. military land conversions: brownfields and biodiversity recreation. The article interrelates the conversion of contaminated land in both cases as waste frontier that facilitates ongoing cycles of land repossession as a form of dispossession and containment or denial of war. The article concludes by advocating for decolonial studies of the global color line entrenched by the U.S. Pentagon’s ongoing climate colonialism, to subvert the compartmentalized harms of war and capitalist extraction, and to galvanize explicitly anticolonial, antiracist land reuse and governance. © 2024 by American Association of Geographers.

Enhancing precision agriculture through cloud based transformative crop recommendation model

Singh, Gurpreet; Sharma, Sandeep (2025.0)

ID do Documento: Enhancing-precision-agriculture-through-cloud

ABSTRACT ORIGINAL

Modern agriculture relies more on technology to boost food production. It aims to improve both the quality and quantity of food. This paper introduces a novel TCRM (Transformative Crop Recommendation Model). It uses advanced machine learning and cloud platforms to give personalized crop recommendations. Unlike traditional methods, TCRM uses real-time data. It includes environmental and agronomic factors to optimize recommendations. The system has SMS alerts for remote farmers. It outperforms baseline algorithms like Logistic Regression, KNN(k-nearest neighbor), and AdaBoost. TCRM empowers farmers with actionable insights, reducing resource wastage while boosting yield. By offering region-specific recommendations, it enhances profitability and promotes sustainable agricultural practices. The model has 94% accuracy, 94.46% precision, and 94% recall. Its F1 score is 93.97%. The fivefold cross-validation score is 97.67%. These findings show that the model can improve precision farming. It can make agriculture more sustainable and efficient. © The Author(s) 2025.

Glyphosate—A love story. Ordinary thoughtlessness and response-ability in industrial farming

Müller, Birgit (2021.0)

ID do Documento: Glyphosate-A-love-story

ABSTRACT ORIGINAL

More than 8.6 billion kilograms of glyphosate have been used worldwide since the 1970s. Herbicide tolerant crops became the lynchpin of the technological revolution for large-scale farming first in the United States and Canada, and now in Europe. Zero-till farming, as a production scheme and a world view, is based on simplifications promoted by a handful of transnational corporations with the complicity of politicians looking for easy solutions for problems, such as climate change, erosion and the hunger in the world. At the same time, the massive use of glyphosate is branded as an endocrine disrupter, causing cancer, male sterility and infertility. It interferes with soil bacteria and acts on the equilibrium of soil fungi. Glyphosate resistant crops connect farmers to far away consumers ingesting the food they grow together with the traces of chemicals. Farmers intra-act with the myriads of life-forms of the soil eco-system. How do they perceive the life in the soil, when they spray chemicals? The article explores the political dimensions of the agency of both humans and non-humans to understand the effects of the modernizing project of zero-till, as well as to identify spaces and scales of possibility from where alternatives can emerge. © 2020 John Wiley & Sons Ltd

Observe It, Draw It: Scaffolding Children's Observations of Plant Biodiversity with an Interactive Drawing Tool

Zhang, Chao; Zhou, Zili; Hu, Yajing; Liu, Lanjing; Wu, Jiayi; Shao, Yaping; Liu, Jianhui; Zhang, Lingyan; Liu, Lijuan; Chen, Hangyue; Ying, Fangtian; Yao, Cheng (2023.0)

ID do Documento: Observe-It-Draw-It-Scaffolding

ABSTRACT ORIGINAL

Observation is common for children to connect with nature, increasing their knowledge and awareness of biodiversity. However, it is challenging for them to make and document their observations due to a lack of observation and drawing skills. Therefore, we designed an interactive drawing tool, Bio Sketchbook, which scaffolds children in systematic observation, observational drawing, and knowledge acquisition. It can recognize plant species and generate contour drawings from children's photographs, guiding them to observe and draw multi-dimensional plant features with a digital magnifier and in-context biological information. Our in-situ user study with 19 children revealed that Bio Sketchbook provided an engaging experience and effectively supported children in recording and retaining biodiversity information and in balancing observations with screen time. Additionally, Bio Sketchbook intervened in children's interaction with plants by prompting observational behaviors, encouraging them to directly touch and establish rapport with plants, and arousing their interest and knowledge of plants.

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Soil {Refusal}: {Thinking} {Earthly} {Matters} as {Radical} {Alterity}

Tironi, Manuel (2020.0)

ID do Documento: Soil-Refusal-Thinking-Earthly-Matters-as-Radical-Alterity

ABSTRACT ORIGINAL

Olivia had been telling me about planting a magnolia tree for a while. She is an enthusiastic and skillful gardener and spends long hours among her plants. Every time I visit her, I return home with seeds, baby plants, lemons, and even small trees. She talks passionately about the cactuses, roses, and lavender bushes she nurses in her garden. The toxicity of La Greda, her hometown, hasn't discouraged her. La Greda is a small hamlet in the Puchuncaví Valley in central Chile, and adjacent to the infamous Centro Industrial Ventanas (CIV), one of the most polluted industrial compounds in Latin America. CIV hosts 27 petrochemical industries, including one copper-smelting complex and four coal-burning energy stations. Stories of ecological collapse abound in the area—and are dramatically evident after spending some time in La Greda. Actually, Olivia and her plants live just across from the now abandoned La Greda elementary school, closed down in 2012 after several children and teachers fainted owing to high levels of sulfur trioxide. Visible from almost all sides of her home, the abandoned building is a ghostly reminder of the chemical violence to which Olivia and her plants are constantly subjected. For the last decade, I have been paying attention to soil–plant–human relations, in Puchuncaví and elsewhere. I'm interested in how people like Olivia establish deep ethico-practical commitments with plants and soils in a context of chronic industrial violence, and in the way these human–soil–plant embroilments allow for a politics of intimate resistance (Tironi 2018). I have also paid attention to how earth scientists relate to inapprehensible geological matter, including topsoil, and to what extent these engagements recompose bio-geo arrangements (Tironi 2019). Spending time with neighbors from Puchuncaví and with geophysicists as they relate, sound, remediate, and work upon soily things has allowed me to get a sense of the kind of alliances articulated among different kin for vital endurance and knowledge production. One week after Olivia got the magnolia tree, I visited her so that we could plant it together. It was perfect timing, she told me, ideal for planting because of the full moon. When I arrived, she had already decided where the magnolia tree would be sited: a

Soil {Refusal}: {Thinking} {Earthly} {Matters} as {Radical} {Alterity}

Tironi, Manuel (2020.0)

ID do Documento: Soil-Theories-Relational-Decolonial-Inhuman

ABSTRACT ORIGINAL

Olivia had been telling me about planting a magnolia tree for a while. She is an enthusiastic and skillful gardener and spends long hours among her plants. Every time I visit her, I return home with seeds, baby plants, lemons, and even small trees. She talks passionately about the cactuses, roses, and lavender bushes she nurses in her garden. The toxicity of La Greda, her hometown, hasn't discouraged her. La Greda is a small hamlet in the Puchuncaví Valley in central Chile, and adjacent to the infamous Centro Industrial Ventanas (CIV), one of the most polluted industrial compounds in Latin America. CIV hosts 27 petrochemical industries, including one copper-smelting complex and four coal-burning energy stations. Stories of ecological collapse abound in the area—and are dramatically evident after spending some time in La Greda. Actually, Olivia and her plants live just across from the now abandoned La Greda elementary school, closed down in 2012 after several children and teachers fainted owing to high levels of sulfur trioxide. Visible from almost all sides of her home, the abandoned building is a ghostly reminder of the chemical violence to which Olivia and her plants are constantly subjected. For the last decade, I have been paying attention to soil–plant–human relations, in Puchuncaví and elsewhere. I'm interested in how people like Olivia establish deep ethico-practical commitments with plants and soils in a context of chronic industrial violence, and in the way these human–soil–plant embroilments allow for a politics of intimate resistance (Tironi 2018). I have also paid attention to how earth scientists relate to inapprehensible geological matter, including topsoil, and to what extent these engagements recompose bio-geo arrangements (Tironi 2019). Spending time with neighbors from Puchuncaví and with geophysicists as they relate, sound, remediate, and work upon soily things has allowed me to get a sense of the kind of alliances articulated among different kin for vital endurance and knowledge production. One week after Olivia got the magnolia tree, I visited her so that we could plant it together. It was perfect timing, she told me, ideal for planting because of the full moon. When I arrived, she had already decided where the magnolia tree would be sited: a

Semantic similarity in geographic information retrieval for decision making; [Similitud semántica en la recuperación de información geográfica para la toma de decisiones]

Machado, Neili; Balmaseda, Carlos; Montoyo, Andres (2014.0)

ID do Documento: Semantic-similarity-in-geographic-information-retrieval-for-decision-making

ABSTRACT ORIGINAL

People use the similarity to store and retrieve information comparing new situations with similar experiences occurred in the past also for learning and concept formation. While equality comparison between two objects can be calculated by computers quickly and accurately similarity is a complex problem to calculate. Even still difficult to determine similarity plays a fundamental role in many applications such as decision-making systems, data mining and pattern recognition. The same applies to the spatial similarity in the processes of recovery and integration of spatial information. In this paper a methodology based on the semantic processing of geospatial information is proposed. It consists of five stages: conceptualization, synthesis, application processing, retrieval and management. It uses ontology of New Genetic Soil Classification of Cuba and applying the measure of semantic similarity Conceptual Distance (DIS-C) combined with an implementation model TDD (Topology-Direction-Distance) to restore and support the user in the selection of spatial scenes. As a case study is considered the town of San Jose de Las Lajas located in the province of Mayabeque in western Cuba. © 2014 AISTI.

Semantic and spatial similarity in geographic information retrieval

Machado, N.; Marrero, Y.; Balmaseda, C.; Montoyo, A. (2014.0)

ID do Documento: Semantic-and-spatial-similarity-in-geographic-information-retrieval

ABSTRACT ORIGINAL

People use the similarity to store and retrieve information, also for learning and concept formation. Similarity plays a fundamental role in many applications such as decision-making systems, data mining and pattern recognition. The same applies to the spatial similarity in the processes of recovery and integration of spatial information. In this paper a methodology based on the semantic processing of geospatial information is proposed. It consists of five stages: conceptualization, synthesis, application processing, retrieval and management. It uses ontology of New Genetic Soil Classification of Cuba and applying the measure of semantic similarity DISC combined with an implementation of model TDD (Topology-Direction-Distance) to restore and support the user in the selection of spatial scenes. As a case study is considered the town of San José de Las Lajas located in the province of Mayabeque in western of Cuba. ©2014 International Conference on Artificial Intelligence, ICAI 2014 - WORLDCOMP 2014. All right reserved.

Permaculture

Chakroun, Leila (2023.0)

ID do Documento: Permaculture

ABSTRACT ORIGINAL

Permaculture offers an ethical, pragmatic, philosophical and technical response to the daunting diagnosis of the Anthropocene. Its comprehension of the Earth draws upon the idea of “working with nature, rather than against it”. In that sense, permaculture radically departs from the fixist and homogenizing logics that prevail in agronomic sciences, and in modern territorial planning. By reengaging with the cycle of life and with our own finitude as living beings, permaculture conveys an earthly vision of human that resists the impertinent idea of an Earth becoming the scene solely for anthropos. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

Permafrost Politics: Toward a Relational Materiality and Design of Arctic Ground

Cho, Leena (2021.0)

ID do Documento: Permafrost-Politics

ABSTRACT ORIGINAL

Permafrost—continuously frozen ground for more than two years—is one of the defining landscape elements in the Arctic. In the U.S., its development as an idea has paralleled mid-twentieth-century territorial struggles in Alaska and is inextricably linked to permafrost science and frontier engineering. Permafrost is more than a scientific category and engineering risk subject to correction and control; it is a foundation for dynamic socioecological and cultural expressions in arctic landscapes. In response, a relational approach is taken to examine permafrost's cold, vibrant and plural materialities, with an aim of generating design possibilities that are attuned to these dynamics. In the Arctic where science continues to play a central role in determining putative futures, the article further suggests to creatively instrumentalise scientific forms of landscape inquiry, highlighting thermo-material interactions and multiplicities of arctic ground. © 2020 Landscape Research Group Ltd.

A Survey on Computational Aptitudes towards Precision Agriculture using Data Mining

Murali, E.; Anuncia, S.Margret (2022.0)

ID do Documento:

A-Survey-on-Computational-Aptitudes-towards-Precision-Agriculture-using-Data-Mining

ABSTRACT ORIGINAL

Precision agriculture is a modern agriculture implementation technique in which analysis of numerous source data takes place for decision-making and operation in the management of crop production. The data for precision agriculture are collected through robots, sensors, satellites, and drones. The two approaches of precision agriculture are the predictive approach which is used for representing the static indicator during the crop cycle whereas the control approach is an updating of information. Ontology is a demonstration of concepts and their shared association. It can be used in a wide range of contexts, including the classification of agricultural information and the development of knowledge bases. The basic steps involved in precision farming are assessing variation, managing variability, and evaluation. The various tools used in precision farming are the internet of things (IoT), a global positioning system (GPS), geographic information system (GIS), remote sensor, proximate sensor technology, grid sampling, etc. With the increase in information technology in the field of agriculture. Consequently, data mining become much essential for decision-making. This paper attempts to emphasize the coherence of data mining approaches toward helping precision agriculture as a valuable venture. © 2022 IEEE.

Gendered Perceptions of Public Services: Quality, Equality of Treatment, and Corruption in Chile

Cazzuffi, Chiara; Retamal, Patricia (2025.0)

ID do Documento: Gendered-Perceptions-of-Public-Services

ABSTRACT ORIGINAL

Public services in Latin America face persistent challenges that affect access, quality, and fairness, particularly for women and marginalized groups. This study examines gendered differences in public service perceptions in Chile, focusing on quality, equality of treatment, and corruption in education, health, and law enforcement. Using data from the Index of Perceptions of Governance Quality at the Regional Level, we investigate whether women perceive public services more negatively than men and whether explanatory factors differ by gender. Employing linear, multilevel, and seemingly unrelated regression models, we analyze individual-level and municipal-level variables. Women report greater perceptions of inequality and corruption in health and law enforcement, with indigenous women expressing the most negative views. Female municipal leadership is associated with lower perceived corruption among women. These findings contribute to public administration scholarship by integrating care theory, representative bureaucracy, and policy feedback to inform gender-sensitive approaches to public service reform. © 2025 John Wiley & Sons Ltd.

Complexities of care in a college microbiology laboratory

El Halwany, Sarah; Alsop, Steve (2023.0)

ID do Documento: Complexities-of-care-in-a-college-microbiology-laboratory

ABSTRACT ORIGINAL

Recognising effects of care departs from previous conceptions of care within science education as an idealised, ethical, and normative dimension of being, thinking and relating. In this article, we hesitate with care's moralistic qualities to render visible disruptive and complicit moments of care in complex ethical and political situated practices. We follow care inside a college microbiology laboratory, documenting three practices: caring for bacteria, caring for logbooks, and caring for a nation. We show how these care practices potentialise/disrupt bacteria as technoscientific commodities and biotechnicians as a labour force for markets and nations organised along racial and gendered lines. Our paper attends to the complexities of care as an assembling force with its ambivalent ethical and political orientations. We conclude by reflecting on the possibilities of 'care-full' science educational analyses. How can we better care for care? What might care offer science education research and practice that other more popular explanatory units (such as concepts and effectiveness) do not?. © 2023 Informa UK Limited, trading as Taylor & Francis Group.

Writing with rocks

Valtonen, Anu; Pullen, Alison (2021.0)

ID do Documento: Writing-with-rocks

ABSTRACT ORIGINAL

Rocks. Geological forces across time and space. Non-human beings. Humans. Affect material encounters with rocks. Connecting. Being. Writing... From the Arctic to Eastern Finland. From Sydney to Kangaroo Island, Australia. From the north to the south, and back again. Corporeal, affective. These rocks live with and through us. Touching rocks—rocks touch us. Bodies—rocks, co-constituted in life. Disrupting. Non-violent. Ethico-political acts of writing. Writing rocks. Humans become geologic forces. Care. Response-able. © 2021 John Wiley & Sons Ltd.

Circles of Care for Safety: A Care Ethics Approach to Safe-by-Design

Baas, Lieke; Metselaar, Suzanne; Klaassen, Pim (2022.0)

ID do Documento: Circles-of-Care-for-Safety

ABSTRACT ORIGINAL

Safe-by-Design is an approach to engineering that aims to integrate the value of safety in the design and development of new technologies. It does so by integrating knowledge of potential dangers in the design process and developing methods to design undesirable effects out of the innovation. Recent discussions have highlighted several challenges in conceptualizing safety and integrating the value into the design process. Therefore, some have argued to design for the responsibility for safety, instead of for safety itself. However, this idea has not been developed further. In this article, we develop an approach to Safe-by-Design, grounded in care ethics, which builds on the idea of designing for responsibility and can deal with the complexity that is inherent to the conceptualization of safety. We describe five ways in which care ethics contributes to the conceptualization of Safe-by-Design: (1) It suggests the development of 'circles of care' in which stakeholders share the responsibility for safety; (2) it recognizes the importance of considering safety as something that is situated in the surroundings of a technology, instead of as a property of the technology itself; (3) it acknowledges that achieving safety is labour that requires an ongoing commitment; (4) it emphasizes that the way in which we relate to technology impacts its safety; and (5) it recognizes the role of emotions in assessing safety. All these elements combined lead to a broader understanding of safety and a philosophically more substantial and practically more appealing conceptualization of Safe-by-Design. © 2022, The Author(s).

Science, Technology and Society: An Introduction

Gobo, Giampietro; Marcheselli, Valentina (2023.0)

ID do Documento: Science-Technology-and-Society

ABSTRACT ORIGINAL

Science, Technology and Society: An Introduction provides students with an accessible overview of the interdisciplinary field of Science and Technology Studies (STS). The discipline breaks down traditional conceptions of knowledge as universal, neutral and ahistorical, and takes a more critical approach to science and technology as social embedded phenomena. This comprehensive textbook makes use of unique examples and case studies to illustrate theoretical debates and concepts. In addition, the reader acquires a unique vision of contemporary issues (such as the power of algorithms, the mystification of fake news, the role of experts within the decision-making process, for example). Each chapter incorporates pedagogically rich features, including interactive discussion points to be used individually or in class as prompts for debate. © The Editor(s) (if applicable) and The Author(s), under exclusive licence to Springer Nature Switzerland AG 2022

Waste, Improvement and Repair on Ireland's Peat Bogs

Bresnihan, Patrick; Brodie, Patrick (2023.0)

ID do Documento: Waste-Improvement-and-Repair-on-Irelands-Peat-Bogs

ABSTRACT ORIGINAL

Since the 1950s, the rural midlands of Ireland have been the site of large-scale industrial peat-cutting. With the phasing out of the peat industry, these post-industrial 'waste' landscapes are now being positioned as central to the country's 'green' and 'smart' futures as sites for renewable energy, carbon sequestration and information services. Contemporary discourses of waste and bog landscapes resonate with discourses in the 18th and 19th centuries, when colonial efforts to drain and reclaim these semi-aqueous territories were driven by moral as well as economic imperatives. In this chapter, we aim to disrupt linear accounts of bog progress and ruination that rely on binaries of waste/value, to surface 'submerged' bog relations that pre-date and endure in spite of their designation as 'un-valuable' within colonial and post-colonial economies. Our contribution is that 'repair', rooted in maintenance, endurance and liveability, while not spectacular, resists the temporality of boom/bust, even if remaining ambivalent in political expression. © 2023 Bristol University Press.

Backyard activisms: Site dance, permaculture and sustainability

Barbour, Karen (2019.0)

ID do Documento: Backyard-activisms-Site-dance-permaculture-and-sustainability

ABSTRACT ORIGINAL

Site dance performance offers creative opportunities for social and environmental activism. Conversely, social and environmental activism is becoming increasingly performative in seeking to evocatively engage a wide range of community members in both local and global concerns. The following pages focus on how research poetry arising in the creative process of developing performances of site dance may support activism. Drawing on interdisciplinary understandings of permaculture and sustainability and new approaches in narrative representation, the research poetry expresses backyard activism in a complementary manner to the site dance itself. The poetry synthesizes creative journal entries relating to somatic improvisation and choreography with dancers, and interviews and discussions with relevant community members. In Aotearoa New Zealand and in this wider neo-liberal era, urban residents are moving away from backyard gardening and local seasonal produce, and towards consumption of mass-produced, regulated and imported foods. In this context, community activism through performance may support local and alternative food movements and speak back to dominant, sociocultural and political power systems and norms relating to food production, consumption and sustainability. © 2019 Intellect Ltd Article.

All we want is the earth: {Land}, labour and movements beyond environmentalism

Bresnihan, Patrick; Millner, Naomi (2023.0)

ID do Documento: All-we-want-is-the-earth

ABSTRACT ORIGINAL

Sixty years ago, an upsurge of social movements protested the ecological harms of industrial capitalism. In subsequent decades, environmentalism consolidated into forms of management and business strategy that aimed to tackle ecological degradation while enabling new forms of green economic growth. However, the focus on spaces and species to be protected saw questions of human work and histories of colonialism pushed out of view. This book traces a counter-history of modern environmentalism from the 1960s to the present day. It focuses on claims concerning land, labour and social reproduction arising at important moments in the history of environmentalism made by feminist, anti-colonial, Indigenous, workers' and agrarian movements. Many of these movements did not consider themselves 'environmental,' and yet they offer vital ways forward in the face of escalating ecological damage and social injustice. © Bristol University Press 2023. All rights reserved.

Strengthening soil taxonomy ontology software for description and classification of USDA soil taxonomy up to soil series

Deb, C.K.; Marwaha, S.; Malhotra, P.K.; Wahi, S.D.; Pandey, R.N. (2015.0)

ID do Documento: Strengthening-soil-taxonomy

ABSTRACT ORIGINAL

Software's using ontology as their knowledge base are of due importance now a days due to their synergism with agents and Semantic Web Architecture. Ontologies provide domain language by defining domain concepts and relationships between them which is ultimately meaningful to both humans and machines. This is IEEE standard Web Ontology Language (OWL). Taxonomies are well-defined hierarchy existing in a standardized form to describe real world concepts in various domains of knowledge. The indispensable role of ontology in Agriculture is to convert the unstructured knowledge into structured one, sharing across application. Das (2010) and Das et al. (2012) developed Soil Ontology for USDA soil taxonomy for orders available in India to only Sub group level. This newly developed Soil Ontology has been strengthened and is now available up to family and series level for orders in India and also for the twelve orders worldwide. The web based application follows N-tier architecture. By mentioning the soil properties one can easily get information related to soil taxonomy and also newly found soils can be classified. Information edition or addition facilities of soil taxonomy are available with domain experts. Advance Search and series navigation keys can be use to easily get the detailed information of taxonomic hierarchy and state wise series description respectively. Its knowledge base is in the form of Ontology. © 2015 IEEE.

Soil aggregates response to tillage and residue management in a double paddy rice soil of the Southern China

Chen, Zhongdu; Ti, Jin-song; Chen, Fu (2017.0)

ID do Documento: Soil-aggregates-response-to-tillage-and-residue-management

ABSTRACT ORIGINAL

Soil aggregate stability is a key indicator of soil quality and environmental sustainability of agroecosystems. The protection of organic material within aggregates against microbial decomposition is regarded as an important process in soil organic carbon stabilization but detailed knowledge about this process is still lacking. The objective of our study was to examine the multiple year effects of plow tillage with residue removed (PT0), plow tillage with residue incorporation (PT), rotary tillage with residue retention (RT), and no-till with residue retention (NT) on soil water stable aggregates (WSA) under a double rice (*Oryza sativa* L.) cropping system in the Southern China. Results showed that the NT system increased the proportion of >2 mm aggregate fraction, and reduced the proportion of <0.053 mm aggregates at 0–5 cm depth in 2011. Compared with PT0 and PT, significantly higher large macroaggregate (>2 mm) associated-C contributions to TOC were observed in the surface layer (0–10 cm depth) under RT and NT. A significant positive correlation between TOC and macroaggregate (>2 and 2–0.25 mm) associated-C was observed at 0–20 cm soil depth in the paddy rice ecosystem. Therefore, conversion to NT, could enhance the formation of stable macroaggregate, macroaggregates associated-C, and total C contents in paddy soil of Southern China. © 2017, Springer Science+Business Media B.V.

Soil aggregates response to tillage and residue management in a double paddy rice soil of the Southern China

Chen, Zhongdu; Ti, Jin-song; Chen, Fu (2017.0)

ID do Documento: Soil-aggregates-southern-China

ABSTRACT ORIGINAL

Soil aggregate stability is a key indicator of soil quality and environmental sustainability of agroecosystems. The protection of organic material within aggregates against microbial decomposition is regarded as an important process in soil organic carbon stabilization but detailed knowledge about this process is still lacking. The objective of our study was to examine the multiple year effects of plow tillage with residue removed (PT0), plow tillage with residue incorporation (PT), rotary tillage with residue retention (RT), and no-till with residue retention (NT) on soil water stable aggregates (WSA) under a double rice (*Oryza sativa* L.) cropping system in the Southern China. Results showed that the NT system increased the proportion of >2 mm aggregate fraction, and reduced the proportion of <0.053 mm aggregates at 0–5 cm depth in 2011. Compared with PT0 and PT, significantly higher large macroaggregate (>2 mm) associated-C contributions to TOC were observed in the surface layer (0–10 cm depth) under RT and NT. A significant positive correlation between TOC and macroaggregate (>2 and 2–0.25 mm) associated-C was observed at 0–20 cm soil depth in the paddy rice ecosystem. Therefore, conversion to NT, could enhance the formation of stable macroaggregate, macroaggregates associated-C, and total C contents in paddy soil of Southern China. © 2017, Springer Science+Business Media B.V.

Tellus-Onto: An ontology for soil classification and inference in precision agriculture; [Tellus-Onto: uma ontologia para classificação e inferência de solos na agricultura de precisão]

Helfer, Gilson Augusto; Da Silva, Adilson Ben; Bavaresco, Rodrigo Simon; Barbosa, Jorge Luis Victória
(2021.0)

ID do Documento: Tellus-Onto-An-ontology-for-soil-classification

ABSTRACT ORIGINAL

Soil analysis laboratories demand large volumes of data used in precision agriculture. Among them, parameters that represent soil fertility such as texture and organic matter guide the fertilization process. However, this process can take time, thus limiting its usefulness. Therefore, this article proposes an ontology called Tellus-Onto that extends the state of the art in the classification of Brazilian soils according to the organic and textural composition. A series of axioms and semantic rules provided consultations and inferences about their instantiated basis. In order to test the ontology, we added 98 soil sample results and their classifications were inferred precisely and automatically. © 2021 ACM.

SoilBR-Onto: An Ontology for Soil Fertility Management and Classification in Brazil

Helfer, Gilson Augusto; da Costa, Adilson Ben; Barbosa, Jorge Luis Victória (2025.0)

ID do Documento: SoilBR-Onto

ABSTRACT ORIGINAL

Soil analysis laboratories require extensive data for precision agriculture, including parameters such as texture, organic matter, phosphorus, and potassium, which guide fertilization and soil acidity correction through pH. This article presents SoilBR-Onto, an ontology that advances the use of fertilizers, liming recommendations and classification of Brazilian soils based on laboratory analysis. A total of 481 axioms and 145 semantic rules enable accurate queries and inferences using real soil sample results. The SoilViewer Mobile application, developed within this study, facilitates access to fertilization and liming recommendations and enables the visualization of soil deficiencies through heat maps. Furthermore, this ontology improves soil data management and decision-making, applicable beyond applied tobacco, to crops such as soybeans, wheat and corn, promoting environmental sustainability and efficiency in agriculture. © The Author(s) 2025

Knowing {Earth}, {Knowing} {Soil}: {Epistemological} {Work}... - {Google} {Acadêmico}

Kearnes, Matthew; Rickards, Lauren (2020.0)

ID do Documento: KnowingEarth-Knowing-Soil-Epistemological-Work-and-the-Political-Aesthetics-of-Regenerative-Agriculture

ABSTRACT ORIGINAL

Soil carbon sequestration and regenerative agriculture have emerged as technologies of eco-social optimism. In the face of the dire warnings of “peak nitrate” and “peak phosphorus” on the one hand and climate change on the other, soil’s capacity to store carbon, reverse climate change, regenerate itself, and breathe life into agricultural landscapes has been presented as something of a revelation (Kearnes and Rickards 2017; Kon Kam King et al. 2018; Granjou and Salazar 2019). Marshaled as a response to a raft of contemporary sociopolitical and environmental problems, the retention and long-term sequestration of soil carbon and the promise of regenerative agriculture encode soil and farming with political atmospherics that are simultaneously speculative and hopeful (Krzywoszynska 2019a). At the same time, the management of soil carbon concentrations is increasingly being depicted in technoscientific terms. In this guise, soil carbon sequestration is cast as an earthly technology, overlain by the technics of remote visualization, greenhouse gas accounting, and scenario modeling. Soil carbon sequestration is heralded as one of a number of strategies for directly manipulating environmental and climatic systems in response to global warming, what Tim Flannery evocatively terms “third-way” technologies designed to “recreate, enhance or restore the processes that created the balance of greenhouse gases which existed prior to human interference, with the aim of drawing carbon, at scale, out of Earth’s atmosphere and/or oceans”(2015: 151). Agricultural soil carbon sequestration simultaneously features as a core rationale for the emergent paradigm of “smart farming,” which harnesses environmental sensing technologies and automated management systems to enable more efficient, sustainable, and profitable farm outcomes and related market opportunities (Rossel and Bouma 2016). This quintessentially modernist take on carbon sequestration and agriculture is imbued with extremely different language and images to that of “alternative” regenerative agriculture, but shares with it a promissory logic and similarly high hopes for a brighter future.

Development of semantic enabled engineering soil classification along with visualisation of particle size distribution curve application

Srinivas Kumar, P.; Raja Reddy, K. (2019.0)

ID do Documento: Development-of-semantic-enabled-engineering-soil-classification

ABSTRACT ORIGINAL

Soil classification is the basic knowledge which a geotechnical engineer needs to have before embarking on the construction of projects like highway or metro construction. With the advent of semantic web technologies it is now possible for the humans and machines to collaborate by way of understanding the underlying meaning of the soil classification concepts. An innovative approach is discussed in this paper where artificial intelligence enabled soil classification is developed along with the visualization of particle size distribution curve using R language and owl technologies. © BEIESP.

Development of semantic enabled engineering soil classification along with visualisation of particle size distribution curve application

Srinivas Kumar, P.; Raja Reddy, K. (2019.0)

ID do Documento: Develpment-of-semantic-enabled-engeneering

ABSTRACT ORIGINAL

Soil classification is the basic knowledge which a geotechnical engineer needs to have before embarking on the construction of projects like highway or metro construction. With the advent of semantic web technologies it is now possible for the humans and machines to collaborate by way of understanding the underlying meaning of the soil classification concepts. An innovative approach is discussed in this paper where artificial intelligence enabled soil classification is developed along with the visualization of particle size distribution curve using R language and owl technologies. © BEIESP.

How does national SOC monitoring on agricultural soils align with the EU strategies? An example using five case studies

Meurer, Katharina H. E.; Hendriks, Chantal M. J.; Faber, Jack H.; Kuikman, Peter J.; van Egmond, Fenny; Garland, Gina; Putku, Elsa; Barancikova, Gabriela; Makovníková, Jarmila; Chenu, Claire; Herrmann, Anke M.; Bispo, Antonio (2024.0)

ID do Documento: How-does-national-SOC-monitoring-on-agricultural-soils

ABSTRACT ORIGINAL

Soil functioning contributes to the delivery of a vast range of ecosystem goods and services, and ecosystem health is therefore reflected by the capacity of the soil to perform underlying functions. Soil organic carbon (SOC) is a key indicator for soil quality as it is an integral driver of many soil functions and associated ecosystem services. Across the globe, SOC stocks are declining due to expanding agriculture and unsustainable practices. Awareness of the fact that soil is a non-renewable resource and its functioning important for all life on Earth is increasing, especially among policymakers. As such, goals for the preservation and restoration of SOC are formulated in policies under the European Green Deal. However, the evaluation of these goals at the European level is hampered by a non-harmonized diversity in national SOC monitoring strategies. While some SOC indicators can be useful for the evaluation of most policy goals (i.e., baseline and potential SOC stocks), additional and contrasting SOC data are often required for the evaluation of the goals formulated by the different EU directives. This study provides an overview of five ongoing SOC monitoring programmes across Europe and discusses how national programmes may be aligned to evaluate goals at the EU level. Five countries with very different soil monitoring programmes were included in a case study to illustrate the potential for harmonization and standardization of SOC assessment. Based on this study, we conclude that SOC monitoring strategies can be harmonized, but not standardized. We further suggest five sampling strategies that have potential for harmonization under the proposed Directive on Soil Monitoring and Resilience. © 2024 The Authors. European Journal of Soil Science published by John Wiley & Sons Ltd on behalf of British Society of Soil Science.

What Is Soil, and What Is It For? Social Ontologies and Social Epistemologies of Soil Affordances and Soil Experiments

Kendig, Catherine (2025.0)

ID do Documento: What-Is-Soil-and-What-Is-It-For

ABSTRACT ORIGINAL

Soil has been defined in numerous ways within the scientific study of soil. How soil has been defined and what it is thought to be for have influenced the design of soil experiments and the development of agricultural technologies aimed at its improvement. Normative goals, like improving soil capabilities, increasing crop yield, and sustainably managing soil health for future generations structure the kinds of questions being pursued in soil science research and in turn shape what can be known from agronomy experiments. I analyse how the interplay of epistemological, normative, and ontological conceptualizations of soil have recursively shaped soil knowledge, what we think soil is for, and how soil is categorized. I focus on three case studies to do this: (1) J. H. Quastel's conceptualization of soil metabolism, (2) the development of the Haber-Bosch process and widespread use of chemical nitrogen fertilizers; and (3) George Washington Carver's long-term soil improvement experiments. These illustrate both how social epistemologies and ontologies shape development of new agricultural biotechnologies and how adoption of these affords different opportunities and burdens to farmers in virtue of how these technologies affect the chemical, physical, and biological interactions of soil-plant-animal interactions. © The Author(s) 2025.

Data from long-term experiments in temperate croplands to evaluate soil organic carbon models

Fujisaki, Kenji; Ferchaud, Fabien; Clivot, Hugues; Bruni, Elisa; Guenet, Bertrand; Pichot, Christian; Versini, Antoine; Baudin, François; Bispo, Antonio; Peylin, Philippe; Martin, Manuel P.; Jensen, Johannes L.; Eriksen, Jørgen; Chenu, Claire; Gregory, Andrew S.; Glendining, Margaret J.; Merbach, Ines; Beaudoin, Nicolas; Mary, Bruno; Mollier, Alain; Tison, Gilles; Montagnier, Christophe; Chabbi, Abad; Vertès, Françoise; Cadéro, Alice; Graux, Anne-Isabelle; Pellerin, Sylvain; Levavasseur, Florent; Gilles, Manon; Morvan, Thierry; Resseguier, Camille; Milesi, Luis; Irizar, Alicia; Andriulo, Adrià; Mistou, Marie-Noël; Butier, Arnaud; Bertrand, Michel; Autret, Bénédicte; Jeuffroy, Marie-Hélène; Grandeau, Gilles; Doré, Thierry; Cellier, Vincent; Berthier, Alain; Darras, Sébastien; Audebert, Guillaume; Pasquier, Ludovic; Ecalte, Fabien; Savoie, Antoine; Schiedung, Marcus; Poeplau, Christopher; Maaroufi, Nadia I.; Kätterer, Thomas; Bolinder, Martin A.; Sanderman, Jonathan; Barré, Pierre (2026.0)

ID do Documento: Data-from-long-term-experiments-in-temperate-croplands

ABSTRACT ORIGINAL

Soil organic carbon (SOC) models need independent evaluation against field measurements, but those latter are rarely publicly available and harmonized. In this study, we collected and shared data from 167 agronomic treatments in 34 agronomic long-term experiments (LTEs) located in temperate croplands, allowing the evaluation of several soil organic C models such as RothC, Century, AMG, MIMICS, ICBM, Millennial, and CTOOL. The dataset includes climate data, soil properties, C inputs from crops (n = 4588 records) and organic amendments, irrigation data, monthly soil cover, as well as SOC stock measurements in the topsoil layer (n = 1328 records). Climate, soil moisture, and soil temperature data were extracted from daily climate databases. Carbon inputs from crops were calculated from observed yields and harvest index, with some harvest index values estimated, combined with crop allometric coefficients from the literature. Descriptions of LTE, agronomic treatments, methodological metadata, and a part of the code, accompanies the dataset. The dataset can be reused to evaluate single SOC models, or to evaluate an ensemble of models. © The Author(s) 2026.

Limitations and future perspectives for satellite-based soil carbon monitoring

Nieto, Luciana; Houborg, Rasmus; Tivet, Florent; Olson, Brad J.S.C.; Prasad, P.V. Vara; Ciampitti, Ignacio A. (2024.0)

ID do Documento: Limitations-and-future-perspectives-for-satellite-based-soil-carbon-monitoring

ABSTRACT ORIGINAL

Soil organic carbon (SOC) plays a crucial role in terrestrial C storage and ecosystem services. Agricultural management practices have the potential to increase C inputs and reduce its losses. However, uniform standard protocols for measuring, monitoring, and assessing changes using remote sensing is lacking for SOC in the scientific literature. In this discussion paper, we present techniques for collecting and analyzing ground samples and employing remote sensing to quantify SOC, along with its limitations and future perspectives. Our analysis identified a number of key limitations to advancing the science for remotely sensed terrestrial C in croplands including i) lack of consensus in sampling depth and density, ii) the absence of a standard (or universally accepted) laboratory procedure and statistical methodology, and iii) lack of details on imagery pre-processing or information on the spectral properties of the targeted soils. Establishing standard protocols for ground-truth data collection and remote sensing approaches, as well as a knowledge of the impacts of diverse soil types, land uses, and landscapes on C assessment, are all required to enhance the accuracy and reliability of future SOC assessments. © 2024 The Author(s)

Nurturing soil-abilities: Growing multispecies justice in therapeutic landscapes

Turner, Bethaney (2023.0)

ID do Documento: Nurturing-soil-abilities

ABSTRACT ORIGINAL

Soil, and its underground multispecies microbiome, has so far been overlooked in work interrogating gardens as therapeutic landscapes. This chapter explores the emplaced, affective, and ethical dimensions of human-soil relations in urban home and community gardens through ethnographic fieldwork, demonstrating that the fundamental interdependence of human and soil health and wellbeing makes soil a fertile focus for investigation. Soil not only shapes the productive capacities of gardens and sustains urban green spaces; biodiverse soil has also been shown to have direct health benefits. On the flipside, the presence of contaminated, toxic soils - most prevalent in lower socio-economic neighbourhoods - exposes inequities in access to safe gardening sites and the realisation of their potential therapeutic benefits. Seizing on the growing emphasis on therapeutic spaces as relationally produced, this chapter contends that the therapeutic benefits of gardens are best realised when human participants cultivate attentive, responsive relationships with the myriad multispecies companions that mutually co-produce these sites. To do this, the chapter explores how a multispecies justice framework can be enacted in therapeutic garden landscapes to maximise human and multispecies health and wellbeing. © 2023 selection and editorial matter, Pauline Marsh and Allison Williams; individual chapters, the contributors.

Living and labouring soils: Metagenomic ecology and a new agricultural revolution?

Granjou, Céline; Phillips, Catherine (2019.0)

ID do Documento: Living-and-labouring-soils-Metagenomic-ecology-and-a-new-agricultural-revolution

ABSTRACT ORIGINAL

Soils teem with life, yet only in the last two decades have genomic approaches begun to reveal the secret liveliness of soil. No longer understood as 'abiotic', soil has become a dense genetic web entangling various forms of life, the majority of which remain unidentified. This paper focuses on the growing hopes to identify, categorise, and harness soil liveliness as it emerges from ongoing scientific research at the juncture of soil ecology, molecular biology, and agricultural development. Drawing upon fieldwork interviews and observations with social ecologists in France, we demonstrate how the advancement of soil metagenomics fosters new conceptions of soil as both 'living' and 'labouring'. With these refigurations of soil, soil ecologists anticipate a new agro-ecological revolution in which agrochemicals will be replaced (or at least reduced) through farmers' future employment of biota for soil services. We argue that new understandings of soil as lively and labouring are not only transforming in the sense of fostering a hoped-for agro-ecological transition, but that insights into how underground realms are becoming understood offer new ways into thinking through more-than-human biopolitics and (agricultural) bioeconomies. © 2018, Springer Nature Limited.

Wilted: Pathogens, chemicals, and the fragile future of the strawberry industry

Guthman, Julie (2019.0)

ID do Documento: Wilted-Pathogens-chemicals-and-the-fragile-future-of-the-strawberry-industry

ABSTRACT ORIGINAL

Strawberries are big business in California. They are the sixth-highest-grossing crop in the state, which produces 88 percent of the nation's favorite berry. Yet the industry is often criticized for its backbreaking labor conditions and dependence on highly toxic soil fumigants used to control fungal pathogens and other soilborne pests. In *Wilted*, Julie Guthman tells the story of how the strawberry industry came to rely on soil fumigants, and how that reliance reverberated throughout the rest of the fruit's production system. The particular conditions of plants, soils, chemicals, climate, and laboring bodies that once made strawberry production so lucrative in the Golden State have now changed and become a set of related threats that jeopardize the future of the industry. © 2019 by Julie Guthman.

Sufficiency and the logic of care: Transforming the relationships, practices and temporalities of businesses

Beyeler, Laura (2024.0)

ID do Documento: Sufficiency-and-the-logic-of-care

ABSTRACT ORIGINAL

Sufficiency in business represents a shift from a logic of profit towards a logic of care. The purpose of sufficiency-oriented businesses is to serve the fulfilment of needs. A logic of care transforms the relationships, practices and temporalities of businesses and represents leverage to enable sufficiency in praxis. © Maike Gossen, Laura Niessen (eds.). All rights reserved.

City infrastructure ontologies

Du, Heshan; Wei, Lijun; Dimitrova, Vania; Magee, Derek; Clarke, Barry; Collins, Richard; Entwisle, David; Eskandari Torbaghan, Mehran; Curioni, Giulio; Stirling, Ross; Reeves, Helen; Cohn, Anthony G. (2023.0)

ID do Documento: City-infrastructure-ontologies

ABSTRACT ORIGINAL

Sustainable urban infrastructure planning and maintenance require an integrated approach that considers various infrastructure assets (e.g., the ground, roads, and buried pipes) and their inter-linkages as a holistic system. To facilitate the usage of this integrated approach, we propose a model of city infrastructure assets and their interdependencies, providing details on how asset properties and processes affect each other. This model is represented as ontologies in OWL 2 Web Ontology Language Manchester Syntax, which can be read and interpreted by machines automatically. These ontologies cover the classifications, properties and processes of the ground, roads and buried water pipes, as well as some related human activities and natural phenomena (e.g., precipitation). The ontologies not only provide a foundation for integrating various types of infrastructure and environmental data, but also for understanding the potential knock-on effects of asset failures. The ontologies have been utilised in a decision support system for integrated urban inter-asset management. © 2023 The Authors

SOWING THE SEEDS OF TRANSFORMATION IN THE COLOMBIAN ANDES: THE ROLE OF THE HUERTA (HOME GARDEN) IN FOSTERING AND SUSTAINING AGROECOLOGICAL PRODUCTION*

Mahecha-Groot, Ana Maria; Leichenko, Robin (2025.0)

ID do Documento: SOWING-THE-SEEDS-OF-TRANSFORMATION-IN-THE-COLOMBIAN-ANDES

ABSTRACT ORIGINAL

The adoption of agroecological and related alternative farming practices is frequently cited as a viable strategy for fostering transformative change within agricultural and food systems. This study investigates the role of agroecology as a pathway of transformation of small-scale agriculture in the Colombian Andes. Drawing upon the “three spheres” model of transformation, which encompasses interactions between the personal, practical, and political spheres, the study explores motivations, actions, and political implications associated with the adoption of agroecology in three sites within the Bogotá region. The study employs a qualitative methodology, including in-depth farmer interviews and participant observation. Study findings highlight a range of economic, cultural, and ecological motivations within the personal sphere, all of which are driving farmers toward adoption of agroecology. These motivations, many of which are rooted in an ethics of care and connection, manifest in the practical sphere through actions including soil care, plant care, and care of seeds. Within the political sphere, these actions are contributing to micropolitical changes that are fostering broader shifts toward sustainable local food systems. Taken together, these findings suggest that agroecological practices can offer a viable pathway for addressing the current challenges in agriculture, promising a more sustainable and socially just future for small-scale farming. © Copyright © 2024 by the American Geographical Society of New York.

Use of NLP Techniques and High-Performance Computing for Automated Knowledge Based Ontology Construction of Saffron Crop

Joshi, Manjushri Sanjay; Pangave, Vijaya Vijay (2025.0)

ID do Documento: use-of-NLP-techniques

ABSTRACT ORIGINAL

The agriculture industry is very different from other industries, as it depends on many natural and climatic factors. Revolutionary changes are shaking up the way farming is done and opening up new opportunities and challenges. The amount of user-generated information about smart farming, green houses, and indoor farming that is available on the web keeps growing. Learning how to represent language and solve tasks from beginning to end without the help of human experts to extract and create features has made approaches more precise and much more challenging in terms of the number of parameters. This means that they need parallelized and distributed resources for high-performance computing. This chapter gives a knowledge-based representation of the most recent problems, algorithms, and models in the field of agriculture that involve natural language processing. Mainly we consider saffron crop cultivation. Additionally, the impact of high-performance computing for natural language processing (NLP) applications is shown by going into detail about deep learning approaches that use large amounts of data to train models before using them to understand text. In the proposed method, NLP techniques are used to prepare the data; rule-based formal concept analysis and mapping are used to analyze and map the tags that have been reduced; and finally weather, pests, and soil subdomain ontologies are built separately from the saffron agricultural domain data. © 2025 selection and editorial matter, Mukesh Raghuwanshi, Pradnya Borkar, Rutvij H. Jhaveri, and Roshani Raut.

Use of NLP Techniques and High-Performance Computing for Automated Knowledge Based Ontology Construction of Saffron Crop

Joshi, Manjushri Sanjay; Pangave, Vijaya Vijay (2025.0)

ID do Documento: use-of-NLP-techniques

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Use of NLP Techniques and High-Performance Computing for Automated Knowledge Based Ontology Construction of Saffron Crop

Joshi, Manjushri Sanjay; Pangave, Vijaya Vijay (2025.0)

ID do Documento: Use-of-NLP-Techniques-and-High-Performance-Computing-for-Automated-Knowledge-Based-Ontology

ABSTRACT ORIGINAL

The agriculture industry is very different from other industries, as it depends on many natural and climatic factors. Revolutionary changes are shaking up the way farming is done and opening up new opportunities and challenges. The amount of user-generated information about smart farming, green houses, and indoor farming that is available on the web keeps growing. Learning how to represent language and solve tasks from beginning to end without the help of human experts to extract and create features has made approaches more precise and much more challenging in terms of the number of parameters. This means that they need parallelized and distributed resources for high-performance computing. This chapter gives a knowledge-based representation of the most recent problems, algorithms, and models in the field of agriculture that involve natural language processing. Mainly we consider saffron crop cultivation. Additionally, the impact of high-performance computing for natural language processing (NLP) applications is shown by going into detail about deep learning approaches that use large amounts of data to train models before using them to understand text. In the proposed method, NLP techniques are used to prepare the data; rule-based formal concept analysis and mapping are used to analyze and map the tags that have been reduced; and finally weather, pests, and soil subdomain ontologies are built separately from the saffron agricultural domain data. © 2025 selection and editorial matter, Mukesh Raghuwanshi, Pradnya Borkar, Rutvij H. Jhaveri, and Roshani Raut.

Worlds of Gray and Green Mineral Extraction as Ecological Practice

Ureta, Sebastián; Flores, Patricio (2022.0)

ID do Documento: Worlds-of-Gray-and-Green-Mineral-Extraction

ABSTRACT ORIGINAL

The Anthropocene has arrived riding a wave of pollution. From "forever chemicals" to oceanic garbage patches, human-made chemical compounds are seemingly everywhere. Concerned about how these compounds disrupt multiple lives and ecologies, environmental scholars, activists, and affected communities have sought to curb the causes of pollution, focusing especially on the extractive industries. In *Worlds of Gray and Green*, authors Sebastián Ureta and Patricio Flores challenge us to rethink extraction as ecological practice. Adopting an environmental humanities analytic lens, Ureta and Flores offer a rich ethnographic exploration of the waste produced by Chile's El Teniente, the world's largest underground mine. Deposited in a massive dam, the waste—known as tailings—engages with human and non-human entities in multiple ways through a process the authors call geosymbiosis. Some of these geosymbioses result in toxicity and damage, while others become the basis of lively novel ecologies. A particular kind of power emerges in the process, one that is radically indifferent to human beings but that affects them in many ways. Learning to live with geosymbioses offers a tentative path forward amid ongoing environmental devastation. © 2022 by Sebastián Ureta & Patricio Flores.

The importance of practices of collective care: Exploring directions for an alternative productive paradigm fitting our times' social, economic and ecological requirements

Rigkos-Zitthen, Ioannis; Kapitsinis, Nikos (2025.0)

ID do Documento: The-importance-of-practices-of-collective-care

ABSTRACT ORIGINAL

The Anthropocene is characterized by multiple crises associated with the infinite accumulation of growth on a planet of finite resources. Productive labour and the 8-h working model contribute to this contradiction. We argue for the reduction of productive labour in favour of reproductive labour accumulated through practices of collective care. The latter can heal the damage capital accumulation produces. Collective care brings into light various social practices often invisible to production, allows for a new understanding of nonhuman agency, and challenges the dominant ethical disposition around work. We advocate for a new power equilibrium between productive and reproductive labour. © The Author(s) 2025.

Development of an Ontology for Bulgarian Soil Types

Stoyanova-Doycheva, Asya; Madanska, Sebiha; Bilyanov, Sheban (2023.0)

ID do Documento: Development-of-an-Ontology-for-Bulgarian-Soil-Types

ABSTRACT ORIGINAL

The article presents the development of an ontology for the Taxonomic List of Soils in Bulgaria (according to the FAO World System). In addition to soil types, the ontology includes knowledge about the different agro-climatic regions in Bulgaria that are suitable for cultivating various agricultural crops. The ontology is part of the knowledge base of the ZEMELA platform for smart agriculture. © 2023 IEEE.

Revitalizing Air: More-than-Human Relations in Urban Health Beyond the Modern-Premodern Binary

Meloni, Maurizio; Maller, Cecily (2024.0)

ID do Documento: Revitalizin-Air

ABSTRACT ORIGINAL

The cleanliness, mobility and quality of urban air has regained political legibility in debates on post-pandemic cities. To contextualize the political and epistemological significance of air in urban contexts, we suggest looking at the under-researched experience of premodern cities and bodies, how they developed a complex ecological imagination and solutions guided by findings from Hippocratic-Galenic medicine. While we do not romanticize these efforts, we argue that they represent an overlooked archive through which the post-Enlightenment mechanization, securitization and abstraction of air can be challenged. Turning to recent findings from both more-than-human thinking and microbiology as applied to air (aerobiome), we acknowledge that microbiome science is a result of laboratory science; however, we argue that findings from microbiome science point to a reanimation of air as something that cannot be fully instrumentalized or securitized as in modernistic programs of biopolitical control. By drawing on the on the experience of the Hippocratic tradition as a catalyst and a proxy for wider ontologies of flows and corporeal porosity across the Eurasian landmass, we suggest arriving at an affirmative reconceptualization of human-environment entanglement based on notions of permeability and a non-binary ontology of flows. This more-than-human approach may not only complicate the alleged simplistic view of the “West” as a dualistic monolith but act also as bridge and companion to Indigenous and Southern ontologies and experiences of life, non-life, matter and nature.

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Revitalizing Air: More-than-Human Relations in Urban Health Beyond the Modern-Premodern Binary

Meloni, Maurizio; Maller, Cecily (2024.0)

ID do Documento: Revitalizing-Air-More-than-Human-Relations

ABSTRACT ORIGINAL

The cleanliness, mobility and quality of urban air has regained political legibility in debates on post-pandemic cities. To contextualize the political and epistemological significance of air in urban contexts, we suggest looking at the under-researched experience of premodern cities and bodies, how they developed a complex ecological imagination and solutions guided by findings from Hippocratic-Galenic medicine. While we do not romanticize these efforts, we argue that they represent an overlooked archive through which the post-Enlightenment mechanization, securitization and abstraction of air can be challenged. Turning to recent findings from both more-than-human thinking and microbiology as applied to air (aerobiome), we acknowledge that microbiome science is a result of laboratory science; however, we argue that findings from microbiome science point to a reanimation of air as something that cannot be fully instrumentalized or securitized as in modernistic programs of biopolitical control. By drawing on the on the experience of the Hippocratic tradition as a catalyst and a proxy for wider ontologies of flows and corporeal porosity across the Eurasian landmass, we suggest arriving at an affirmative reconceptualization of human-environment entanglement based on notions of permeability and a non-binary ontology of flows. This more-than-human approach may not only complicate the alleged simplistic view of the “West” as a dualistic monolith but act also as bridge and companion to Indigenous and Southern ontologies and experiences of life, non-life, matter and nature.

© 2024 American Association of Geographers Published by Taylor & Francis Group, LLC.

MORE-THAN-HUMAN PARTICIPATORY DESIGN

Heitlinger, Sara; Light, Ann; Akama, Yoko; Lindström, Kristina; Stahl, Asa (2025.0)

ID do Documento: MORE-THAN-HUMAN-PARTICIPATORY-DESIGN

ABSTRACT ORIGINAL

The ecological crises we face are fed by a disconnect from interspecies interdependencies and misguided human-exceptionalism (the belief we are somehow distinct from nature). Given that design may contribute to our worsening socio-ecological conditions, this is seen as a pivotal moment to question and expand Participatory Design's traditional tenets, such as those grounded in democracy, rights, fairness, inclusion and empowerment, to consider what a different philosophical starting point and commitment to more-than-human Participatory Design might bring. This chapter explores the different traditions underpinning existing Participatory Design work oriented towards sustainability. We show the beginnings of a Participatory Design drawn from an appreciation of the innately interdependent and entangled nature of the world, called here the more-than-human. Two lineages of Participatory Design scholarship are traced: Sustainability (with branches in Modernist and rights-based thinking) and Entanglement (with branches in care-based and co-ontological being). The chapter presents three cases to ground our explorations and bring to light how human-nature separations in systems, structures, values and mindsets inadvertently condition our practice. Our cases ground the theories and paradigms identified in the four branches and show how commitments play out and evolve through practice. This allows us to critically examine existing tenets of Participatory Design and reinterpret the visions that these hold to explore what more-than-human relationality might entail. Our discussion reveals the tensions, politics, paradoxes and difficulties in turning towards more-than-human relationalities and in working across and between worldviews. The chapter closes with the questions that the ambitions of more-than-human Participatory Design pose for our practice. At best, we hope they present a means to tread more gently and responsibly on the earth we share. © 2025 selection and editorial matter, Rachel Charlotte Smith, Daria Loi, Heike Winschiers-Theophihis, Liesbeth Huybrechts and Jesper Simonsen.

Care and conveyance: Buying baladi bread in Cairo

Barnes, Jessica; Taher, Mariam (2019.0)

ID do Documento: Care-and-conveyance-Buying-baladi-bread-in-Cairo

ABSTRACT ORIGINAL

The Egyptian government has long subsidized bread as part of its program of social support. Most Egyptians eat this subsidized bread, known as baladi bread, daily. We examine everyday practices of handling baladi bread as casual care. Scholarship on care has drawn attention to the multiple ways in which care can be practiced around a range of human and nonhuman others. Casual care signals a mode of practice that constitutes care but that is not necessarily performed by its practitioners as such. It is intentional but also habitual, deliberate but also perfunctory. We look at people's casual care for baladi bread during the understudied phase of conveyance, between the moment of purchase and arrival at home. Practices of handling and selecting bread help produce the quality of the baladi bread that people are eating. While past bread riots have underscored the political salience of this bread, casual care elucidates what goes on between these moments of crisis, revealing the role of people's day-to-day actions in producing a general level of satisfaction with subsidized bread. This analysis draws theoretical attention to how care is practiced, opening up space for thinking about kinds of care that are not all that careful in their enactment. © American Anthropological Association 2019.

Predatory Data: Eugenics in Big Tech and Our Fight for an Independent Future

Chan, Anita Say (2025.0)

ID do Documento: Predatory-Data-Eugenics-in-Big-Tech

ABSTRACT ORIGINAL

The first book to draw a direct line between the datafication and prediction techniques of past eugenicists and today's often violent and extractive "big data" regimes. *Predatory Data* illuminates the throughline between the nineteenth century's anti-immigration and eugenics movements and our sprawling systems of techno-surveillance and algorithmic discrimination. With this book, Anita Say Chan offers a historical, globally multisited analysis of the relations of dispossession, misrecognition, and segregation expanded by dominant knowledge institutions in the Age of Big Data. While technological advancement has a tendency to feel inevitable, it always has a history, including efforts to chart a path for alternative futures and the important parallel story of defiant refusal and liberatory activism. Chan explores how more than a century ago, feminist, immigrant, and other minoritized actors refused dominant institutional research norms and worked to develop alternative data practices whose methods and traditions continue to reverberate through global justice-based data initiatives today. Looking to the past to shape our future, this book charts a path for an alternative historical consciousness grounded in the pursuit of global justice. A free ebook version of this title is available through Luminos, University of California Press's Open Access publishing program. Visit www.luminosoa.org to learn more. © 2025 by Anita Say Chan.

Calling for a Plurality of Perspectives on Design Futuring: An Un-Manifesto

Howell, Noura; Schulte, Britta F.; Twigger Holroyd, Amy; Fatás Arana, Rocío; Sharma, Sumita; Eden, Grace (2021.0)

ID do Documento: Calling-for-a-Plurality-of-Perspectives-on-Design-Futuring-An-Un-Manifesto

ABSTRACT ORIGINAL

The Futures Cone, a prominent model in design futuring, is useful for promoting discussions about possible, plausible, probable, and preferable futures. Yet this model has limitations, such as representing diverse human experiences as a singular point of "the present" and implicitly embedding notions of linear progress. Responding to this, we argue that a plurality of perspectives is needed to engage imaginations that depict a diverse unfolding of potential futures. Through reflecting on our own cultural and professional backgrounds, we offer five perspectives for design futuring as a contribution to this plurality: Parallel Presents, "I Am Time", Epithelial Metaphors, the Uncertainties Cone, and Meet (with) "Speculation". These perspectives open alternative approaches to design futuring, move outside prevalent notions of technological progress, and foreground interdependent, relational agencies. © 2021 Owner/Author.

Thinking-with {Soils}: {An} {Introduction}

Salazar, Juan Francisco; Granjou, Celine; Krzywoszynska, Anna; Tironi, Manuel; Kearnes, Matthew
(2020.0)

ID do Documento: Thinking-with-Soils-An-Introduction

ABSTRACT ORIGINAL

The geneses of this book go back to two parallel encounters. First, the international workshop “Going to Ground,” convened in October 2016 in Sydney by Matthew Kearnes, with Céline Granjou and Juan Francisco Salazar, which was designed as an opportunity to think both creatively and earnestly about the dirt we live on and off. The workshop brought together many of the contributors in this book to discuss how soil conservation and improvement practices are being marshaled in response to concerns over climate change, food security, and rural livelihoods, and how these might be indicative of the deep connections between soil and social processes. Second, the book also has its beginnings in a series of panels at the Knowledge/Culture/Ecologies International Conference held in Santiago, Chile, in November 2017, convened by Juan Francisco Salazar and Céline Granjou, where all five co-editors outlined the initial analytical coordinates for this book. These panels also served as a catalyst to start a broader interdisciplinary discussion that had been brewing, and which we see as having been largely initiated by Maria Puig de la Bellacasa through her pioneering work in recent years. This discussion aimed to engage scholars from not only the humanities and social sciences but also the ecological and soil sciences, as well as soil practitioners, particularly those proponents of integrative science frameworks and social-ecological systems thinking. A premise of these conference panels was that, despite notable contemporary reconceptualizations of soil as a matter of care and concern, it is striking to observe how soil, and its manifold entanglements with plants, fungi, bacteria, and other forms of life, remains largely undertheorized or ignored in contemporary social theory. Despite soil's vital ecological importance, its significance as a belowground three-dimensional living world remains elusive in social and cultural research. This book is about developing work that is attuned and attentive to generating more ethical relations with nonhumans who both pervade and create livable environments, such as soil biota (Krzywoszynska 2019).

Studying the Implementation of Equity Projects in Science Education in Divisive Political Contexts

Tran, Trang C.; Penuel, William R.; Singleton, Corinne; Bell, Philip; Leonhart, Sarah (2025.0)

ID do Documento: Studying-the-Implementation-of-Equity-Projects

ABSTRACT ORIGINAL

The implementation of equity-oriented reforms is never simply a technical matter: it involves directly engaging with the norms and politics responsible for reproducing inequitable opportunities and outcomes, and with efforts to promote educational justice. To date, there has been little research on how leaders in science education navigate the political environments of schooling to engage in equity work in their local contexts. The current political divisions within and among states regarding teaching about racial equity provide an important and timely context for such study. This study examines equity projects that science education leaders engage in and how these relate to recently passed legislation in several states regarding teaching about race and its ongoing role in shaping American society and institutions. It relies on survey data from science education leaders in 33 states, focusing on their familiarity with and involvement in different kinds of equity projects in science education, along with their perceptions of what supports and what hinders their equity-focused work. Employing a mixed methods approach, including descriptive analysis, hierarchical linear modeling, and thematic analysis, we found that engagement in equity projects varied widely across and within states. However, in states with laws promoting equity, leaders were engaged in more racial equity-related projects than in states with legislation that restrict discussion of matters of race, so-called “gag orders”. These findings underscore the significance of the broader political environments in shaping science education leadership and present opportunities for researchers in the field to support leaders in navigating them.

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Qin, Jingjing; Liu, Yuxue; He, Lili; Wang, Yuying; Lü, Haohao; Chen, Litian; Yang, Shengmao (2022.0)

Mechanisms-of-interactions-between-soil-minerals-and-dissolvable-biochar-components

The interactions between dissolvable biochar components and soil minerals were explored from the perspective of minerals to reveal the mechanism of biochar stability in soil. Batch experiments on the adsorption of dissolvable biochar components by soil minerals were carried out. Biochar was prepared from rice straw under different carbonization temperatures (RS300, RS500, and RS700). Kaolinite, montmorillonite, and illite were selected as three representative soil minerals. The results showed that the adsorption capacity of soil minerals gradually increased as the carbon concentration of dissolvable biochar components increased. The order of adsorption capacity of the three soil minerals, montmorillonite>illite>kaolinite, was related directly to their structures. The proportions by which each mechanism contributed to the stability of the representative soil minerals also differed. The binding mechanisms of both kaolinite and montmorillonite to the dissolvable biochar components, whose contribution proportions were 3.4%~87.0% and 32.0%~82.0%, respectively, were a result of the van der Waals force. The binding of illite to the dissolvable components of RS300 was primarily conducted by Ca^{2+} , at a contribution ratio of 60.4%~70.6%; its binding to RS500 and RS700, with contribution ratios of 18.7%~65.0% and 53.0%~67.6%, respectively, was a result of the van der Waals force. Comprehensive comparative analysis shows that RS500 can inhibit the dissolution of dissolvable components to the greatest extent by interactive combination with montmorillonite, which is conducive to utilizing the carbon sequestration and emissions reduction advantages of biochar. © 2022 Editorial Board of Journal of Agro-Environment Science. All rights reserved.

Intergenerational transdisciplinary knowing toward stewarding the land of refuge: learning through the pandemic

Takeuchi, Miwa Aoki; Dadkhahfard, Shima; Kopparla, Mahati; Elhowari, Raneem (2024.0)

ID do Documento: Intergenerational-transdisciplinary-knowing-toward

ABSTRACT ORIGINAL

The knowledge of historically marginalized learners, including racially and linguistically minoritized learners, tends to be obscured in institutionalized learning contexts and by the dominant discourse of “learning loss,” which was reinforced during the COVID-19 pandemic. Based on critical ethnography and the methodology of shared walks, this article highlights intergenerational ways of knowing embodied and emplaced in the context of traditional farming sustained and mobilized by a Syrian refugee family. We illustrate what children were indeed learning in the land of refuge during the pandemic, with their family, beyond narrowly defined in-school learning. We conceptualize the recentering of intergenerational ways of knowing, often overlooked in colonial institutionalized learning spaces, as transdisciplinary acts for disrupting the hegemonic disciplinary formation of science, technology, engineering, and mathematics (STEM). © The Author(s), under exclusive licence to Springer Nature B.V. 2024.

Long-term sustainability of ecological monitoring: perspectives from the Multi-Agency Rocky Intertidal Network

Gilbane, Lisa; Ambrose, Richard F.; Burnaford, Jennifer L.; Helix, Mary Elaine; Miner, C. Melissa; Murray, Steven; Sullivan, Kathleen M.; Whitaker, Stephen G. (2022.0)

ID do Documento: Long-term-sustainability-of-ecological-monitoring

ABSTRACT ORIGINAL

The Multi-Agency Rocky Intertidal Network (MARINe) is structured around a thoughtfully crafted set of sustainable practices, or culture of care, which includes equal partnership among its members and member organizations and a collaborative decision-making process. MARINe prioritizes the use of rigorous, standardized protocols, maintains a shared central database, and keeps costs low and consistent through a diversified funding approach. Because long-term sustainability was MARINe's goal from the beginning, every aspect of the program was established to ensure continuity. MARINe's founders understood the importance of long-term, continuous ecological data, and today this thirty-year-old network is invaluable for informing management decisions and alerting scientists to changing conditions in rocky shore habitats along the Pacific coast of North America. © 2022 Elsevier Inc. All rights reserved.

MORE POSTHUMAN GLOSSARY

Jones, Emily; Klumbyt■, Goda; Braidotti, Rosi (2022.0)

ID do Documento: MORE-POSTHUMAN-GLOSSARY

ABSTRACT ORIGINAL

The notion of the posthuman continues to both intrigue and confuse, not least because of the huge number of ideas, theories and figures associated with this term. More Posthuman Glossary provides a way in to the dizzying array of posthuman concepts, providing vivid accounts of emerging terms. It is much more than a series of definitions, however, in that it seeks to imagine and predict what new terms might come into being as this exciting field continues to expand. A follow-up volume to the brilliant interventions of Posthuman Glossary (2018), this book extends and elaborates on that work, particularly focusing on concepts of race, indigeneity and new ideas in radical ecology. It also includes new and emerging voices within the new humanities and multiple modes of communicating ideas. This is an indispensable glossary for those who are exploring what the non-human, inhuman and posthuman might mean in the 21st century. © Natascha Unkart, 2019 "Untitled (Salvaged)". All rights reserved.

MORE POSTHUMAN GLOSSARY

Jones, Emily; Klumbyt■, Goda; Braidotti, Rosi (2022.0)

ID do Documento: MORE-POSTHUMAN-GLOSSARY

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The notion of the posthuman continues to both intrigue and confuse, not least because of the huge number of ideas, theories and figures associated with this term. More Posthuman Glossary provides a way in to the dizzying array of posthuman concepts, providing vivid accounts of emerging terms. It is much more than a series of definitions, however, in that it seeks to imagine and predict what new terms might come into being as this exciting field continues to expand. A follow-up volume to the brilliant interventions of Posthuman Glossary (2018), this book extends and elaborates on that work, particularly focusing on concepts of race, indigeneity and new ideas in radical ecology. It also includes new and emerging voices within the new humanities and multiple modes of communicating ideas. This is an indispensable glossary for those who are exploring what the non-human, inhuman and posthuman might mean in the 21st century. © Natascha Unkart, 2019 "Untitled (Salvaged)". All rights reserved.

Ecological Transition: What It Is and How to Do It

Ghelfi, Andrea; Papadopoulos, Dimitris (2021.0)

ID do Documento: Ecological-Transition-What-It-Is-and-How-to-Do-It

ABSTRACT ORIGINAL

The paper examines different practices, imaginaries and programs of ecological transitions whose articulation points towards a more-than-local and less-than-global green eco-social transformation. Translocal ecological transitions bring together climate action politics, environmental justice, and the everyday ecologism of experimental community-led technoscience. Within transition projects we see the emergence of new more-than-human political constituencies, the making of broad eco-social coalitions, and the implementation of innovative forms of reparative governance. Ecological transitions foster a new political space, green democracy, as an alternative to both regressive nationalism and green globalism that dominate contemporary politics. © 2021 STS Italia. All Rights Reserved.

Multiple worlds and strange objects: environmental education research as an additive practice

Rautio, Pauliina; Hohti, Riikka; Tammi, Tuure; Ylirisku, Henrika (2022.0)

ID do Documento: Multiple-worlds-and-strange-objects

ABSTRACT ORIGINAL

The paper offers three examples of passionate immersion with strange objects and working with peculiar multispecies assemblages, such as the assemblage of a dove called Romeo and the technology to humidify a greenhouse called 'Princess', or the experiment of orienteering in forests for years, accounting for slips, scratches and tumbles as being taught by the forest - and prioritising these over the more commonplace educational narratives. The paper is structured in a nonconventional way in that most space is reserved for reports from these ongoing inquiries. The authors will each discuss how they situate themselves in relation to strangeness in research and how they proceed methodologically, locating their approaches as postqualitative. The questions each example addresses are: What is a strange object? How do we come across them? What do we begin to do/produce with them? The additive orientation described in the research stories is proposed to be an important constituent for new survival knowledge especially relevant for environmental education, addressing environmental problems as wicked, and demanding approaches that reach beyond methodological divides. © The Author(s), 2022. Published by Cambridge University Press.

Towards Transformative Soil Science: Who am I in relation to my research?

Wiersma, W.; Lopez, D.E. (2024.0)

ID do Documento: Towards-Transformative-Soil-Science

ABSTRACT ORIGINAL

The paradigm of modernity is associated with human detachment from soil as the agricultural and affective foundation of modern societies. The focus on soil as a resource has created increasingly exhausted landscapes. Soil science is well placed to (re)build the urgently needed connection between humans and nature by generating evidence for soil functioning and, perhaps most importantly, by establishing linkages with diverse types of soil knowledge and integrating them in collaborative solutions. Such transdisciplinary solutions can be achieved by transforming current ways of doing soil science, which begin with soil scientists. Anchored in feminist and critical thinking, we used the 'reflecting and doing' framework (Lopez et al., 2023) to demonstrate how reflexivity can help soil science to become a transformative space. By partially dissolving the mind–heart and human–soil dualisms, we can foster a way of generating soil knowledge that emerges from the relations within and beyond the human Self. Such an approach is relatively new in academic soil science, which relies heavily on the scientific method as a detached form of knowing that privileges mind over body, reason over emotion, culture over nature and production over reproduction. We argue that this journey starts with (re)positioning ourselves as soil scientists in relation to our research and advance a Transformative Soil Science that recognizes soil knowledge as situated and embodied and that identifies soil as the main ally for a sustainable world. © 2024 British Society of Soil Science.

Algorithmic food justice: Co-designing more-than-human blockchain futures for the food commons

Heitlinger, Sara; Houston, Lara (2021.0)

ID do Documento:

Algorithmic-food-justice-Co-designing-more-than-human-blockchain-futures-for-the-food-commons

ABSTRACT ORIGINAL

The relationships that constitute the global industrial food system tend towards two dominant values that are creating unsustainable social and environmental inequalities. The first is a human-centered perspective on food that privileges humans over all other species. The second is a view of food as a commodity to be traded for maximum economic value, rewarding a small number of shareholders. We present work that explores the unique algorithmic affordances of blockchain to create new types of value exchange and governance in the food system. We describe a project that used roleplay with urban agricultural communities to co-design blockchain-based food futures and explore the conditions for creating a thriving multispecies food commons. We discuss how the project helped rethink algorithmic food justice by reconfiguring more-than-human values and reconfiguring food as more-than-human commons. We also discuss some of the challenges and tensions arising from these explorations. © 2021 ACM.

A Smart Device for Monitoring Porosity and Moisture Levels of Various Soils

Kavitha, A.; Aarthi, P.; Chandika, N.; Gowri, R.; Kaviyavardhini, S. (2024.0)

ID do Documento: A-Smart-Device-for-Monitoring-Porosity-and-Moisture

ABSTRACT ORIGINAL

The sustainability of agriculture may be threatened by continuous cropping if soil nutrients are not properly measured and supplied. Measuring the nutrients in the soil is crucial for healthy plant development and efficient fertilization. Determining the amount of nutrients in the soil is important for formulated fertilization. Recommendations for nutrient requirements and fertilization tailored to the individual site come next. Plant development requires the three essential nutrients: potassium, phosphorus, and nitrogen. The present study involves the development of an electrochemical sensor capable of differentiating various soil nutrients, including N, P, and K. The sensor operates by absorbing ions from the soil's aqueous solution. The integrated circuits of the sensor and Arduino Microcontroller are designed to detect and pinpoint any deficiencies in the soil's components. It has been noted that the suggested algorithm performs better overall than alternative techniques regarding F-Score, recall, accuracy, and precision. With just a little amount of training, the farmer may effortlessly manage his field soil data in the cloud service by use of the IoT SNA-CR gadget on his mobile Smartphone. It also boosts production and lowers the cost of balancing the minerals in the soil. © 2024 IEEE.

A {Toolkit} for {Hope} ({ASHA}): {Farmers}' {Sovereignty} and {Holistic} {Agriculture} in {India}

Pandey, Poonam; Kuruganti, Kavitha (2024.0)

ID do Documento: A-toolkit-for-hope-ASHA-Farmers-Sovereignty-India

ABSTRACT ORIGINAL

The sustainability of agri-food systems is a major concern worldwide in relation to the changing climate, resource depletion and biodiversity loss (FAO, 2018). Civil Society Organisations (CSO) are often seen in polar opposition to the state-corporate coalition when it comes to concerns related to ownership of resources such as land and seeds (Peschard & Randeria, 2020). There is a lot of growing scholarship on the ways in which civil society groups and networks, especially in the global south, engage in long, confrontational, legal battles with corporate entities (Peschard & Randeria, 2020; Silva Garzón & Gutiérrez Escobar, 2020). However, within this scholarship and the activist circles, there is a growing concern that legal battles and confrontational protests are not enough to engage with the skewed power dynamics in favour of big corporations and neo-liberal policies (Pottinger, 2017; Silva Garzón & Gutiérrez Escobar, 2020). As a result, CSO groups are increasingly engaging in a number of grassroots activities and social innovations that extract agriculture from the exploitative calculation of productivity and economic growth and embed it into a post-growth imaginary of culturally rich practices and relationships of care (McGreevy et al., 2022; Puig de la Bellacasa, 2015). In this chapter, we will discuss the vision of holistic and sustainable agriculture developed and materialised through the network – Alliance for Sustainable and Holistic Agriculture (ASHA). ASHA is a large, volunteer-driven informal network of civil society groups, farmers and like-minded individuals that are concerned about the ongoing crisis in agriculture, the plight of farmers, the destruction of biodiversity and environment and the corporate takeover of agriculture enabled by neo-liberal policies. Based in India, the network came into being in 2010 to collectively build a platform to engage with these pressing concerns. In this chapter, we will discuss three innovative ways – Kisan Swaraj Yatra (Farmers' Sovereignty March), Kisan Swaraj Niti (Policy for Farmers' Sovereignty) and Beej Utsav (Seed Festivals) – through which ASHA is co-crafting and mobilising the vision of a sustainable and holistic agricultural future.

TEMPORALITIES OF CARE IN MORE-THAN-HUMAN DESIGN

Oktaý, Gizem; Lee, Minha; Barati, Bahareh; Wakkary, Ron (2024.0)

ID do Documento: TEMPORALITIES-OF-CARE-IN-MORE-THAN-HUMAN-DESIGN

ABSTRACT ORIGINAL

The traditional way of considering time as a linear trajectory toward infinite human progress limits how our relationship with more-than-human entities such as animals, plants, land and waters provides alternative perceptions of time, which can help tackle environmental injustices and climate change. We provide a concept of temporalities of care to unlock plural expressions of time as informed by more-than-human care practices, things we do to maintain the more-than-human world we cohabit. The concept is exemplified through an example study we conducted with humans and their dog companions, which revealed a set of sensitising concepts (synchrony, sequence, repetition, latency and asynchrony) that shed light on care time relations, which are informative of care practices. We contribute with definitions and examples for sensitising concepts. © 2025 selection and editorial matter, Anton Poikolainen Rosén, Antti Salovaara, Andrea Botero and Marie Louise Juul Søndergaard; individual chapters, the contributors.

Do we have time for democracy? Climate action and the problem of time in the Anthropocene

Ejsing, Mads; Tønder, Lars; Jensen, Ingrid Helene Brandt; Hansen, Janus (2025.0)

ID do Documento: Do-we-have-time-for-democracy

ABSTRACT ORIGINAL

The urgency of climate change has brought democracy to a critical juncture. Existing democratic systems struggle to address the pressing time frames required for effective climate action. This article explores a fundamental shift in temporal orientation caused by climate change. Democracy's linear and progressive image of time clashes with the expanding scales of temporality, encompassing both planetary and microscopic processes. More specifically, the article examines the theoretical links between democracy, climate change, and time in the Anthropocene through empirical engagements with the climate politics associated with deliberative assemblies, community organizing, and activist movements. The core argument is that within these democratic innovations, new future-oriented responses to the crises are beginning to emerge. These innovations experiment with extending democracy's temporal image beyond its conventional boundaries, paving the way for a multi-trajectory democratic politics. The article concludes by discussing the tensions and potential consequences of these democratic practices, ultimately arguing in favor of a multiple and layered understanding of democratic time that extends beyond distinctly human temporalities. © The Author(s) 2024.

KIASOntoRec: A Knowledge Infused Approach for Socially Aware Ontology Recommendation

Valecha, Aastha; Deepak, Gerard; Surya, Deep ak (2023.0)

ID do Documento: KIASOntoRec

ABSTRACT ORIGINAL

The World Wide Web is progressively adopting web 3.0 standards, and there is a demand for semantically compatible recommendation techniques in today's world. This paper presents a framework to facilitate Knowledge infused automatic Ontology recommendation. User queries and user clicks data are preprocessed and Structural Topic Modelling is employed to identify the latent semantic structure found in the corpus and generate topics. RDF triplets are generated based on the topics extracted using OntoCollab. RDF data along with the Google's Knowledge Base API used to obtain the subgraphs and relevant topics. Bagging algorithm based on Random Forest Classification Technique with Decision Trees Algorithm is used to classify the terms. A knowledge base is constructed by merging the subgraphs. Semantic similarity among entities of the knowledge graph and the top 25% of the classified results is computed using SemantoSim and ANOVA. The topic set is extracted and subjected to axiomatization. Finally, the most relevant Ontology to the user query is suggested to the user. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

The Wetland Is Disappearing: Conservation and Care on Turkey'S Kizilirmak Delta

Scaramelli, Caterina (2018.0)

ID do Documento: The-Wetland-Is-Disappearing-Conservation-and-Care-on-TurkeyS-Kizilirmak-Delta

ABSTRACT ORIGINAL

This article analyzes the transformation of the Kizilirmak Delta on the Black Sea coast of Turkey into a Turkish wetland. This production involved the transformation of international categories of wetlands into national imaginaries, as well as the material remaking of landscapes themselves. Population and agro-economic shifts concurrent to the formation of the Turkish nation-state transformed the delta into an agricultural landscape, and subsequently into a contested conservation area whose use is informed by changing Turkish and international notions of wetlands. I focus on the situated, local processes and practices through which wetlands are produced and become relevant to different social groups as subjects of scientific knowledge and environmental imaginations. These, I argue, have rendered the wetland an open-air laboratory and an object of care for environmental advocates, scientists, and residents. © 2018 Cambridge University Press.

Rhythms of care: Medical travels beyond the borders of Bangladesh

McNamara, Karen M. (2020.0)

ID do Documento: Rhythms-of-care-Medical-travels-beyond-the-borders-of-Bangladesh

ABSTRACT ORIGINAL

This article examines the experiences of Bangladeshi patients and their families as they travel transnationally within Asia for medical care. I explain how failures of biomedicine in Bangladesh feed into idealized expectations of care abroad. This medical imaginary is fueled by the hope that more expensive treatment in wealth-ier countries will result in better care, and it is sustained by the way the medical tourism industry operates and the way Bangladeshi patients and their families make choices and engage in the doing of care abroad. A detailed case study of a Bangladeshi cancer patient's prolonged care in Singapore illustrates the tensions and ambivalences in the quest for the best treatment. These tensions are exacerbated by the linguistic, monetary, and emotional challenges faced in traveling back and forth between countries. While patients feel at times betrayed by experiences of care that do not meet their expectations, they also feel compelled to carry on. I capture this dynamic in the term rhythms of care, understanding these as the way the medical imaginary shapes care practices that become a scaffolding for hope to be maintained and further travel to be undertaken. I also reflect on how I become part of these rhythms by acting as the family's interpreter as they navigate health care in Singapore. © 2020 Ministry of Science and Technology, Taiwan.

Ethnography of DIY revisionism: unfinished reburials and right-wing populism in Hungary

Szafonova, Tatyjana (2025.0)

ID do Documento: Ethnography-of-DIY-revisionism

ABSTRACT ORIGINAL

This article examines the phenomenon of reburials as a manifestation of historical revisionism in Hungary, with a particular focus on their role in shaping national identity and iconography. The article explores the transformative nature of reburials, highlighting their role in reclaiming individuals as martyrs and as a tool for historical revisionism. The study draws on ethnographic research to provide insights into vernacular reburials and alternative commemorations, demonstrating how reburials can challenge established orders. Furthermore, the author introduces a new approach that emphasizes the active role of local agents and more broadly shows reburials and revisionism as contested processes involving multiple participants, including non-humans such as soil. © 2025 Informa UK Limited, trading as Taylor & Francis Group.

What Was Soil in Antiquity? A View from Ritual

Blidstein, Moshe (2025.0)

ID do Documento: What-Was-Soil-Antiquity

ABSTRACT ORIGINAL

This article examines the ritual uses and cultural meanings of soil in the ancient Mediterranean (c. 8th century BCE-6th century CE). It analyzes rituals involving the physical manipulation of soil across Greek, Roman, Jewish, and early Christian traditions. Through a thematic examination of practices such as burial, mourning, purification, healing, and symbolic transfer, the article identifies two primary, often overlapping, symbolic dimensions: soil signifying lowliness, humility, dirt, and purification (often 'placeless' soil), and soil representing specific, potent locations such as homeland, sacred sites, or burial grounds ('placed' soil). Significantly, rituals manipulating soil rarely centered directly on agricultural fertility, diverging from interpretations focused solely on earth goddesses. This provides some insights for contemporary relationships with soils. © Moshe Blidstein, 2025.

Grasping onto land: the emotional political ecology of land on the Tonle Sap Lake Wetlands

Chann, Sopheak; Beban, Alice; Wisesjindawat-Fink, Wisa (2025.0)

ID do Documento: Grasping-onto-land

ABSTRACT ORIGINAL

This article examines the socio-ecological disruptions faced by floating communities on Tonle Sap Lake, Cambodia, due to environmental changes and state interventions. Through the lens of emotional political ecology, we draw on multidisciplinary research methods, including participatory mapping, photovoice, and ethnography, to study people's relationships with changing waterscapes and their desire for land. As a response to discussions of 'land grabbing', we introduce the concept of land 'grasping'—attempting to secure land amid increasing precarity—to highlight the emotional dimensions of land politics. This study reveals how grasping for land results in significant costs, including debt and exposure to violence. © 2025 Informa UK Limited, trading as Taylor & Francis Group.

Ackern gegen das {Anthropozän}. {Der} kleinbäuerliche {Weg} der {Revolution}

von Redecker, Sophie; Herzig, Christian (2024.0)

ID do Documento: Ackern-gegen-das-Anthropozan

ABSTRACT ORIGINAL

This article explores how the peasants organized in La Via Campesina, the world's biggest social movement, fight the multiple crises of our time. Drawing on theories of prefigurative social change, we highlight how they enact the future in the interstices of the hegemonic now. With their everyday practices, the peasants grow not only food but also examples of less anthropocentric encounters with the non-human. We show that their farming against the neoliberal, postcolonial and extractive logics of the Anthropocene resonates with decolonial and indigenous critiques of the concept and that their care work for the non/human offers transformative examples of worlds yet to establish. © 2024 Mattersburger Kreis für Entwicklungspolitik. All rights reserved.

Mother-Led Science: Rethinking Science-Based Social Movements Through Temporality and Care

Smith, Geneva M.; Smith, Lindsay A. (2025.0)

ID do Documento: Mother-Led-Science-Rethinking-Science

ABSTRACT ORIGINAL

This article explores the subjective and temporal modes of organizing underlying science-based social movements through an analysis of two mother-led movements in Argentina. The Abuelas de Plaza de Mayo appropriated forensic anthropology and nascent DNA technologies to identify their disappeared grandchildren during the 1976–83 dictatorship. The Madres de Ituzaingó Anexo developed community-led epidemiology studies and agrochemical contamination mapping to argue for a causal relation between intensive pesticide use and high rates of childhood illnesses. We focus on how these movements delineate visions of time, responsibility, and collective action by closely examining the underlying histories and practices of each social movement as they politicized motherhood and appropriated scientific practice. We offer the concept of mother-led science through the temporal registers of constancy, desgaste, and durability to illuminate the iterative relationship between care work, creating sustainable communities and institutions, and the fragile processes of stabilizing facts. Mother-led science, with its dual claims to scientific authority in an epistemic register and maternal authority in an affective register, articulates a potent form of scientific organizing. We suggest that these affective temporalities may be present across all science-based social movements but can be obscured by narratives of linear progress toward objective truths. © The Author(s) 2024.

“A matter of time”: Evidence-making temporalities of vaccine development in the COVID-19 media landscape

Harrison, Mia; Lancaster, Kari; Rhodes, Tim (2022.0)

ID do Documento: A-matter-of-time-Evidence-making-temporalities-COVID-19

ABSTRACT ORIGINAL

This article investigates how evidence of the safety and efficacy of COVID-19 vaccines is enacted in news media via a focus on the temporality of vaccine development. We argue that time constitutes a crucial object of and mechanism for knowledge production in such media and investigate how time comes to matter in vaccine evidence-making communication practices. In science communication on vaccine development, the vaccine object (along with the practices through which it is produced) undergoes a material-discursive shift from an imagined “rushed” product to being many years in the making and uninhibited by unnecessarily lengthy processes. In both these enactments of vaccine development, time itself is constituted as evidence of vaccine efficacy and safety. This article traces how time (performed as both calendar time and as a series of relational events) is materialized as an affective and epistemic object of evidence within public science communication by analyzing the material-discursive techniques through which temporality is enacted within news media focused on the timeline of COVID-19 vaccine development. We contend that time (as evidence) is remade through these techniques as an ontopolitical concern within the COVID-19 vaccine assemblage. We furthermore argue that science communication itself is an important actor in the hinterland of public health practices with performative effects and vital evidence-making capacities. © The Author(s) 2021.

Developing earthly attachments in the anthropocene

Huijbens, Edward H. (2021.0)

ID do Documento: Developing-earthly-attachments-in-the-anthropocene

ABSTRACT ORIGINAL

This book explores the development and significance of an Earth-oriented progressive approach to fostering global wellbeing and inclusive societies in an era of climate change and uncertainty. *Developing Earthly Attachments in the Anthropocene* examines the ways in which the Earth has become a source of political, social, and cultural theory in times of global climate change. The book explains how the Earth contributes to the creation of a regenerative culture, drawing examples from the Netherlands and Iceland. These examples offer understandings of how legacies of non-respectful exploitative practices culminating in the rapid post-war growth of global consumption have resulted in impacts on the ecosystem, highlighting the challenges of living with planet Earth. The book familiarizes readers with the implied agencies of the Earth which become evident in our reliance on the carbon economy - a factor of modern-day globalized capitalism responsible for global environmental change and emergency. It also suggests ways to inspire and develop new ways of spatial sense making for those seeking earthly attachments. Offering novel theoretical and practical insights for politically active people, this book will appeal to those involved in local and national policy making processes. It will also be of interest to academics and students of geography, political science, and environmental sciences.

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Geophagiatic: {Art}, {Food}, {Dirt}

Kelley, Lindsay (2020.0)

ID do Documento: Geophagiatic-Art-Food-Dirt

ABSTRACT ORIGINAL

This book presents a novel and systematic social theory of soil, and is representative of the rising interest in 'the material' in social sciences.

Food and Faith A Theology of Eating

Wirzba, Norman (2018.0)

ID do Documento: Food-and-Faith-A-Theology-of-Eating

ABSTRACT ORIGINAL

This book provides a comprehensive theological framework for assessing significance of eating, employing a Trinitarian theological lens to evaluate food production and consumption practices as they are being worked out in today's industrial food systems. Norman Wirzba combines the tools of ecological, agrarian, cultural, biblical and theological analyses to draw a picture of eating that cares for creatures and that honors God. Unlike books that focus on vegetarianism or food distribution as the key theological matters, this book broadens the scope to include discussions on the sacramental character of eating, eating's ecological and social contexts, the meaning of death and sacrifice as they relate to eating, the Eucharist as the place of inspiration and orientation, the importance of saying grace and whether or not there will be eating in Heaven. Food and Faith demonstrates that eating is of profound economic, moral and theological significance. © Norman Wirzba 2011 Cambridge University Press 2019.

Taking {Soil} {Care} {Seriously}: {A} {Proposition}

Krzywoszynska, Anna (2023.0)

ID do Documento: Taking-Soil-Care-Seriously

ABSTRACT ORIGINAL

This chapter argues that care theories can usefully inform research and governance which aims to create better relations between soils and societies. Firstly, care theories are relevant to the framing of soil-related situations, such as soil research. Applying care theories brings into high relief the political nature of deciding what constitutes a soil or what constitutes soils' needs. Secondly, care theories are relevant to the question of what it means to be responsible for soils, and to the societal distribution of this responsibility. By stressing inter-dependence, the care perspective encourages a shift in soil governance from a focus on individual agents towards broader socio-material relations. Thirdly, care is both sparked into being and propelled by attentiveness, which is both an individual and collective practice. Cultivating soil attentiveness requires the creation of new collective framings which would link attention to soils with societally valued ways of living off and with the land. Overall, the chapter argues for a relational approach to soil research. Further, it shows that applying care to soil research shifts the question from what the soils need to do to support human societies, to what human societies need to do to support soils. Care theories can thus contribute to understanding soils as independent of but entangled with human activities, and understanding human well-being as intrinsically bound with the well-being of soils. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

Mapping careful engagement: From reflexivity to response-ability

Pihkala, Suvi; Karasti, Helena (2024.0)

ID do Documento: Mapping-careful-engagement

ABSTRACT ORIGINAL

This chapter contributes to the theorisation of 'careful engagement' by exploring the affordances of the notion and praxis of response-ability for thinking about our engagements with the worlds of our enquiry. The present work stems from the recent upsurge of feminist new materialist and posthuman approaches that have spurred the reimagination of the ethics and politics of knowledge production and the rethinking of questions such as how we understand our role in shaping the world and with whom and what (else) we are engaging. These questions are underpinned by the understanding that knowing is a material doing with ethicality and care at its marrow. Inspired by this idea, this chapter employs the notion of response-ability to consider the ethics and politics of engagement in the authors' participatory research and development process. Returning to their efforts to engage well and with reflexive sensitivity-and to articulate this effort as they have done over the years-the authors draw on response-ability to better capture the intricacies and trouble at stake in enacting and embodying care in their engaged practice. With the help of response-ability, the chapter brings forth the layers of careful engagement that become unveiled when the focus is shifted from the autonomous, reflexively aware researcher to the intra-active dynamisms of the world. For careful engagement, slowing down with this unfolds as a matter of attentive curiosity, responsiveness and commitment to stay with the trouble of the complexities of engaging in and as part of the world. © Springer Nature Switzerland AG 2023. All rights reserved.

No dig, no ride: Repairing and caring for DIY-designed mountain bike and BMX trails

Healy, Liam (2024.0)

ID do Documento: No-dig-no-ride-Repairing-and-caring-for-DIY-designed-mountain-bike

ABSTRACT ORIGINAL

This chapter draws on research produced with DIY bike trail builders to unpack the forms of repair and care that they employ. I first outline the practices involved in keeping trails running, conceptualising the squatted spaces they occupy as 'ruins'. Second, I draw from literature around care and commoning to argue that these spots are not neglected due to being overlooked, but that this is cultivated, and is a requirement of their subsistence. I find that in these spaces, the mundane practices of shovelling and watering, to the forms of governance and exclusion enacted, illustrate forms of repair and maintenance that are multiple and layered, and that this can lead to a 'repair' of the macho, hetero-normative culture that has previously pervaded in these spots. © 2024 selection and editorial matter, Jim Cherrington; individual chapters, the contributors. All rights reserved.

Rethinking the (Wool) Economy

Smith, Tone; Ehrnström-Fuentes, Maria; Hagolani-Albov, Sophia E.; Klepp, Ingun Grimstad;
Tobiasson, Tone Skårdal (2022.0)

ID do Documento: Rethinking-the-Wool-Economy

ABSTRACT ORIGINAL

This chapter examines the limits to changing the current economic system through policy measures like green growth and the circular economy. We examine the biophysical aspects of the economy and the huge amounts of materials and energy the global economy consumes to achieve growth. Thus far, governmental responses have been incapable of addressing the underlying structural issues of the global textile industry and the accompanying exploitation of nature and peoples. While the necessary deep structural transformations are difficult to achieve through governmental policy change, we suggest that re-localisation of wool production-consumption networks is an expression of how engaged citizens can build more sustainable textile and fibre alternatives in place. Drawing on local food research, this chapter highlights the dangers of conflating local solutions with sustainability. Instead, we argue that assessing these emergent wool ventures based on how they are organised in the living landscape in specific places will enhance the understanding of what kind of socioecological impacts they can achieve. This includes how organising/connecting the activities and visions of wool entrepreneurs in place is essential if these ventures are going to be able to overcome the barriers set by the dominant growth-based system of global trade. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2022, corrected publication 2022.

Deep Learning-Based Soil Nutrient Content Prediction for Crop Yield Estimation

Sharma, Iti; Kumar, Nimish; Verma, Himanshu (2024.0)

ID do Documento: Deep-learning-based-soil

ABSTRACT ORIGINAL

This chapter proposes a deep learning-based approach for predicting soil nutrient content and its impact on crop yield. The objective is to develop an accurate model that can assist farmers in making informed decisions about nutrient management and improving crop productivity. The proposed approach employs a combination of a convolutional neural network (CNN) architecture and long short-term memory (LSTM) networks for analyzing soil samples and forecasting nutrient content. Subsequently, the trained model is harnessed to assess the influence of soil nutrient content on crop yield, taking into account factors like climate, water availability, and soil type. The approach was tested on publicly available soil nutrient and crop yield datasets of soil samples collected from different regions and crops. The findings illustrate that the suggested model surpasses conventional approaches and attains remarkable precision in forecasting soil nutrient levels and crop yield. © 2024. IGI Global. All rights reserved.

The Creative Arts in Early Childhood Teacher Education as a Site of Resistance to Neoliberalism

Grimes, Jessica; Gordon, Paddy; Jobson, Sarah (2024.0)

ID do Documento: The-Creative-Arts-in-Early-Childhood-Teacher-Education

ABSTRACT ORIGINAL

This chapter will examine the claim that early childhood pedagogy is structured by neoliberal political and social normativity, where the processes of creative exploration are secondary to an ultimately arbitrary and pre-determined “outcome”. An “outcomes-focussed” approach to early childhood arts pedagogy is incompatible with the engagement necessary for early childhood teachers to meaningfully co-create with children. However, despite neoliberalism’s hegemony—which today is being contested by often alarming reactionary tendencies—we can find instances of play-based and process-focussed approaches that inform our early childhood education courses. Neoliberalism as a theoretical framework is inimical to transformative teaching approaches, and a tension emerges between early childhood arts pedagogy and teacher preparation within the neoliberal university. Early childhood arts pedagogy, therefore, can potentially produce sites of resistance to neoliberal social and political structures. The challenges of teaching processual creativity to preservice teachers within a university setting that is itself structured by neoliberal norms will be considered, in conjunction with a discussion of the epistemic possibilities that emerge when neoliberalism’s “economization of heretofore non-economic domains” (Brown, *Undoing the demos: Neoliberalism’s stealth revolution*, Zone Books, 2015, p. 31) can be undone. Ultimately, we will argue that early childhood arts pedagogies can open spaces and places that are sites of resistance to neoliberal norms, despite early childhood education’s imbrication with institutions structured by neoliberalism. © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

History, anthropogenic soil, and unbecoming human

Domańska, Ewa (2019.0)

ID do Documento: History-anthropogenic-soil-and-unbecoming-human

ABSTRACT ORIGINAL

This essay avows Chakrabarty's work on the humanities in the age of the Anthropocene, which shows us how climate change collapses the conventional dichotomy between natural and human history, forcing scholars to rethink basic categories of historical thought. It is concerned not with what happens in the air, above ground as it were, but with what is below the ground, the soil or the earth – anthropogenic soil change. Specifically, the essay focuses on the decomposition of human bodies and remains. Now, soil is one of the most diversified environments on the planet, a complex and living organism; and in cemeteries human remains are a part of this organism. Thus, while decomposing cadavers represent the loss of humanness from one perspective, from another perspective these dead bodies are, or become, elements of symbiotic collectivities made up of organic and inorganic being that are constantly commingling. The decomposition of human remains into humus not only marks a departure from the world of human collectivities, but also an entry into a collectivity of diverse forms of organic and inorganic existence. If Dipesh Chakrabarty shows us the complexity of what it means to be human in the era of the Anthropocene, this essay seeks to show us that humanity is a temporary state and a matter of degree, ever followed by unbecoming human.

The swarming life of pastures: living with vole outbreaks in the French Jura uplands

Meulemans, Germain (2020.0)

ID do Documento: The-swarming-life-of-pastures

ABSTRACT ORIGINAL

This paper addresses the joint becoming of landscapes, agricultural tasks and prairie rodents in the French Jura uplands, where the development of hay monoculture triggered outbreaks of water voles that reduce pastures to dust. I explore links between processual landscape anthropology and contemporary scholarship on more-than-human entanglements in order to follow how ecological disruptions called for the development of new arts of noticing towards multispecies life. I first describe the relationships between Jura farmers, voles, fields and agricultural modernisation programmes, and suggest that vole outbreaks bring these together around shared tasks. I then consider how disputes over how to control voles led to changes in farmers' ways of caring for their cows and tending the fields. I argue that these underlined changes in their ways of understanding and responding to the rhythms of the landscape's more-than-human activities. Finally, I draw on the example of conflicts between farmers over whether cows or pastures should be more central to their work. I make the case that to be attentive to fields as a landscape in the Jura is ultimately to define the (in)appropriateness of certain actions and tasks. It becomes constitutive of what 'good farming' should be, and precipitates new identities. © 2020 European Association of Social Anthropologists

Legibility and the {Legacy} of {Racialized} {Dispossession} in {Digital} {Agriculture}

Liu, Jen; Sengers, Phoebe (2021.0)

ID do Documento:

Let-me-show-you-A-caring-ethnography-of-embodied-knowledge-in-weaving-and-engineering

ABSTRACT ORIGINAL

This paper examines the causes and consequences of legibility as an organizing principle in the design of digital agriculture (DA) systems in the United States. Legibility refers to systems of governance that use simplified understandings of a situation to control and direct action upon it. Legibility in digital agriculture systems occurs at the confluence of two traditions of legibility: the data-driven model common in the design of digital systems, and tactics for the control of nature and labor that have developed in the United States since the foundation of the colonies. Our argument draws from (1) a historical analysis of broader patterns of agricultural technology and racialized land dispossession in what is now the United States and (2) empirical fieldwork that examines the adoption and maintenance of digital agriculture systems in rural New York State. We describe the role that legibility historically has played in the development of agricultural systems in the US, and their consequences for who is able to farm and how. This history raises the questions: What is made legible to whom? In that process, what becomes illegible? While legibility promises transparent and environmentally beneficial control, in our fieldwork we find that the demands of legibility are also restructuring the physical landscape, creating additional invisible labor, producing systems that are brittle to real-world conditions on farms, and creating opaque systems that block people from adapting to their circumstances. In reading our fieldwork together with the historical case, we demonstrate the pressures that are shaping the stakes, subject, and objects of legibility in agricultural technology. As more data-driven systems are used for environmental contexts, the CSCW community needs to extend its ways to understand how data-driven systems impact land, labor, and resources.

Governing pathological markets: Microbes, banana export markets, and speculative farming practices

Wang, Chi-Mao; Chien, Ker-Hsuan (2024.0)

ID do Documento: Governing-pathological-markets-Microbes-banana-export-markets

ABSTRACT ORIGINAL

This paper examines the making and remaking of Taiwan's banana export market in response to the devastation caused by an outbreak of a novel infectious plant disease, Fusarium wilt disease Tropical Race 4 (TR4, *Fusarium oxysporum* f. sp. *Cubense*). Taiwan was the world's fourth-largest exporter of bananas in the 1960s before the collapse of the market in the early 1970s. While scholars have drawn on actor-network theory-inspired performativity approach to understand the role of non-human actants in market-making, insufficient attention has been given to the distinct impacts of microbes on cases such as that of Taiwan's banana export market. Microbes' creative and ever-evolving qualities constantly present challenges related to the control and containment of such non-human entities, for which no pre-existing or universally applicable solutions exist. Consequently, there is a lack of research that provides useful frameworks to understand such disease-plagued markets. To bridge this gap in the literature, we examine the remaking of Taiwan's banana export market in the aftermath of the TR4 crisis using a case study approach and develop the notion of pathological markets. Inspired by recent scholarship on biosecurity and related care practices, we outline two characteristics that shape pathological markets: (a) speculative and probiotic care practices and (b) the rescaling of market organisations. The results of the fieldwork conducted as part of the present study in laboratories, government offices and on banana farms lead us to contend that the growth and development of particular microbes in multispecies environments such as Taiwan's banana farms constantly pose significant challenges for market farming. Moreover, to co-exist with the threats posed by the growth and development of microbes such as those which cause Fusarium wilt disease TR4, growers in Taiwan's banana export market rely heavily on probiotic and speculative care practices. © The Author(s) 2023.

A troubling view of food: tensions across matters of fact, concern, and care through the case of raw milk

Michael, Carolan (2025.0)

ID do Documento: A-troubling-view-of-food

ABSTRACT ORIGINAL

This paper explores the relationships, tensions, and contradictions inherent in the phrase “food is political.” The concepts of “matters of fact,” “matters of concern,” and “matters of care” play critical roles in the argument. Arguing that agrifood debates arise from diverse relationalities that sit in tensions according to their “style” of fact and concerns, a thought experiment is conducted, proposing a figure and four archetypes. While the exercise may seem contrived, its value lies in surfacing the various ways these matters—of fact, concern, and care, respectively—interrelate, often in tension. To convey this value and further illuminate each archetype, the conceptual framework is employed to unpack the raw milk debate, which has recently gained national (US) attention. The paper concludes by suggesting the abandonment of the view that agrifood systems must be “fixed,” proposing recommendations that embrace ambiguity and tension. We must stay with the trouble, representing a significant departure from agrifood imaginaries that continue to present phenomena such as food, health, disease, and nutrition in black-and-white (e.g., good/bad, safe/unsafe) terms. © The Author(s), under exclusive licence to Springer Nature B.V. 2025.

Dawn of the lively dead: Living queerly with rot in the sustainable city

Shcheglovitova, Mariya (2022.0)

ID do Documento: Dawn-of-the-lively-dead

ABSTRACT ORIGINAL

This paper presents stories of living with dead trees in Baltimore, MD, USA to explore where, how, and with whom tree remains become lively through art, farming, and sustainability practices. These stories serve as a lens into the diverse ways practitioners produce knowledge about death, healing, disease, and decay as they encounter and transform dead nonhuman bodies. I find that practitioners who work with dead trees develop a circular understanding of death, and propose that decay is needed to heal bodies and remediate land. However, when these everyday embodied knowledges are translated from the level of body to the city, death does not become lively for all. When circular ways of knowing death are applied to the city, death is displaced into vulnerable human and tree bodies showing that the sustainable city both relies on and (re)produces the binaries of life and death. As a contrast, practitioner knowledges of tree death suggest sustainabilities that trouble established notions of life and death, bodies and environments, health and disease, and decay and beauty. © 2020 Informa UK Limited, trading as Taylor & Francis Group.

Mobility, milieu and the ethos of care: Exploring the rabari camel-pastoralist kinship of Western India

Shah, Rhea; Maru, Natasha (2025.0)

ID do Documento: Mobility-milieu-and-the-ethos-of-care

ABSTRACT ORIGINAL

This paper throws light on how mobility is practiced as a unique form of care by the Rabari pastoralists of Gujarat. The Rabari-camel pastoralist communities traverse the arid lands of Western India to ensure the health of the camels, the vitality of the environment and the economic and social sustenance of the Rabari. It shows the ways that mobility is co-constituted, embodied and enacted as an inextricable relationship between the pastoralists, the animals and the environment. Through the practice of multi-species - mobility the conceptual and sensually experienced landscapes merge into the 'milieu'. It shows how the dynamic, sensory and material Rabari 'milieu' is threatened by mainstream spatial politics, and sedentist, dualist imaginaries. Finally, it examines how reciprocity and care is leveraged as a adaptation and survival mechanism under growing pressures. © 2025 Landscape Research Group Ltd.

Diverse methodologies of care: Thinking with and practising (soil) in situated, affective and enactive ways

Sharp, Emma L.; Yee, Kenzi; Makey, Leane; Fisher, Karen (2025.0)

ID do Documento: Diverse-methodologies-of-care

ABSTRACT ORIGINAL

This research article outlines a provocation for diverse and experimentally open, situated approaches to exploring care and caring. The diversely positioned authors discuss this idea using the subject of soil, in the place and context of Aotearoa New Zealand. Little is known about the diversity of ways that everyday people value, or, have caring relationships for/with soil, among a plethora of research that positions soil 'care' around, for example, commercial food production, waste-sinking, or property land value. To study diverse care in relation to soil, as with many relational subjects, requires equivalent diversity in the ways in which we might explore it. Here we outline the basis for diverse, situated methodologies that necessarily lead to a diversity of methods. This paper looks at the methodological imperatives that lead to exploring care, and discusses a variety of methods that generate different forms of 'data' with different forms of representation of that care. We observe that to holistically observe care relations with soil requires a diversity of methodologies, inherently ontological and epistemological – worldmaking. We discuss situated and enactive, affective approaches of Kaupapa Māori enquiry, monitoring and arts-based approaches to 'measure' soil care taking place, in place, and contextualise this with our own author positionality. We discuss this suite of experimental, reflexive, affective and responsive ways to measure soil care that are contingent on that being cared about, for, with and by, and which reciprocally give care. © 2024 Victoria University of Wellington and John Wiley & Sons Australia, Ltd.

Nature, care, and (re)productivity: feminist perspectives on sustainability and the bioeconomy

Hackfort, Sarah; Saave, Anna; Wichterich, Christa (2025.0)

ID do Documento: Nature-care-and-reproductivity

ABSTRACT ORIGINAL

This special issue addresses the bioeconomy, currently one of the most prominent concepts for transitioning toward sustainable economies. In the face of the rapidly accelerating climate crisis, the notion of the bioeconomy seems to offer an elegant solution for “greening” the economy while simultaneously maintaining economic growth and high productivity. Considering the existing feminist research on society-nature relations and the bioeconomy more specifically, it becomes clear that many bioeconomy policies fail to consider logics of care and lack a consistent focus on social justice and the preservation, promotion, and interaction of human and non-human reproductive capacities. Feminist research also provides a foundation for problematizing narratives of ecological modernization and green productivism. It highlights the need for a critique of sustainability narratives, which tend to gloss over the challenges of the bioeconomy rather than address them. This special issue considers this need, bringing together and into dialogue different feminist approaches and key concepts from feminist political ecology, ecofeminism, and feminist science and technology studies to critically analyze bioeconomy-related transformation processes. It also aligns with critical literature on society-nature relations in challenging the prevailing techno-optimism of bioeconomy policies and projects, which risk not just remaining oblivious to social and economic power asymmetries and (re)productivity but even opposing the latter. Focusing on the core concepts of care and (re)productivity contributes to the analysis of the (re)productive dimension of the bioeconomy as counterpart and in contrast to its inherent green productivism. © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Recipes for Technoutopia: On hospitality and infrastructure as experimental performance

Jordan, Stephanie Beth (2021.0)

ID do Documento: Recipes for Technoutopia

ABSTRACT ORIGINAL

Through the performance of a radical brunch, this piece explores questions of inclusion, responsibility, labor, and care in the development of infrastructure, mirroring the many ways in which the promises of big data will solve pressing problems of participation, climate, and culture. By conceiving of infrastructure as experimental performance, I assert that infrastructure is experiential, sensorial, political, and situated. Through the performance, new rich understanding arises that connects infrastructural moments of breakdown, maintenance, and repair to the politics of the personal, orientation, and disorientation, uncovering a dissonance between the rhetoric of future-building and the sociomaterial realities of the present. If the recipe for the future is a technoutopia, this experimental performance asks: what are its ingredients and the labor of its production? Who is the chef, the served, the hungry? What spoils utopias? © 2022 selection and editorial matter, Hannah Star Rogers, Megan K. Halpern, Dehlia Hannah, and Kathryn de Ridder-Vignone individual chapters, the contributors.

Development of soil mineral classification using ontology mining

Elumalai, Murali; Anuncia, S. Margret (2021.0)

ID do Documento: Development-of-soil-mineral-classification

ABSTRACT ORIGINAL

To maintain the quality of agriculture, soil minerals are playing an important role, and it is considered an unavoidable factor for crop growth. Soil is an organic matter together with mineral particles that help to regulate water, air, and other nutrients to the plants. For growing a quality crop, the composition of minerals needs to be appropriate to provide the sufficient level of nutrients. The composition of minerals is varied depending upon the types of crop being grown. Usually the knowledge about the composition is known to the experienced farmers and elders who try to designate the details to the followers. However, there is no guarantee for maximum level of knowledge transfer. Thus there arrives a demand for capturing these valuable knowledge for the agricultural novice. Due to the advent of new technology, capturing and storing these knowledge becomes easy and effective. Thus an attempt is made to capture the mineral content and its composition of soil to make it available as a single knowledge base. The designed knowledge base is expected to be available for novice farmers to guide and help them in the various agricultural processes and decision-making. In this paper, an ontology-based knowledge base is created for storing the details of soil composition with different minerals. The ontology provides a structured formalized knowledge which is mined for different patterns that subsequently concludes with recommendation of crops and the soil composition suitable for crops. A mining methodology is derived to fulfill the abovementioned objective. © 2021, Saudi Society for Geosciences.

Soils and Subsoils in STS: Technosciences and Underground Entities

Laurent, Brice; Merlin, Julien; Meulemans, Germain (2025.0)

ID do Documento: Soils-and-Subsoils-in-STS

ABSTRACT ORIGINAL

Today's ambitions for ecological and energy transition highlight the central role of the sites of production and extraction of natural resources, and connect soils and subsoils with technical, scientific, economic and political questions. As a result, soils and subsoils have been the subject of recent interest in science and technology studies (STS). This chapter begins by discussing the relationships between the critique of soil and subsoil sciences and (post)colonial issues. It presents works that have analyzed the transformation of soils and subsoils into resources. The chapter discusses the questions raised by the use of soils and subsoils in times of transition. Echoing Latour's invitation to develop a policy of the "terrestrial", soils and subsoils open up stimulating avenues of research for STS, consisting of exploring the making of the territory, in all its material, epistemic, economic and social thickness, as well as attachments to it. © ISTE Ltd 2025.

Soils and Subsoils in STS: Technosciences and Underground Entities

Laurent, Brice; Merlin, Julien; Meulemans, Germain (2025.0)

ID do Documento: Soils-and-Subsoils-STS

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Today's ambitions for ecological and energy transition highlight the central role of the sites of production and extraction of natural resources, and connect soils and subsoils with technical, scientific, economic and political questions. As a result, soils and subsoils have been the subject of recent interest in science and technology studies (STS). This chapter begins by discussing the relationships between the critique of soil and subsoil sciences and (post)colonial issues. It presents works that have analyzed the transformation of soils and subsoils into resources. The chapter discusses the questions raised by the use of soils and subsoils in times of transition. Echoing Latour's invitation to develop a policy of the "terrestrial", soils and subsoils open up stimulating avenues of research for STS, consisting of exploring the making of the territory, in all its material, epistemic, economic and social thickness, as well as attachments to it. © ISTE Ltd 2025.

Affect beyond the human: {Indian} agriculture in a multispecies world

Münster, Daniel (2019.0)

ID do Documento: Affect-beyond-the-human-Indian-agriculture-in-a-multispecies-world

ABSTRACT ORIGINAL

Transcultural perspectives have been effective in decolonizing epistemologies and representational strategies of the humanities and social sciences in their engagements with historical and contemporary dynamics of our globalizing world. As many chapters in this volume demonstrate, transcultural studies have destabilized many of the epistemological binaries and purities that underlie persistent constructions of race, culture, civilization, religion, language and nation (see König, Krämer, Maran and Tontini, all in this volume). This anti-essentialist stance has redirected humanities and social sciences, and, in particular, area and cultural studies, towards the study of encounters between people, ideas and practices, building on the awareness that actual encounters challenge putative purities of identity and belonging. Transcultural perspectives, focusing on encounters, have effectively highlighted the cultural dynamics at the interstices and in-betweens of entities – cultures, nations, religions or languages – that are central to humanistic studies, but have shaky ontological foundations. Transculturality is thus ultimately also a world-making project (see also Juneja, in this volume), it is a quest for adequate knowledge practices for the study of a world that is made up of relations rather than entities and essences. But does the relational perspective of transcultural studies and its study of encounters have to stop at the boundaries of the human? Are transcultural perspectives equipped to study relations made up of humans, technologies, plants, animals and other non-humans? What vocabulary and concepts do anthropological studies have available to talk, in a non-essentialist way, about more-than-human relationality?

Regenerating soil, regenerating soul: an integral approach to understanding agricultural transformation

Gosnell, Hannah (2022.0)

ID do Documento: Regenerating-soil-regenerating-soul

ABSTRACT ORIGINAL

Understanding what motivates farmers to adopt “climate-smart” regenerative practices is critical for developing the right policies, incentives, outreach, and support mechanisms. This article explores factors that motivated farmers in NSW Australia to transition from conventional to regenerative agriculture (RA), focusing on the role that their perceptions of agrochemicals and the microbiome played. Drawing on integral theory, the article takes a holistic approach to analyzing how farmer interiorities in personal and collective realms interacted with external behavior and the larger social-ecological system in which food and fiber is produced. A key finding is that negative experiences with agrochemicals associated with increasing costs and declining results were an important driver of change. Conversely, positive experiences learning about the microbiome and practicing ecological approaches to fertilization and pest control engendered enthusiasm and commitment to a transition away from high-input agriculture and a transformation in mindset. Further, conviviality associated with communities of practice, e.g. microscope groups, played an important role in the transition process, as farmers solidified new identities and participated in ongoing social learning. Based on these results, I argue that farmers’ feelings of kinship with nature (animals, plants, microbes) resulting from learning about and working with soil are underappreciated drivers of behavioral change and powerful leverage points for larger-scale social-ecological transformation. The integral model facilitates recognition of the connections between soil condition, farmers’ perceptions of and feelings about its condition, ensuing behavior including participation in new networks, and the creation of new norms, all of which create space for the emergence of institutional and systemic change. © 2021, The Author(s), under exclusive licence to Springer Japan KK, part of Springer Nature.

Web based software for the study of USDA soil taxonomy and classification of newly found soil

Das, Manoranjan; Malhotra, P.K.; Marwaha, Sudeep; Pandey, R.N. (2014.0)

ID do Documento: Web-based-software-for-the-study-of-USDA-soil-taxonomy

ABSTRACT ORIGINAL

United States Department of Agriculture (USDA) Soil Taxonomy is based on soil properties that can be objectively observed and measured in the natural conditions as they exist today. There are many soil classification systems but USDA Soil Taxonomy is most accepted worldwide. Ontologies are the new form of knowledge representation that acts in synergy with agents and Semantic Web Architecture. Soil ontology developed for USDA soil taxonomy has been used to develop a query interface that will help in detailed study of soil taxonomy, classification of new soil as well as exchange knowledge between software agents and systems. This is a web based application having N-tier architecture. Application development environment is NetBeans 6. 9 editor and Protégé. Web development technology is Java Server Pages (JSP). Programming languages JAVA and SPARQL are used for querying. Client interface is developed with Hyper Text Markup Language (HTML), Cascading Style Sheet (CSS) and JavaScript. Third tier of software consist of database which is in MS-SQL server 2005. Other two layers are Web Ontology Language (OWL) Ontology layer and Semantic Web Framework layer. OWL layer contains soil taxonomy information in the form of Ontology. Semantic Web Framework layer is implemented using JENA. In the search panel user can search anything related to USDA Soil Taxonomy, which comprises of twelve orders. However, this software contains information about seven soil orders reported in India. Domain experts can see and edit the knowledge base (i. e. Soil Ontology) or can suggest anything related to the creation of Soil Taxonomy Ontology through WebProtégé.

Web based software for the study of USDA soil taxonomy and classification of newly found soil

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ID do Documento: Web-based-software-USDA

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Studying a women's collective from a politics of commoning perspective: the case of Skouries, Greece

Rigkos-Zitthen, Ioannis; Kapitsinis, Nikos (2024.0)

ID do Documento: Studying-a-womens-collective-from-a-politics

ABSTRACT ORIGINAL

We live in a period of multiple crises, including economic, environmental, and social emergencies. The prioritisation of capitalist economic growth by most political institutions is among the most critical drivers of these crises in a never-ending loop. Against this context, commoning, perceived as collective care, presents an alternative political strategy to respond to those crises. In this paper, we investigate how an anti-mining community in Skouries, Greece, opposes the imposition of a large-scale mining project by forming a commons. Our commons is a women's collective that sought a sustainable economic alternative to large-scale mining and a model that could influence different political values much needed for politics in a crisis-ridden framework. We argue that commoning can enrich the politics of our times. Commoning highlights the importance of caring collectively for developing collective identities and relations between humans and the environment, which can benefit local, national, and international politics. © 2024 Informa UK Limited, trading as Taylor & Francis Group.

Connected seeds and sensors: Co-designing internet of things for sustainable smart cities with urban food-growing communities

Heitlinger, Sara; Bryan-Kinns, Nick; Comber, Rob (2018.0)

ID do Documento: Connected-seeds-and-sensors-Co-designing-internet-of-things-for-sustainable-smart-cities-with-urban-food-growing-communities

ABSTRACT ORIGINAL

We present a case study of a participatory design project in the space of sustainable smart cities and Internet of Things. We describe our design process that led to the development of an interactive seed library that tells the stories of culturally diverse urban food growers, and networked environmental sensors from their gardens, as a way to support more sustainable food practices in the city. This paper contributes to an emerging body of empirical work within participatory design that seeks to involve citizens in the design of smart cities and Internet of Things, particularly in the context of marginalised and culturally diverse urban communities. It also contributes empirical work towards non-utilitarian approaches to sustainable smart cities through a discussion of designing for urban diversity and slowness. © 2018 ACM.

The Sympoietic Orchard: Everyday Ways of Co-creating an Orchard

Louw, Bronwyn (2023.0)

ID do Documento: The-Sympoietic-Orchard

ABSTRACT ORIGINAL

What can sympoiesis look like in the dappled light of an urban orchards' everyday life? This chapter is an agripoetic inquiry in search of a sympoietic approach to orchard-making in a twenty-first century urban context. "Sympoiesis" is a twice-born neologism first forged by Dempster B: In: Allen JK and Wilby J (ed) Proceedings of the world congress of the systems sciences and ISSS, (2000) and then Haraway D: Staying With the Trouble. Making Kin in the Chthulucene, (2016) from the prefix sym (with) and the Greek root of poetry, poiesis (making). Sympoiesis is a way to name multispecies creation or (world)making. I will describe three approaches to sympoietic orchard-making I experimented in a communal orchard south of Paris: visiting, prolonging, and gardening possibilities. These immersive methods are not protocols I applied during fieldwork, but rather practices, or even postures, I came to embody in contact with the orchard, and then to name, with the help of orchard writing from poetry collections and gardening books, and in conversation with landscape thinkers such as Gilles Clément or Jean-Marc Besse and multispecies theorists in biology, ethology, philosophy, critical geography, and literature, including Donna Haraway, Vinciane Despret, Isabelle Stengers, Emanuele Coccia, Frédérique Aït Touati, and Marielle Macé. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2023.

Exploring {Democracy} in the {Anthropocene}: {The} {Case} of {Practices} of {Collective} {Care}

Rigkos-Zitthen, Ioannis; Granberg, Mikael (2024.0)

ID do Documento: Exploring-Democracy-in-the-Anthropocene

ABSTRACT ORIGINAL

What is philosophically interesting about how soil is managed and categorized? This paper begins by investigating how different soil ontologies develop and change as they are used within different social communities. Analyzing empirical evidence from soil science, ethnopedology, sociology, and agricultural extension reveals that efforts to categorize soil are not limited to current scientific soil classifications but also include those based in social ontologies of soil. I examine three of these soil social ontologies: (1) local and Indigenous classifications farmers and farming communities use to conceptualize their relationships with soil in their fields; (2) categorizations ascribed to farmers in virtue of their agricultural goals and economic priorities relied upon in sociological research; and (3) federal agency classifications used to manage soil. Using a perspectival realism framework, I argue that these social ontologies of soil are all partially successful in capturing aspects of soil properties, functions, and processes while also being limited in what they can represent. I use this analysis to advocate for a non-reductive interactionist approach to understanding soil ontologies where scientific and social ontologies of soil interact to improve soil management.

Human-managed soils and soil-managed humans: An interactive account of perspectival realism for soil management

Kendig, Catherine (2024.0)

ID do Documento: Human-managed-Soils

ABSTRACT ORIGINAL

What is philosophically interesting about how soil is managed and categorized? This paper begins by investigating how different soil ontologies develop and change as they are used within different social communities. Analyzing empirical evidence from soil science, ethnopedology, sociology, and agricultural extension reveals that efforts to categorize soil are not limited to current scientific soil classifications but also include those based in social ontologies of soil. I examine three of these soil social ontologies: (1) local and Indigenous classifications farmers and farming communities use to conceptualize their relationships with soil in their fields; (2) categorizations ascribed to farmers in virtue of their agricultural goals and economic priorities relied upon in sociological research; and (3) federal agency classifications of land capability employed by agricultural scientists. Studying the interplay of these social ontologies shows how assessing soil properties and capabilities are the result of previous agricultural strategies informed by culture, agroecological history, weather, soil biodiversity, crop rotation, and the goals held by decision-makers. The paper then identifies the soil relationships and interactions that constitute ontology-making activities. Building on recent work, I outline a novel interactive account of perspectival realism grounded in agricultural extension research and ethnopedological data that captures the haptic nature of farmers' soil strategies. This interactive account explains how ontologies are chosen, why they are chosen, and how they interact and inform soil management decision-making. The paper concludes by examining the values laden in these ontologies and those which are causally implicated in the choice of soil management strategies. © 2024 Author(s) published by the Journal of Social Ontology.

From coping strategy to hopeful everyday practice: Changing interpretations of food self-provisioning

Daněš, Petr; Sovová, Lucie; Jehlička, Petr; Vávra, Jan; Lapka, Miloslav (2022.0)

ID do Documento: From-coping-strategy-tohopeful-everyday-practice

ABSTRACT ORIGINAL

While alternative food networks (AFNs) have become the leading conceptualisation of sustainable food systems, vibrant scholarship on food self-provisioning (FSP) in Central and Eastern Europe has remained confined to the geopolitical region it investigates. This article brings these two bodies of thought closer together in two steps. First, we trace four framings of FSP deployed over the last three decades—coping strategy, cultural practice, hobby and source of good food and reading FSP as transformative practice—to demonstrate its progressive affinity with AFNs. Second, we follow the most recent framing in highlighting the material reality of local food production as a feature shared by both FSP and AFNs. From this perspective, FSP can be understood as a more radical variant of AFNs given its more substantial environmental and social impact (FSP is more widespread and socially inclusive and less dependent on market transactions). By uncovering the epistemological underpinnings of these different framings of FSP and exploring their implications for food practices on the ground, this article draws general lessons for scholarship aiming to advance food system transformation. © 2022 European Society for Rural Sociology.

Nonhuman labor and the making of resources making soils a resource through microbial labor

Krzywoszynska, Anna (2020.0)

ID do Documento: Nonhuman-labor-and-the-making-of-resources-making-soil

ABSTRACT ORIGINAL

With soils increasingly seen as living ecosystems, the understanding of the relationship between soils and agricultural labor is changing. A shift from working the soil to working with the soil is hoped to deliver a true ecological modernization of capitalist agriculture, making the production of ever-growing yields and the maintenance of healthy ecosystems co-constitutive. Drawing on ethnographic data from English farming, this article argues that the current trends are in fact a continuation of the logic of capitalist soil improvement in which soils are made into economic resources. By proposing a new conceptualization of labor as a material process of transformation oriented toward the generation of capital value, the author establishes a dialogue between hitherto separate literatures on the making of economic resources and on nonhuman labor. This approach transforms the debates on the relationship between nature and capital by productively collapsing the distinction between labor and resources. The author argues that acknowledging the material co-constitution of (any form of) labor and resource making allows us to better analyze the processes through which natures are rolled into capital. Today's enrollment of soil biota as labor thus opens up the whole biosphere to the logic of improvement, and to the operations of capital. © 2020 Anna Krzywoszynska.

Towards a semantic structure for classifying IoT agriculture sensor datasets : An approach based on machine learning and web semantic technologies

Lynda, Djakhdjakha; Brahim, Farou; Hamid, Seridi; Hamadoun, Cissé (2023.0)

ID do Documento: Towards-a-semantic-structure-for-classifying-IoT-agriculture

ABSTRACT ORIGINAL

With the increase in the number of IoT farming datasets, it has become so difficult to identify the right data for IoT agriculture applications. Therefore, a meaningful structure is needed to well understand, interpret and index IoT farming datasets. This paper proposes a new IoT farming ontology that allows the organization, the understanding, and the classification of IoT agriculture datasets knowledge as well as meta-data storage. For this, we have developed a new IoT agriculture taxonomy that helps to identify an IoT agriculture application based on the combination of various IoT agriculture sensors. The evaluation of the semantic IoT agriculture datasets classification, based on the background knowledge provided by the proposed ontology, was achieved using Machine Learning algorithms, including Logistic Regression, Decision Tree Classifier, K-Neighbors Classifier, Linear Discriminant Analysis, Gaussian NB, SVM, and Random Forest Regressor. The obtained results clearly show the effectiveness of the proposed ontology to classify IoT agriculture datasets with high performances and accuracy (0.98), (0.99) using Decision tree classifier and SVM respectively. © 2023 The Author(s)

Plants: Crop diversity pre-breeding technologies as agrarian care co-opted?

Graddy-Lovelace, Garrett (2020.0)

ID do Documento: Plants-Crop-diversity-pre-breeding-technologies

ABSTRACT ORIGINAL

Within the realm of international agricultural biodiversity conservation, there has been a surge of funding for “pre-breeding” of plant genetic resources for food and agriculture. Molecular high-throughput analysis, among other techniques, attempts to discern, document, and digitise the genomic traits of farmer/landrace varieties and crop wild relatives stored in gene banks to render them legible fodder for professional breeding. But pre-breeding necessitates thorough phenotypic evaluation and characterisation to understand the physiological attributes, heritable traits, and responses of a plant through its life cycle, under various growing and climactic conditions. This paper explores the irony that a range of surveillance technologies have been developed and deployed to mimic the agrarian work and skills of observing plants and attending to how they are faring, what they like and do not like over many seasons and contexts. These calls and technologies acknowledge the need for heedful attention to crops, even as they further displace actual farmers and their longstanding modes of selecting and saving open-pollinated seeds each harvest. Here, attending to crops entails remembering and communicating collectively gathered information of and from the plant. Such agrarian expertise of caring for plants has been systematically devalued and de-intellectualised, with gendered implications. Drawing on feminist geographies and political ecology, a landscape of care framework discloses the matrix of human and beyond-human care at work in cultivating agricultural biodiversity. Rather than ushering in a new valuation of this expertise, new pre-breeding technologies and trainings continue to ignore on-farm, plant-based care work and the farmers who do it. Calling out this contradiction could help re-centre such agrarian care skills as the crux to effective agricultural biodiversity utilisation. The proliferation of pre-breeding technologies could signify the co-optation of agrarian care skills or the opportunity to re-centre and revalue them. The information, practices and views in this article are those of the author(s) and do not necessarily reflect the opinion of the Royal Geographical Society (with IBG). © 2020 Royal Geographical Society (with the Institute of British Geographers).